

LLM Evaluation — Experimental Record

Appendix: Per-Case Outputs and Scores

Prepared by: **Mingtong Du**
Date: May 20, 2026

Purpose of this appendix. This document is the experimental record appendix for the Master’s thesis: “*Explainable AI for Household Transmission of SARS-CoV-2 and Identifying Risk Factors for Susceptibility in Sweden*”. It records the full LLM-generated outputs and evaluation scores for **all five frontier models** evaluated in this study, Claude Sonnet 4.5, GPT-5.3, Grok 4.2, DeepSeek V3.2, and Llama 4 across **ten household cases**. Evaluation uses a two-tier framework: four qualitative metrics (A1–A4, scored 0–2) and eight quantitative metrics (B1–B8, computed against the source `report.txt`). The **Plausibility** metric (A3) requires epidemiological expertise and was assessed by a qualified domain expert for **Claude Sonnet 4.5 only**; the automated LLM evaluator scored A3 for all five models. All other metrics were assessed for every model. Per-sample scoring tables are located at the end of each sample section.

Contents

1	Study Background and Task Description	4
1.1	Research context	4
1.2	Models evaluated	4
2	Evaluation Framework	5
2.1	Part A: Qualitative Metrics	5
2.2	Part B: Quantitative Metrics	6
3	Source Data Reports (Input to All Models)	8
3.1	Sample 1	8
3.2	Sample 2	12
3.3	Sample 3	17
3.4	Sample 4	22
3.5	Sample 5	24
3.6	Sample 6	26
3.7	Sample 7	30
3.8	Sample 8	32
3.9	Sample 9	35
3.10	Sample 10	39

4	Model Outputs and Evaluation Scores	41
4.1	Sample 1	42
4.1.1	Claude Sonnet 4.5	42
4.1.2	GPT-5.3	45
4.1.3	Grok 4.2	48
4.1.4	DeepSeek V3.2	50
4.1.5	Llama 4	52
4.2	Sample 2	56
4.2.1	Claude Sonnet 4.5	56
4.2.2	GPT-5.3	59
4.2.3	Grok 4.2	62
4.2.4	DeepSeek V3.2	64
4.2.5	Llama 4	66
4.3	Sample 3	70
4.3.1	Claude Sonnet 4.5	70
4.3.2	GPT-5.3	73
4.3.3	Grok 4.2	76
4.3.4	DeepSeek V3.2	78
4.3.5	Llama 4	80
4.4	Sample 4	85
4.4.1	Claude Sonnet 4.5	85
4.4.2	GPT-5.3	87
4.4.3	Grok 4.2	90
4.4.4	DeepSeek V3.2	91
4.4.5	Llama 4	94
4.5	Sample 5	97
4.5.1	Claude Sonnet 4.5	97
4.5.2	GPT-5.3	99
4.5.3	Grok 4.2	102
4.5.4	DeepSeek V3.2	104
4.5.5	Llama 4	106
4.6	Sample 6	110
4.6.1	Claude Sonnet 4.5	110
4.6.2	GPT-5.3	112
4.6.3	Grok 4.2	115
4.6.4	DeepSeek V3.2	117
4.6.5	Llama 4	120
4.7	Sample 7	124
4.7.1	Claude Sonnet 4.5	124
4.7.2	GPT-5.3	126
4.7.3	Grok 4.2	129
4.7.4	DeepSeek V3.2	130
4.7.5	Llama 4	133
4.8	Sample 8	137
4.8.1	Claude Sonnet 4.5	137

4.8.2	GPT-5.3	139
4.8.3	Grok 4.2	142
4.8.4	DeepSeek V3.2	143
4.8.5	Llama 4	146
4.9	Sample 9	150
4.9.1	Claude Sonnet 4.5	150
4.9.2	GPT-5.3	153
4.9.3	Grok 4.2	156
4.9.4	DeepSeek V3.2	157
4.9.5	Llama 4	160
4.10	Sample 10	165
4.10.1	Claude Sonnet 4.5	165
4.10.2	GPT-5.3	167
4.10.3	Grok 4.2	170
4.10.4	DeepSeek V3.2	171
4.10.5	Llama 4	174

1. Study Background and Task Description

1.1 Research context

This appendix is part of a Master’s thesis project investigating whether large language models can serve as interpretive intermediaries between machine-learning predictions and domain experts in public health.

The underlying study focuses on **within-household COVID-19 secondary transmission** in Sweden during the 2020 pandemic wave. A machine-learning system was built to predict two related outcomes:

1. **Household-level:** Will at least one secondary case of COVID-19 occur within a given household after an index case is detected?
2. **Individual-level:** Among the non-index members of a flagged household, which specific individuals does the model identify as most susceptible to infection?

The model used is **TabPFN** (Tabular Prior-Fitted Network), a transformer-based in-context learner trained on fold-1 of a 5-fold cross-validation split and evaluated on held-out data. Explainability is provided through KernelSHAP. Each model receives a structured `report.txt` containing household composition, SHAP attributions, and predicted probabilities, together with a chain-of-thought (CoT) prompting strategy, and produces a five-section policy-oriented plain-language explanation.

Important scope constraint. Reports describe *model behaviour*, not clinical ground truth. Decision thresholds are empirical ML values (F1-score optimised on the validation set) with no clinical meaning. Ground truth is intentionally excluded from all reports, reflecting prospective deployment conditions.

1.2 Models evaluated

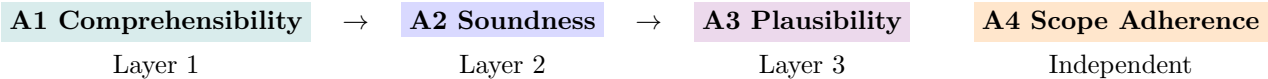
Five frontier LLMs are evaluated across all ten household cases: **Claude Sonnet 4.5**, **GPT-5.3**, **Grok 4.2**, **DeepSeek V3.2**, and **Llama 4**. Each model receives the same `report.txt` input and CoT prompt per case.

2. Evaluation Framework

A **two-tier evaluation framework** is used. Part A comprises four qualitative metrics assessed holistically (scored 0–2 each). Part B comprises eight quantitative metrics computed objectively against the source `report.txt`. The two tiers capture complementary failure modes: qualitative metrics detect subtle causal overclaims or imprecise mechanisms, while quantitative metrics provide reproducible, objective measurements of factual coverage and numerical fidelity.

2.1 Part A: Qualitative Metrics

The four qualitative metrics form a *layered dependency chain*: a score of zero on any lower layer caps all subsequent layers at zero.



Part A: Qualitative scoring rubric (0–2 per metric, max total 8)

Metric	Score	Criterion
A1 Comprehensibility No domain knowledge required. Is the text readable?	2	Unambiguous, well-structured; a general reader can follow the argument without difficulty.
	1	Mostly clear but contains vague, ambiguous, or poorly phrased statements.
	0	Incomprehensible, incoherent, or self-contradictory at the level of expression.
A2 Soundness Basic logic required. Is the reasoning consistent?	2	All reasoning steps are internally coherent; no logical contradictions.
	1	No direct contradictions but reasoning contains unexplained gaps or unsupported leaps.
	0	Contains a logical contradiction or internal inconsistency; <i>or</i> A1 = 0.
A3 Plausibility Epidemiological expertise required. Are causal claims valid?	2	Causal direction and mechanism fully consistent with established epidemiological knowledge.
	1	Causal direction correct but mechanism described imprecisely or too generically.
	0	Causal direction or mechanism contradicts domain knowledge; <i>or</i> A2 = 0.
A4 Scope Adherence Assessed independently. Does the model stay in scope?	2	All probability statements framed as model outputs; threshold described as empirical ML value; no prediction-correctness claims; all statements appropriately hedged.
	1	Minor scope slips: one sentence implying prediction correctness or treating probability as true infection risk.
	0	Explicit ground-truth claim, clinical threshold interpretation, or strong causal overclaim.

2.2 Part B: Quantitative Metrics

Quantitative metrics are computed objectively against the SHAP tables in the source `report.txt`. Let HH_top_k denote the set of top- k household SHAP features by $|\text{SHAP}|$, $\text{IND}_{i,\text{top}_k}$ the top- k individual SHAP features for member i , M_{HH} the household features mentioned, M_{IND_i} the features mentioned for member i , and N_{mem} the number of non-index members with individual predictions.

Part B: Quantitative metric definitions

Label	Name	Definition
B1	HH Top-2 Recall	$ M_{\text{HH}} \cap \text{HH_top}_2 / 2$
B2	HH Top-3 Recall	$ M_{\text{HH}} \cap \text{HH_top}_3 / 3$
B3	IND Top-2 Recall	Per-member: $ M_{\text{IND}_i} \cap \text{IND}_{i,\text{top}_2} / 2$
B4	IND Top-3 Recall	Per-member: $ M_{\text{IND}_i} \cap \text{IND}_{i,\text{top}_3} / 3$
B5	Mean IND Top-2 Recall	$\frac{1}{N_{\text{mem}}} \sum_i \text{IND}_{i,\text{Top2}}$
B6	Mean IND Top-3 Recall	$\frac{1}{N_{\text{mem}}} \sum_i \text{IND}_{i,\text{Top3}}$
B7	Direction Accuracy	Fraction of features for which effect direction stated correctly (sign of SHAP value)
B8	Faithfulness	Fraction of cited numerical values matching source within rounding precision

Coverage degradation flag: raised when any member’s B3 (Top-2 Recall) falls more than 33 percentage points below the mean B5.

3. Source Data Reports (Input to All Models)

Each subsection contains the model inference report (`report.txt`) for one of the ten household cases, exactly as provided to all five LLMs. The models also received a system prompt and a CoT user prompt (not reproduced here). Ground truth is excluded from all reports.

3.1 Sample 1

Household inference report - Sample 1

Household ID : 1215809
 Household size : 4 persons
 Index cases : ['10305057']
 Non-index mems : ['11039582', '535501', '7242083']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.6245
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : TRANSMISSION FLAGGED

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	index_to_household_ratio_std	0.2500	+0.0165	HIGHER	↑
2.	mean_DisplnkFam04_log_std	5613.0000	+0.0159	HIGHER	↑
3.	household_size	4	+0.0155	HIGHER	↑
4.	age_18_64_count	2	+0.0066	HIGHER	↑
5.	age_range_std	96.0000	+0.0049	HIGHER	↑
6.	min_age_2020_std	7.0000	+0.0043	HIGHER	↑
7.	max_DisplnkFam04_std	7484.0000	+0.0042	HIGHER	↑
8.	index_mean_age_2020_std	37.0000	+0.0033	HIGHER	↑
9.	median_Displnk04_std	3530.0000	+0.0018	HIGHER	↑
10.	min_Displnk04_std	0.0000	+0.0012	HIGHER	↑
11.	sd_DisplnkFam04_std	3742.0000	+0.0000	lower	↓
12.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
13.	proportion_children_std	0.2500	+0.0000	lower	↓
14.	three_generation	-	+0.0000	lower	↓
15.	total_Boarea_log_std	69.2000	+0.0000	lower	↓
16.	sd_Displnk04_std	1902.7590	+0.0000	lower	↓
17.	proportion_male_std	0.2500	+0.0000	lower	↓

18.	range_DisplnkFam04_std	7484.0000	+0.0000	lower	↓
19.	range_Displnk04_std	4166.0000	+0.0000	lower	↓
20.	total_Boarea_std	69.2000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (3 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (3 members) ...

MEMBER 11039582

P(susceptible) : 0.7383 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	7484.0000	+0.1147	HIGHER	↑
2.	Age in 2020	1984 (→ age 36)	+0.0697	HIGHER	↑
3.	log(disposable income)	4166.0000	+0.0538	HIGHER	↑
4.	hh_age_variance_std	-	+0.0081	HIGHER	↑
5.	hh_any_TRYGG_1	-	+0.0072	HIGHER	↑
6.	Boendeform_FlerHy	FlerBo	+0.0031	HIGHER	↑
7.	hh_male_count	-	+0.0030	HIGHER	↑
8.	hh_median_Displnk04_std	-	+0.0021	HIGHER	↑
9.	hh_Fodelseland_diversity	-	+0.0006	HIGHER	↑
10.	hh_three_generation	-	+0.0004	HIGHER	↑
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 535501

P(susceptible) : 0.6357 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	7484.0000	+0.1068	HIGHER	↑
2.	Age in 2020	1917 (→ age 103)	+0.0439	HIGHER	↑
3.	log(disposable income)	3742.0000	+0.0399	HIGHER	↑
4.	hh_male_count	-	+0.0271	HIGHER	↑
5.	hh_mean_DisInk04_std	-	-0.0257	lower	↓
6.	hh_min_age_2020_std	-	-0.0256	lower	↓
7.	hh_mean_age_2020_std	-	-0.0223	lower	↓
8.	hh_age_variance_std	-	+0.0166	HIGHER	↑
9.	hh_index_to_household_ratio_std	-	-0.0124	lower	↓
10.	hh_index_has_elderly	-	+0.0115	HIGHER	↑
11.	hh_three_generation	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_N06A	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓
MEMBER 7242083					
P(susceptible) : 0.2165 (threshold=0.30)					
Model flags : not flagged					
Top 20 features by model influence (SHAP):					
#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	0.0000	-0.0822	lower	↓
2.	Age in 2020	2013 (→ age 7)	-0.0777	lower	↓
3.	hh_max_DisInk04_std	-	-0.0181	lower	↓
4.	hh_crowding_index_std	-	-0.0164	lower	↓
5.	log(disposable income)	0.0000	-0.0131	lower	↓
6.	hh_mean_DisInk04_log_std	-	-0.0129	lower	↓
7.	hh_has_any_foreign_background	-	-0.0120	lower	↓
8.	hh_index_to_household_ratio_std	-	-0.0116	lower	↓
9.	hh_index_max_age_2020_std	-	-0.0082	lower	↓
10.	hh_proportion_children_std	-	-0.0070	lower	↓
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_R03A	-	+0.0000	lower	↓
13.	lmed_N06A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓

15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓
SUMMARY					
Household transmission P = 62.5% []					
Model flags household : YES - TRANSMISSION FLAGGED					
Person ID	Role	P(susceptible)		Model flag	
10305057	Index case	-		-	
11039582	Non-index member	[]	73.8%	SUSCEPTIBLE	
535501	Non-index member	[]	63.6%	SUSCEPTIBLE	
7242083	Non-index member	[]	21.6%	not flagged	

3.2 Sample 2

Household inference report - Sample 2

Household ID : 2432745
 Household size : 6 persons
 Index cases : ['10646251', '646375']
 Non-index mems : ['2470031', '197687', '11108007', '12177548']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.7886
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : TRANSMISSION FLAGGED

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	household_size	6	+0.0849	HIGHER	↑
2.	index_mean_age_2020_std	32.5000	+0.0297	HIGHER	↑
3.	Boendeform_SmahusBo	Ovr	+0.0249	HIGHER	↑
4.	index_cases_count	-	+0.0240	HIGHER	↑
5.	index_max_age_2020_std	45.0000	+0.0212	HIGHER	↑
6.	index_to_household_ratio_std	0.3333	+0.0169	HIGHER	↑
7.	mean_DisplnkFam04_log_std	4259.5000	+0.0139	HIGHER	↑
8.	Boarea_Person_std	14.5000	+0.0099	HIGHER	↑
9.	male_count	4	+0.0078	HIGHER	↑
10.	proportion_with_TRYGG_std	0.0000	+0.0050	HIGHER	↑
11.	proportion_elderly_std	0.0000	+0.0000	lower	↓
12.	proportion_female_std	0.3333	+0.0000	lower	↓
13.	sd_DisplnkFam04_std	3119.1669	+0.0000	lower	↓
14.	sd_Displnk04_std	2507.6246	+0.0000	lower	↓
15.	range_DisplnkFam04_std	6999.0000	+0.0000	lower	↓
16.	range_Displnk04_std	6562.0000	+0.0000	lower	↓
17.	proportion_male_std	0.6667	+0.0000	lower	↓
18.	total_Boarea_std	87.0000	+0.0000	lower	↓
19.	three_generation	-	+0.0000	lower	↓
20.	total_Boarea_log_std	87.0000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (4 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (4 members) ...

MEMBER 2470031

P(susceptible) : 0.5068 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	1991 (→ age 29)	+0.0384	HIGHER	↑
2.	hh_crowding_index_std	-	-0.0370	lower	↓
3.	hh_index_cases_count	-	+0.0354	HIGHER	↑
4.	log(disposable income)	2555.0000	+0.0329	HIGHER	↑
5.	hh_age_18_64_count	-	-0.0292	lower	↓
6.	lmed_R06A	-	+0.0186	HIGHER	↑
7.	hh_has_member_75plus	-	-0.0122	lower	↓
8.	lmed_N06A	-	-0.0115	lower	↓
9.	hh_total_Boarea_std	-	-0.0107	lower	↓
10.	hh_sd_DisplnkFam04_std	-	+0.0063	HIGHER	↑
11.	hh_three_generation	-	+0.0000	lower	↓
12.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
13.	lmed_N05C	-	+0.0000	lower	↓
14.	lmed_N02B	-	+0.0000	lower	↓
15.	lmed_M01A	-	+0.0000	lower	↓
16.	lmed_J01C	-	+0.0000	lower	↓
17.	lmed_R03A	-	+0.0000	lower	↓
18.	hh_proportion_elderly_std	-	+0.0000	lower	↓
19.	hh_proportion_children_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 197687

P(susceptible) : 0.5299 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	hh_index_to_household_ratio_std	-	+0.0416	HIGHER	↑
2.	Age in 2020	1978 (→ age 42)	+0.0375	HIGHER	↑
3.	hh_male_count	-	-0.0333	lower	↓
4.	log(disposable income)	2005.0000	+0.0299	HIGHER	↑

5.	hh_crowding_index_std	-	-0.0194	lower	↓
6.	hh_index_proportion_female_std	-	+0.0115	HIGHER	↑
7.	hh_prop_born_sweden_std	-	+0.0109	HIGHER	↑
8.	lmed_N02B	-	-0.0103	lower	↓
9.	hh_median_DispInk04_std	-	-0.0084	lower	↓
10.	hh_proportion_children_std	-	-0.0059	lower	↓
11.	hh_sd_DispInkFam04_std	-	+0.0000	lower	↓
12.	hh_three_generation	-	+0.0000	lower	↓
13.	lmed_R06A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_R03A	-	+0.0000	lower	↓
16.	lmed_N06A	-	+0.0000	lower	↓
17.	lmed_N05C	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓

MEMBER 11108007

P(susceptible) : 0.2213 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	2014 (+ age 6)	-0.1216	lower	↓
2.	log(family income)	0.0000	-0.1101	lower	↓
3.	hh_index_cases_count	-	+0.0541	HIGHER	↑
4.	log(disposable income)	0.0000	-0.0397	lower	↓
5.	Immigration background (encoded)	12.0000	-0.0132	lower	↓
6.	hh_proportion_children_std	-	-0.0061	lower	↓
7.	hh_index_proportion_foreign_std	-	-0.0060	lower	↓
8.	hh_min_DispInkFam04_std	-	-0.0050	lower	↓
9.	hh_age_18_64_count	-	-0.0047	lower	↓
10.	hh_prop_born_sweden_std	-	-0.0019	lower	↓
11.	hh_three_generation	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_N06A	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓

20. hh_proportion_with_TRYGG_std - +0.0000 lower ↓
 Waterfall plot → TabPFN_Inference_Output/2432745/individual_11108007_waterfall.png

MEMBER 12177548

P(susceptible) : 0.6229 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	6999.0000	+0.0698	HIGHER	↑
2.	Age in 2020	2002 (→ age 18)	+0.0403	HIGHER	↑
3.	hh_index_cases_count	-	+0.0303	HIGHER	↑
4.	hh_index_max_age_2020_std	-	+0.0250	HIGHER	↑
5.	hh_crowding_index_log_std	-	-0.0206	lower	↓
6.	hh_has_any_foreign_background	-	-0.0161	lower	↓
7.	hh_has_child_under_6	-	+0.0134	HIGHER	↑
8.	hh_median_DispInk04_std	-	+0.0042	HIGHER	↑
9.	hh_three_generation	-	+0.0015	HIGHER	↑
10.	hh_index_to_household_ratio_std	-	-0.0010	lower	↓
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

SUMMARY

Household transmission P = 78.9% []

Model flags household : YES - TRANSMISSION FLAGGED

Person ID	Role	P(susceptible)	Model flag
10646251	Index case	-	-
646375	Index case	-	-
2470031	Non-index member	[] 50.7%	SUSCEPTIBLE
197687	Non-index member	[] 53.0%	SUSCEPTIBLE

11108007	Non-index member	☐ 22.1%	not flagged
12177548	Non-index member	☐ 62.3%	SUSCEPTIBLE

3.3 Sample 3

Household inference report - Sample 3

Household ID : 7052823
 Household size : 5 persons
 Index cases : ['6697015']
 Non-index mems : ['1159155', '555428', '11162391', '6072977']

HOUSEHOLD-LEVEL PREDICTION

Model output : $P(\text{transmission}) = 0.8133$
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : TRANSMISSION FLAGGED

Computing household SHAP ...

Top 20 features by model influence ($|SHAP|$):

#	Feature	Value	SHAP	Pushes	risk
1.	household_size	5	+0.0522	HIGHER	↑
2.	index_max_age_2020_std	57.0000	+0.0481	HIGHER	↑
3.	mean_DisInkFam04_log_std	9350.0000	+0.0452	HIGHER	↑
4.	index_min_age_2020_std	20.0000	+0.0334	HIGHER	↑
5.	index_mean_age_2020_std	38.5000	+0.0206	HIGHER	↑
6.	sd_DisInkFam04_std	0.0000	+0.0180	HIGHER	↑
7.	age_65plus_count	0	+0.0171	HIGHER	↑
8.	index_cases_count	-	+0.0147	HIGHER	↑
9.	mean_DisInkFam04_std	9350.0000	+0.0134	HIGHER	↑
10.	age_IQR_std	37.0000	+0.0000	HIGHER	↑
11.	proportion_elderly_std	0.0000	+0.0000	lower	↓
12.	proportion_female_std	0.2000	+0.0000	lower	↓
13.	sd_DisInk04_std	1470.7370	+0.0000	lower	↓
14.	range_DisInkFam04_std	0.0000	+0.0000	lower	↓
15.	range_DisInk04_std	3332.0000	+0.0000	lower	↓
16.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
17.	proportion_male_std	0.8000	+0.0000	lower	↓
18.	total_Boarea_std	240.0000	+0.0000	lower	↓
19.	total_Boarea_log_std	240.0000	+0.0000	lower	↓
20.	three_generation	-	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (4 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (4 members) ...

MEMBER 1159155

P(susceptible) : 0.8523 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	9350.0000	+0.1051	HIGHER	↑
2.	Age in 2020	1963 (→ age 57)	+0.0851	HIGHER	↑
3.	hh_index_cases_count	-	+0.0443	HIGHER	↑
4.	hh_index_to_household_ratio_std	-	+0.0388	HIGHER	↑
5.	log(disposable income)	3152.0000	+0.0359	HIGHER	↑
6.	hh_index_max_age_2020_std	-	+0.0343	HIGHER	↑
7.	hh_crowding_index_log_std	-	+0.0114	HIGHER	↑
8.	hh_index_has_child	-	+0.0097	HIGHER	↑
9.	hh_gender_diversity	-	+0.0059	HIGHER	↑
10.	hh_prop_foreign_background_std	-	+0.0057	HIGHER	↑
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_proportion_children_std	-	+0.0000	lower	↓

MEMBER 555428

P(susceptible) : 0.6950 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	9350.0000	+0.1276	HIGHER	↑
2.	Age in 2020	1961 (→ age 59)	+0.1039	HIGHER	↑
3.	log(disposable income)	392.0000	+0.0199	HIGHER	↑

4.	AntalBarnUnder18_std	0.0000	-0.0171	lower	↓
5.	hh_mean_DispInk04_std	-	-0.0110	lower	↓
6.	hh_mean_DispInkFam04_log_std	-	+0.0093	HIGHER	↑
7.	hh_index_to_household_ratio_std	-	-0.0086	lower	↓
8.	Kon_std	1.0000	-0.0069	lower	↓
9.	hh_crowding_index_std	-	+0.0057	HIGHER	↑
10.	hh_proportion_children_std	-	-0.0032	lower	↓
11.	hh_three_generation	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_N06A	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓

MEMBER 11162391

P(susceptible) : 0.8438 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	9350.0000	+0.1360	HIGHER	↑
2.	Age in 2020	1967 (+ age 53)	+0.0788	HIGHER	↑
3.	hh_index_to_household_ratio_std	-	+0.0456	HIGHER	↑
4.	hh_prop_born_sweden_std	-	+0.0323	HIGHER	↑
5.	hh_index_cases_count	-	+0.0309	HIGHER	↑
6.	log(living area per person)	48.0000	+0.0230	HIGHER	↑
7.	hh_mean_DispInkFam04_log_std	-	+0.0207	HIGHER	↑
8.	hh_female_count	-	+0.0151	HIGHER	↑
9.	hh_mean_DispInkFam04_std	-	-0.0146	lower	↓
10.	hh_three_generation	-	+0.0000	lower	↓
11.	hh_sd_DispInkFam04_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_N06A	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓

19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓

MEMBER 6072977

P(susceptible) : 0.8027 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	hh_mean_DispInkFam04_log_std	-	+0.0874	HIGHER	↑
2.	hh_max_age_2020_std	-	+0.0481	HIGHER	↑
3.	hh_index_to_household_ratio_std	-	+0.0433	HIGHER	↑
4.	Age in 2020	2000 (→ age 20)	+0.0413	HIGHER	↑
5.	hh_three_generation	-	+0.0396	HIGHER	↑
6.	hh_has_elderly_65plus	-	+0.0342	HIGHER	↑
7.	hh_index_cases_count	-	+0.0285	HIGHER	↑
8.	Boendeform_SmahusAg	SmahusAg	+0.0276	HIGHER	↑
9.	hh_proportion_female_std	-	-0.0160	lower	↓
10.	hh_median_DispInk04_std	-	-0.0071	lower	↓
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	hh_sd_DispInk04_std	-	+0.0000	lower	↓
13.	hh_total_Boarea_std	-	+0.0000	lower	↓
14.	lmed_R06A	-	+0.0000	lower	↓
15.	lmed_M01A	-	+0.0000	lower	↓
16.	lmed_R03A	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_male_std	-	+0.0000	lower	↓
19.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓
20.	hh_range_DispInk04_std	-	+0.0000	lower	↓

SUMMARY

Household transmission P = 81.3% []

Model flags household : YES - TRANSMISSION FLAGGED

Person ID	Role	P(susceptible)	Model flag
6697015	Index case	-	-
1159155	Non-index member	[] 85.2%	SUSCEPTIBLE
555428	Non-index member	[] 69.5%	SUSCEPTIBLE

11162391	Non-index member	☐ 84.4%	SUSCEPTIBLE
6072977	Non-index member	☐ 80.3%	SUSCEPTIBLE

3.4 Sample 4

Household inference report - Sample 4

Household ID : 9878768
 Household size : 2 persons
 Index cases : ['6134964']
 Non-index mems : ['6775956']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.0671
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : no transmission flagged

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	1080.0000	-0.1334	lower	↓
2.	index_to_household_ratio_std	0.5000	-0.0869	lower	↓
3.	index_max_age_2020_std	28.0000	-0.0514	lower	↓
4.	index_mean_age_2020_std	28.0000	-0.0471	lower	↓
5.	household_size	2	-0.0372	lower	↓
6.	index_proportion_female_std	1.0000	-0.0349	lower	↓
7.	mean_DispInkFam04_std	1080.0000	-0.0274	lower	↓
8.	min_age_2020_std	3.0000	-0.0227	lower	↓
9.	median_DispInkFam04_std	1080.0000	-0.0215	lower	↓
10.	age_0_17_count	1	-0.0206	lower	↓
11.	range_DispInk04_std	2160.0000	+0.0000	lower	↓
12.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
13.	sd_DispInkFam04_std	1527.3506	+0.0000	lower	↓
14.	three_generation	-	+0.0000	lower	↓
15.	sd_DispInk04_std	1527.3506	+0.0000	lower	↓
16.	range_DispInkFam04_std	2160.0000	+0.0000	lower	↓
17.	proportion_male_std	0.5000	+0.0000	lower	↓
18.	proportion_female_std	0.5000	+0.0000	lower	↓
19.	total_Boarea_std	58.6000	+0.0000	lower	↓
20.	total_Boarea_log_std	58.6000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (1 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (1 members) ...

MEMBER 6775956

P(susceptible) : 0.0492 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	2017 (→ age 3)	-0.2172	lower	↓
2.	log(family income)	0.0000	-0.1129	lower	↓
3.	log(disposable income)	0.0000	-0.0570	lower	↓
4.	Fodelseland_freq_std	SVERIGE	-0.0108	lower	↓
5.	hh_min_age_2020_std	-	-0.0064	lower	↓
6.	Boendeform_SmahusAg	FlerHy	-0.0061	lower	↓
7.	hh_mean_DispInkFam04_log_std	-	-0.0058	lower	↓
8.	Immigration background (encoded)	21.0000	-0.0053	lower	↓
9.	lmed_R03A	-	-0.0026	lower	↓
10.	hh_proportion_with_TRYGG_std	-	-0.0024	lower	↓
11.	hh_three_generation	-	+0.0000	lower	↓
12.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
13.	lmed_R06A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

SUMMARY

Household transmission P = 6.7% []

Model flags household : no

Person ID	Role	P(susceptible)	Model flag
6134964	Index case	-	-
6775956	Non-index member	[] 4.9%	not flagged

3.5 Sample 5

Household inference report - Sample 5

Household ID : 13054736
 Household size : 2 persons
 Index cases : ['8327762']
 Non-index mems : ['12361865']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.2340
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : no transmission flagged

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	1610.0000	-0.0958	lower	↓
2.	household_size	2	-0.0509	lower	↓
3.	index_max_age_2020_std	26.0000	-0.0424	lower	↓
4.	mean_age_2020_std	26.0000	-0.0404	lower	↓
5.	is_spacious	-	-0.0299	lower	↓
6.	total_Boarea_std	58.0000	-0.0197	lower	↓
7.	Boendeform_FlerHy	FlerHy	-0.0186	lower	↓
8.	index_proportion_female_std	0.0000	-0.0112	lower	↓
9.	mean_DispInk04_log_std	1610.0000	-0.0073	lower	↓
10.	Fodelseland_diversity	1	+0.0000	HIGHER	↑
11.	proportion_elderly_std	0.0000	+0.0000	lower	↓
12.	proportion_female_std	0.0000	+0.0000	lower	↓
13.	sd_DispInk04_std	1356.2308	+0.0000	lower	↓
14.	range_DispInkFam04_std	1918.0000	+0.0000	lower	↓
15.	range_DispInk04_std	1918.0000	+0.0000	lower	↓
16.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
17.	proportion_male_std	1.0000	+0.0000	lower	↓
18.	total_Boarea_log_std	58.0000	+0.0000	lower	↓
19.	three_generation	-	+0.0000	lower	↓
20.	sd_DispInkFam04_std	1356.2308	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (1 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)					
Computing individual SHAP (1 members) ...					
MEMBER 12361865					
P(susceptible) : 0.3532 (threshold=0.30)					
Model flags : SUSCEPTIBLE					
Top 20 features by model influence (SHAP):					
#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	651.0000	-0.0333	lower	↓
2.	hh_mean_DispInkFam04_log_std	-	-0.0176	lower	↓
3.	hh_gender_diversity	-	-0.0161	lower	↓
4.	hh_min_age_2020_std	-	-0.0151	lower	↓
5.	Immigration background (encoded)	12.0000	-0.0134	lower	↓
6.	Kon_std	1.0000	-0.0075	lower	↓
7.	hh_male_count	-	-0.0068	lower	↓
8.	AntalBarnUnder18_std	0.0000	-0.0060	lower	↓
9.	hh_has_child_6_17	-	-0.0046	lower	↓
10.	hh_three_generation	-	-0.0023	lower	↓
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓
SUMMARY					
Household transmission P = 23.4% []					
Model flags household : no					
Person ID	Role	P(susceptible)	Model flag		
8327762	Index case	-	-		
12361865	Non-index member	[] 35.3%	SUSCEPTIBLE		

3.6 Sample 6

Household inference report - Sample 6

Household ID : 14260863
 Household size : 4 persons
 Index cases : ['3389243']
 Non-index mems : ['13538493', '9105013', '12108918']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.3756
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : TRANSMISSION FLAGGED

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	1383.2500	-0.0687	lower	↓
2.	index_max_age_2020_std	32.0000	-0.0221	lower	↓
3.	index_min_age_2020_std	32.0000	-0.0141	lower	↓
4.	mean_age_2020_std	18.7500	-0.0136	lower	↓
5.	has_elderly_65plus	-	-0.0132	lower	↓
6.	has_child_under_6	-	-0.0116	lower	↓
7.	total_Boarea_log_std	64.0000	-0.0102	lower	↓
8.	median_DispInk04_std	1335.5000	-0.0084	lower	↓
9.	Boendeform_SmahusAg	FlerBo	-0.0072	lower	↓
10.	index_proportion_female_std	1.0000	-0.0055	lower	↓
11.	proportion_elderly_std	0.0000	+0.0000	lower	↓
12.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
13.	sd_DispInkFam04_std	1599.1417	+0.0000	lower	↓
14.	sd_DispInk04_std	1599.1417	+0.0000	lower	↓
15.	range_DispInk04_std	2862.0000	+0.0000	lower	↓
16.	range_DispInkFam04_std	2862.0000	+0.0000	lower	↓
17.	proportion_male_std	0.7500	+0.0000	lower	↓
18.	proportion_female_std	0.2500	+0.0000	lower	↓
19.	three_generation	-	+0.0000	lower	↓
20.	total_Boarea_std	64.0000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (3 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (3 members) ...

MEMBER 13538493

P(susceptible) : 0.5490 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	1994 (→ age 26)	+0.0549	HIGHER	↑
2.	log(disposable income)	2862.0000	+0.0413	HIGHER	↑
3.	hh_age_variance_std	-	-0.0211	lower	↓
4.	hh_max_age_2020_std	-	+0.0158	HIGHER	↑
5.	hh_index_cases_count	-	-0.0155	lower	↓
6.	hh_multigenerational	-	-0.0140	lower	↓
7.	hh_prop_foreign_background_std	-	-0.0117	lower	↓
8.	AntalBarnUnder18_std	2.0000	+0.0086	HIGHER	↑
9.	hh_proportion_male_std	-	+0.0086	HIGHER	↑
10.	hh_mean_age_2020_std	-	+0.0062	HIGHER	↑
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	hh_sd_DispInk04_std	-	+0.0000	lower	↓
13.	hh_three_generation	-	+0.0000	lower	↓
14.	lmed_R06A	-	+0.0000	lower	↓
15.	lmed_R03A	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_with_TRYGG_std	-	+0.0000	lower	↓
20.	hh_range_DispInk04_std	-	+0.0000	lower	↓

MEMBER 9105013

P(susceptible) : 0.2676 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	0.0000	-0.0774	lower	↓
2.	log(disposable income)	0.0000	-0.0386	lower	↓
3.	hh_mean_DispInkFam04_log_std	-	-0.0199	lower	↓

4.	Boendeform_SmahusAg	FlerBo	-0.0188	lower	↓
5.	hh_index_max_age_2020_std	-	-0.0156	lower	↓
6.	hh_crowding_index_std	-	-0.0140	lower	↓
7.	hh_total_Boarea_log_std	-	-0.0103	lower	↓
8.	hh_index_has_child	-	-0.0078	lower	↓
9.	hh_TRYGG_1_sum	-	-0.0076	lower	↓
10.	hh_has_child_under_6	-	+0.0018	HIGHER	↑
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_N02B	-	+0.0000	lower	↓
15.	lmed_M01A	-	+0.0000	lower	↓
16.	lmed_J01C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 12108918

P(susceptible) : 0.0514 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	2015 (→ age 5)	-0.1836	lower	↓
2.	log(family income)	0.0000	-0.1051	lower	↓
3.	log(disposable income)	0.0000	-0.0412	lower	↓
4.	hh_index_max_age_2020_std	-	-0.0269	lower	↓
5.	hh_has_child_under_6	-	-0.0259	lower	↓
6.	hh_mean_DisplnkFam04_log_std	-	-0.0176	lower	↓
7.	hh_prop_foreign_background_std	-	-0.0074	lower	↓
8.	hh_three_generation	-	-0.0068	lower	↓
9.	hh_TRYGG_1_per_elderly_std	-	-0.0057	lower	↓
10.	hh_TRYGG_total_sum	-	-0.0042	lower	↓
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓

19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_children_std	-	+0.0000	lower	↓
SUMMARY					
Household transmission P = 37.6% []					
Model flags household : YES - TRANSMISSION FLAGGED					
Person ID	Role	P(susceptible)		Model flag	
3389243	Index case	-		-	
13538493	Non-index member	[]	54.9%	SUSCEPTIBLE	
9105013	Non-index member	[]	26.8%	not flagged	
12108918	Non-index member	[]	5.1%	not flagged	

3.7 Sample 7

Household inference report - Sample 7

Household ID : 14392369
 Household size : 2 persons
 Index cases : ['11841419']
 Non-index mems : ['12708868']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.2917
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : no transmission flagged

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	1423.5000	-0.0621	lower	↓
2.	household_size	2	-0.0553	lower	↓
3.	index_to_household_ratio_std	0.5000	-0.0471	lower	↓
4.	index_mean_TRYGG_total_std	0.0000	-0.0264	lower	↓
5.	sd_DispInk04_std	2011.7188	-0.0197	lower	↓
6.	age_variance_std	544.5000	-0.0142	lower	↓
7.	Boendeform_FlerHy	FlerHy	-0.0115	lower	↓
8.	mean_DispInkFam04_std	1423.5000	-0.0115	lower	↓
9.	gender_diversity	-	-0.0070	lower	↓
10.	AntalBarnUnder18_std	-	-0.0040	lower	↓
11.	proportion_elderly_std	0.0000	+0.0000	lower	↓
12.	proportion_female_std	1.0000	+0.0000	lower	↓
13.	sd_DispInkFam04_std	2013.1330	+0.0000	lower	↓
14.	range_DispInkFam04_std	2847.0000	+0.0000	lower	↓
15.	range_DispInk04_std	2845.0000	+0.0000	lower	↓
16.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
17.	proportion_male_std	0.0000	+0.0000	lower	↓
18.	total_Boarea_std	77.0000	+0.0000	lower	↓
19.	three_generation	-	+0.0000	lower	↓
20.	total_Boarea_log_std	77.0000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (1 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (1 members) ...

MEMBER 12708868

P(susceptible) : 0.5288 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	0.0000	-0.1091	lower	↓
2.	Age in 2020	2007 (→ age 13)	+0.0705	HIGHER	↑
3.	hh_index_to_household_ratio_std	-	+0.0441	HIGHER	↑
4.	hh_male_count	-	+0.0385	HIGHER	↑
5.	hh_age_variance_std	-	-0.0322	lower	↓
6.	hh_TRYGG_1_sum	-	+0.0185	HIGHER	↑
7.	hh_crowding_index_log_std	-	+0.0136	HIGHER	↑
8.	hh_index_proportion_foreign_std	-	+0.0106	HIGHER	↑
9.	hh_crowding_index_std	-	-0.0014	lower	↓
10.	lmed_R03A	-	+0.0000	lower	↓
11.	lmed_N06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_N02B	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	hh_proportion_female_std	-	+0.0000	lower	↓
18.	hh_proportion_elderly_std	-	+0.0000	lower	↓
19.	hh_proportion_children_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

SUMMARY

Household transmission P = 29.2% []

Model flags household : no

Person ID	Role	P(susceptible)	Model flag
11841419	Index case	-	-
12708868	Non-index member	[] 52.9%	SUSCEPTIBLE

3.8 Sample 8

Household inference report - Sample 8

Household ID : 18583221
 Household size : 3 persons
 Index cases : ['878991']
 Non-index mems : ['10795703', '12472756']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.2056
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : no transmission flagged

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	513.3333	-0.1790	lower	↓
2.	index_max_age_2020_std	28.0000	-0.0419	lower	↓
3.	has_member_75plus	-	-0.0207	lower	↓
4.	has_child_under_6	-	-0.0204	lower	↓
5.	index_min_age_2020_std	28.0000	-0.0178	lower	↓
6.	proportion_children_std	0.6667	-0.0160	lower	↓
7.	index_mean_Boarea_std	29.3000	-0.0155	lower	↓
8.	prop_born_sweden_std	-	-0.0119	lower	↓
9.	mean_DispInkFam04_std	513.3333	-0.0110	lower	↓
10.	Boarea_Person_std	27.7000	-0.0105	lower	↓
11.	proportion_female_std	0.3333	+0.0000	lower	↓
12.	range_DispInkFam04_std	1540.0000	+0.0000	lower	↓
13.	sd_DispInkFam04_std	889.1194	+0.0000	lower	↓
14.	three_generation	-	+0.0000	lower	↓
15.	range_DispInk04_std	1540.0000	+0.0000	lower	↓
16.	sd_DispInk04_std	889.1194	+0.0000	lower	↓
17.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
18.	proportion_male_std	0.6667	+0.0000	lower	↓
19.	total_Boarea_std	83.1000	+0.0000	lower	↓
20.	total_Boarea_log_std	83.1000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (2 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (2 members) ...

MEMBER 10795703

P(susceptible) : 0.1522 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	2014 (→ age 6)	-0.1562	lower	↓
2.	log(family income)	0.0000	-0.0785	lower	↓
3.	hh_index_max_age_2020_std	-	-0.0334	lower	↓
4.	hh_has_child_under_6	-	-0.0175	lower	↓
5.	hh_mean_DispInkFam04_log_std	-	-0.0144	lower	↓
6.	log(disposable income)	0.0000	-0.0096	lower	↓
7.	hh_age_range_std	-	-0.0072	lower	↓
8.	lmed_R03A	-	-0.0032	lower	↓
9.	hh_proportion_male_std	-	-0.0018	lower	↓
10.	hh_any_TRYGG_1	-	-0.0016	lower	↓
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_N05C	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 12472756

P(susceptible) : 0.0476 (threshold=0.30)

Model flags : not flagged

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	2019 (→ age 1)	-0.1627	lower	↓
2.	log(family income)	0.0000	-0.1317	lower	↓
3.	log(disposable income)	0.0000	-0.0487	lower	↓
4.	hh_mean_DispInkFam04_log_std	-	-0.0444	lower	↓

5.	hh_mean_DispInkFam04_std	-	-0.0128	lower	↓
6.	hh_gender_diversity	-	-0.0095	lower	↓
7.	hh_index_cases_count	-	-0.0081	lower	↓
8.	hh_age_0_17_count	-	-0.0079	lower	↓
9.	hh_range_DispInk04_std	-	-0.0016	lower	↓
10.	hh_all_foreign_background	-	-0.0008	lower	↓
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_elderly_std	-	+0.0000	lower	↓

SUMMARY

Household transmission P = 20.6% []

Model flags household : no

Person ID	Role	P(susceptible)	Model flag
878991	Index case	-	-
10795703	Non-index member	[] 15.2%	not flagged
12472756	Non-index member	[] 4.8%	not flagged

3.9 Sample 9

Household inference report - Sample 9

Household ID : 22954508
 Household size : 4 persons
 Index cases : ['10601784']
 Non-index mems : ['9806355', '420689', '265342']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.6564
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : TRANSMISSION FLAGGED

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	index_min_age_2020_std	16.0000	+0.0276	HIGHER	↑
2.	proportion_children_std	0.2500	+0.0189	HIGHER	↑
3.	mean_DisInkFam04_log_std	5428.0000	+0.0164	HIGHER	↑
4.	Boendeform_FlerHy	SmahusAg	+0.0130	HIGHER	↑
5.	total_Boarea_std	140.0000	+0.0098	HIGHER	↑
6.	TRYGG_1_sum	24	+0.0064	HIGHER	↑
7.	median_DisInkFam04_std	6104.0000	+0.0056	HIGHER	↑
8.	has_elderly_65plus	-	+0.0042	HIGHER	↑
9.	index_proportion_foreign_std	-	+0.0020	HIGHER	↑
10.	index_mean_DisInk04_log_std	106.0000	+0.0019	HIGHER	↑
11.	sd_DisInkFam04_std	3208.4956	+0.0000	lower	↓
12.	range_DisInkFam04_std	6570.0000	+0.0000	lower	↓
13.	proportion_elderly_std	0.0000	+0.0000	lower	↓
14.	three_generation	-	+0.0000	lower	↓
15.	range_DisInk04_std	1457.0000	+0.0000	lower	↓
16.	sd_DisInk04_std	717.3722	+0.0000	lower	↓
17.	proportion_with_TRYGG_std	0.2500	+0.0000	lower	↓
18.	proportion_male_std	0.5000	+0.0000	lower	↓
19.	proportion_female_std	0.5000	+0.0000	lower	↓
20.	total_Boarea_log_std	140.0000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (3 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)

Computing individual SHAP (3 members) ...

MEMBER 9806355

P(susceptible) : 0.5848 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	8037.0000	+0.0681	HIGHER	↑
2.	hh_index_max_age_2020_std	-	-0.0603	lower	↓
3.	Age in 2020	2001 (→ age 19)	+0.0260	HIGHER	↑
4.	log(disposable income)	509.0000	+0.0178	HIGHER	↑
5.	hh_index_has_child	-	+0.0169	HIGHER	↑
6.	hh_min_DispInkFam04_std	-	+0.0160	HIGHER	↑
7.	Homecare alarms	0	+0.0096	HIGHER	↑
8.	lmed_M01A	-	+0.0070	HIGHER	↑
9.	hh_index_min_age_2020_std	-	+0.0064	HIGHER	↑
10.	hh_sd_DispInkFam04_std	-	+0.0015	HIGHER	↑
11.	hh_total_Boarea_log_std	-	+0.0000	lower	↓
12.	lmed_R06A	-	+0.0000	lower	↓
13.	lmed_N05C	-	+0.0000	lower	↓
14.	lmed_N02B	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_R03A	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 420689

P(susceptible) : 0.7214 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	1969 (→ age 51)	+0.1128	HIGHER	↑
2.	log(family income)	4171.0000	+0.0753	HIGHER	↑
3.	log(disposable income)	1563.0000	+0.0463	HIGHER	↑

4.	hh_mean_DispInk04_std	-	-0.0212	lower	↓
5.	lmed_R06A	-	+0.0121	HIGHER	↑
6.	hh_index_has_child	-	+0.0070	HIGHER	↑
7.	hh_crowding_index_log_std	-	+0.0049	HIGHER	↑
8.	hh_min_DispInkFam04_std	-	+0.0041	HIGHER	↑
9.	log(living area per person)	35.0000	+0.0035	HIGHER	↑
10.	hh_has_child_under_6	-	+0.0008	HIGHER	↑
11.	hh_three_generation	-	+0.0000	lower	↓
12.	lmed_R03A	-	+0.0000	lower	↓
13.	lmed_N06A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_prop_foreign_background_std	-	+0.0000	lower	↓

MEMBER 265342

P(susceptible) : 0.6302 (threshold=0.30)

Model flags : SUSCEPTIBLE

Top 20 features by model influence (|SHAP|):

#	Feature	Raw value	SHAP	Pushes	risk
1.	Age in 2020	1976 (→ age 44)	+0.0925	HIGHER	↑
2.	AntalBarnUnder18_std	0.0000	-0.0234	lower	↓
3.	hh_prop_born_sweden_std	-	+0.0232	HIGHER	↑
4.	hh_has_child_under_6	-	+0.0147	HIGHER	↑
5.	hh_crowding_index_log_std	-	+0.0135	HIGHER	↑
6.	hh_max_DispInkFam04_std	-	+0.0104	HIGHER	↑
7.	hh_proportion_female_std	-	+0.0087	HIGHER	↑
8.	hh_age_IQR_std	-	+0.0070	HIGHER	↑
9.	Kon_std	2.0000	+0.0041	HIGHER	↑
10.	hh_age_range_std	-	+0.0037	HIGHER	↑
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_N05C	-	+0.0000	lower	↓
13.	lmed_R03A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	hh_total_Boarea_std	-	+0.0000	lower	↓
17.	lmed_N06A	-	+0.0000	lower	↓
18.	hh_proportion_male_std	-	+0.0000	lower	↓

19.	hh_proportion_elderly_std	-	+0.0000	lower	↓
20.	hh_proportion_children_std	-	+0.0000	lower	↓
SUMMARY					
Household transmission P = 65.6% []					
Model flags household : YES - TRANSMISSION FLAGGED					
Person ID	Role	P(susceptible)		Model flag	
10601784	Index case	-		-	
9806355	Non-index member	[]	58.5%	SUSCEPTIBLE	
420689	Non-index member	[]	72.1%	SUSCEPTIBLE	
265342	Non-index member	[]	63.0%	SUSCEPTIBLE	

3.10 Sample 10

Household inference report - Sample 10

Household ID : 23495841
 Household size : 2 persons
 Index cases : ['13545356']
 Non-index mems : ['7618294']

HOUSEHOLD-LEVEL PREDICTION

Model output : P(transmission) = 0.1264
 Decision threshold : 0.256 (empirical ML value - not a clinical cutoff)
 Model prediction : no transmission flagged

Computing household SHAP ...

Top 20 features by model influence (|SHAP|):

#	Feature	Value	SHAP	Pushes	risk
1.	mean_DispInkFam04_log_std	1240.0000	-0.0909	lower	↓
2.	TRYGG_total_sum	0	-0.0760	lower	↓
3.	index_to_household_ratio_std	0.5000	-0.0616	lower	↓
4.	max_age_2020_std	27.0000	-0.0464	lower	↓
5.	TRYGG_total_per_capita_std	0.0000	-0.0417	lower	↓
6.	mean_DispInkFam04_std	1240.0000	-0.0380	lower	↓
7.	Boendeform_SmahusBo	FlerHy	-0.0328	lower	↓
8.	proportion_children_std	0.5000	-0.0327	lower	↓
9.	range_DispInkFam04_std	2480.0000	-0.0038	lower	↓
10.	mean_age_2020_std	16.5000	-0.0003	lower	↓
11.	proportion_female_std	1.0000	+0.0000	lower	↓
12.	proportion_elderly_std	0.0000	+0.0000	lower	↓
13.	sd_DispInkFam04_std	1753.6248	+0.0000	lower	↓
14.	three_generation	-	+0.0000	lower	↓
15.	range_DispInk04_std	2480.0000	+0.0000	lower	↓
16.	sd_DispInk04_std	1753.6248	+0.0000	lower	↓
17.	proportion_with_TRYGG_std	0.0000	+0.0000	lower	↓
18.	proportion_male_std	0.0000	+0.0000	lower	↓
19.	total_Boarea_std	72.0000	+0.0000	lower	↓
20.	total_Boarea_log_std	72.0000	+0.0000	lower	↓

INDIVIDUAL-LEVEL PREDICTIONS (1 non-index members)

Decision threshold : 0.192 (empirical ML value - not a clinical cutoff)					
Computing individual SHAP (1 members) ...					
MEMBER 7618294					
P(susceptible) : 0.1744 (threshold=0.30)					
Model flags : not flagged					
Top 20 features by model influence (SHAP):					
#	Feature	Raw value	SHAP	Pushes	risk
1.	log(family income)	0.0000	-0.1085	lower	↓
2.	Age in 2020	2014 (→ age 6)	-0.1082	lower	↓
3.	log(disposable income)	0.0000	-0.0398	lower	↓
4.	hh_mean_DispInkFam04_log_std	-	-0.0266	lower	↓
5.	hh_has_child_6_17	-	+0.0252	HIGHER	↑
6.	hh_index_max_age_2020_std	-	-0.0213	lower	↓
7.	hh_all_foreign_background	-	-0.0110	lower	↓
8.	Homecare alarms	0	-0.0058	lower	↓
9.	hh_mean_DispInk04_log_std	-	-0.0041	lower	↓
10.	hh_max_DispInk04_std	-	-0.0013	lower	↓
11.	lmed_R06A	-	+0.0000	lower	↓
12.	lmed_R03A	-	+0.0000	lower	↓
13.	lmed_N06A	-	+0.0000	lower	↓
14.	lmed_M01A	-	+0.0000	lower	↓
15.	lmed_J01C	-	+0.0000	lower	↓
16.	lmed_N05C	-	+0.0000	lower	↓
17.	lmed_N02B	-	+0.0000	lower	↓
18.	hh_proportion_female_std	-	+0.0000	lower	↓
19.	hh_proportion_male_std	-	+0.0000	lower	↓
20.	hh_proportion_elderly_std	-	+0.0000	lower	↓
SUMMARY					
Household transmission P = 12.6% []					
Model flags household : no					
Person ID	Role	P(susceptible)	Model flag		
13545356	Index case	-	-		
7618294	Non-index member	[] 17.4%	not flagged		

4. Model Outputs and Evaluation Scores

Each subsection presents the outputs and full evaluation scores for one household case. **Claude Sonnet 4.5** outputs are reproduced in full (the model used for expert Plausibility assessment). Outputs for the remaining four models are reserved for insertion. All scores are pre-recorded; A3 (Expert) is scored only for Claude Sonnet 4.5.

4.1 Sample 1

4.1.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 1

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 1215809 | Model Analysis

1. CASE SUMMARY

This household comprises four individuals, including one confirmed COVID-19 index case and three exposed members. The model assigned a transmission score of 62% to this household—well above the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission likely. Among the three non-index members, the model flagged two as susceptible: a 36-year-old adult (member 11039582, score 74%) and a 103-year-old elderly person (member 535501, score 64%). The third member, a 7-year-old child (member 7242083), received a score of 22% and was not flagged. The model’s assessment appears heavily influenced by household income patterns and age composition.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 62%, substantially exceeding the 30% threshold, suggesting the model considers this household at elevated risk for within-household transmission. The assessment appears driven by a combination of household structure, income levels, and demographic composition.

Key factors driving the model’s household-level score:

Index case proportion (25% of the household—one of four members) was the single most influential factor, pushing the model’s score higher. This ratio suggests moderate but not severe index case density within the household.

Mean family income (approximately 561,000 SEK/year) emerged as the second-strongest driver, elevating the model’s score. This pattern is counterintuitive from a traditional epidemiological perspective, as higher income typically correlates with protective factors.

Household size (4 persons) contributed to the elevated score, consistent with the general principle that more household members create more transmission opportunities.

Age distribution features played notable roles: having two adults aged 18–64, a wide age range (96 years—spanning from a child to a centenarian), and a minimum household age of 7 years all pushed the score higher. The index case’s age (37 years) also contributed modestly.

Maximum family income (approximately 748,000 SEK/year) and median personal income (approximately 353,000 SEK/year) further elevated the score, reinforcing the pattern of income as a risk-increasing factor in the model’s assessment.

The pattern presents an interpretive challenge. While structural factors like household size and multigenerational composition align with known transmission risks, the model’s consistent weighting of higher income as a risk-increasing factor conflicts with typical health gradient expectations. This may indicate the model has learned correlations specific to Swedish households

with confirmed index cases in 2020 that do not reflect general causative mechanisms—for example, higher-income households might have had different occupational exposure patterns or testing access during that period.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 11039582 (Age 36)

The model assigned a susceptibility score of 74%, well above the flagging threshold, marking this as the highest-risk individual in the household according to the model’s assessment. Family income (approximately 748,000 SEK/year) was by far the dominant driver, contributing substantially to the elevated score. Age (36 years) was the second-strongest factor, pushing the score higher—the model appears to weight adults in their thirties as relatively higher risk. Personal disposable income (approximately 417,000 SEK/year) also elevated the score significantly. Additional factors included household age variance (reflecting the multigenerational structure), the presence of homecare alarm usage somewhere in the household, and residence in a multi-family dwelling (apartment building). The model’s strong emphasis on high income for this individual is epidemiologically puzzling, though the age and household structure patterns appear more conventional.

Member 535501 (Age 103)

The model assigned a susceptibility score of 64%, above the threshold, flagging this elderly household member as susceptible. As with the previous member, family income (approximately 748,000 SEK/year) was the strongest driver elevating the score. Age (103 years) contributed substantially—though notably less than income—consistent with known vulnerability patterns for the very elderly during the early pandemic period. Personal disposable income (approximately 374,000 SEK/year) also pushed the score higher. Household male count (number of males in the household) and age variance further elevated the score. Working against flagging were household mean income, minimum age, and mean age—all of which reduced the model’s score. The presence of an elderly index case in the household increased susceptibility. This individual’s profile presents competing signals: very advanced age points toward genuine vulnerability, while the protective influence of certain household averaging features suggests the model may be uncertain about how to weight individual versus aggregate characteristics.

Member 7242083 (Age 7)

The model assigned a susceptibility score of 22%, well below the threshold, and did not flag this child. The absence of personal family income (recorded as zero) was the strongest protective factor, substantially driving the score downward. Age (7 years) was nearly as influential, with the model strongly weighting young children as lower risk. Multiple household-level income features (maximum personal income, mean log-transformed income) further reduced the score, as did crowding index, the absence of foreign background in the household, index case proportion, index case age, and the proportion of children in the household. The model appears highly confident in not flagging this individual, with the score sitting far from the threshold. The protective effect assigned to young age aligns with early pandemic observations about lower

documented transmission risk among young children, though this pattern varied by viral variant and study design.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s protective weighting of very young age (7-year-old child) aligns with documented early-pandemic transmission patterns. The multigenerational household structure (age range of 96 years, spanning child to centenarian) as a risk-increasing factor is epidemiologically coherent, as such households face greater challenges in maintaining effective isolation between members.

Uncertain or problematic patterns: The model’s consistent treatment of higher income as risk-increasing across all flagged members contradicts established health gradient patterns and lacks obvious mechanistic explanation. This suggests the model may have learned spurious correlations from the 2020 Swedish training data—perhaps related to differential testing access, occupational patterns among higher earners during lockdown, or registry reporting biases. Domain experts should treat income-based flagging with skepticism.

Proxy features: Income variables may be capturing unmeasured factors beyond socioeconomic status—possibly occupation type, mobility patterns, or household contact networks that differed systematically during Sweden’s 2020 response. The presence of homecare services (TRYGG) likely proxies for frailty and care dependency, though its influence was modest in this case.

Information gaps: Experts would benefit from knowing: actual spatial layout and room usage patterns within the dwelling; timing of index case confirmation relative to symptom onset; occupational exposures and daily mobility patterns for adult members; and any pre-existing isolation measures. The very elderly member’s (103 years) functional status and care needs would be particularly relevant for assessing practical isolation feasibility.

Threshold proximity: The two flagged adults (74% and 64%) sit comfortably above the 30% threshold, while the child (22%) sits well below. No members occupy the model’s uncertain zone near the threshold, suggesting relatively clear-cut assessments within this household.

5. MODEL CONFIDENCE NOTE

The household score of 62% sits well above the 30% threshold, indicating the model is expressing a reasonably strong preference for flagging this household as high-risk—though this reflects the model’s internal pattern matching, not a verified transmission probability. Among individuals, the model shows clear confidence in all three assessments: the two adults are flagged well above threshold (74% and 64%), and the child sits far below it (22%), with no scores in the uncertain zone. However, the model’s reliance on income as a primary risk driver introduces interpretive uncertainty that cannot be resolved from these outputs alone. These predictions represent a limited reference point for qualified experts to integrate with clinical judgment and contextual knowledge—they are not standalone clinical decisions, and the model’s actual predictive accuracy for this household remains unknown.

4.1.2 GPT-5.3

GPT-5.3 — Sample 1

1. CASE SUMMARY

This household consists of four people: one identified index case and three non-index household members. The household-level model assigned a transmission score of 62.5%, which is well above the model’s internal flagging threshold of 25.6%, meaning the model classified this household as higher-risk for within-household transmission according to its learned statistical patterns.

At the individual level, the model assigned relatively high susceptibility scores to two non-index members: one adult aged 36 (73.8%) and one very elderly adult aged 103 (63.6%). A third member, a 7-year-old child, received a substantially lower score (21.6%) and was not flagged by the model. These outputs reflect the model’s internal assessment patterns rather than verified infection outcomes or true biological risk.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level model score of 62.5% was substantially above the empirical threshold of 25.6%, indicating that the model viewed this household profile as relatively consistent with patterns associated with secondary household spread in the training data.

Several household characteristics contributed most strongly to the model’s elevated score:

- **Household size (4 persons)** increased the model score. Larger households provide more opportunities for close interpersonal contact, which is epidemiologically plausible as a structural exposure factor.
- **Presence of two adults aged 18–64 and a very wide age range (96 years)** also pushed the score upward. This suggests the model associated multigenerational structure with increased transmission potential. The large age spread likely reflects the coexistence of a young child and a very elderly adult in the same household.
- The model also weighted the **youngest household member being age 7** as a factor increasing risk. This may indicate that the model learned associations involving school-age children and household mixing patterns.
- Multiple **income-related variables** strongly increased the model score, including mean household disposable income and maximum family income. These findings are harder to interpret directly. Income variables in registry-based models often function as indirect proxies for housing conditions, occupational exposure, social mobility, or healthcare access rather than income itself being causally related to transmission.

The overall pattern is partially coherent from a public health perspective. Household size, multigenerational structure, and the presence of both children and elderly individuals align with known epidemiological mechanisms that can increase within-household exposure opportunities. However, the strong positive contribution from higher income measures is less straightforward and should be interpreted cautiously as a possible proxy effect rather than a direct risk factor. Notably, crowding and homecare-use variables contributed little to this specific household

prediction, suggesting the model’s assessment was driven more by household composition and demographic structure than by apparent frailty or housing-density indicators.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Non-index member 11039582 (age 36)

The model assigned this person a susceptibility score of 73.8%, well above the individual flagging threshold.

The strongest influences on the model’s score were:

- **Higher household/family income measures**, which strongly increased the model score. As above, these variables likely operate as indirect markers of social or structural conditions rather than direct biological risk.
- **Adult age (36 years)** also substantially increased the score. The model appears to associate middle adulthood with higher susceptibility within this household context.
- **Household age diversity and multigenerational structure** further increased the score, suggesting the model viewed this person as embedded within a complex intergenerational exposure environment.

The pattern is moderately plausible from a domain perspective because adults in multigenerational households may occupy central contact roles between children and elderly relatives. However, the prominence of income-related variables makes interpretation less certain.

Non-index member 535501 (age 103)

The model assigned this individual a susceptibility score of 63.6%, also above threshold.

Key factors increasing the score included:

- **Advanced age (103 years)**, which substantially raised the model score. This is epidemiologically plausible because advanced age is often associated with frailty and increased vulnerability in infectious disease models.
- **Higher income-related variables** again pushed the score upward.
- The model also weighted the household’s **age variability** and the presence of an **elderly index-related structure** as increasing susceptibility.

At the same time, some household-level age features reduced the score slightly, suggesting competing influences within the model’s reasoning. Overall, the age-related pattern appears domain-consistent, though the contribution of income remains difficult to interpret mechanistically.

Non-index member 7242083 (age 7)

This child received a susceptibility score of 21.6%, below the model threshold, so the model did not flag this person.

The strongest influences all reduced the model score:

- **Young age (7 years)** strongly lowered the score.

- **Very low or absent individual income values** also lowered the score, likely reflecting the expected economic profile of a child.
- Household-level factors such as lower crowding influence and child composition variables further reduced the model assessment.

This pattern is broadly plausible because the model appears to distinguish children from adults and elderly individuals in its learned susceptibility structure. However, because the score lies only modestly below threshold, this should still be viewed as an uncertain rather than definitive separation.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear epidemiologically coherent, particularly the emphasis on:

- larger household size,
- multigenerational structure,
- coexistence of children and elderly members,
- and advanced age in one household member.

These factors align with plausible exposure and vulnerability mechanisms in household transmission settings.

At the same time, the repeated importance of income-related variables suggests the model may be using socioeconomic status as a proxy for unmeasured factors such as occupation, mobility patterns, housing quality, care access, or social network structure. Without additional contextual variables, these associations are difficult to interpret causally.

The apparent lack of influence from crowding measures in this case is somewhat notable, since crowding is often expected to matter in household spread. This could indicate that the model prioritised demographic composition over physical housing density for this specific household profile.

Before acting on these outputs, a domain expert would likely want:

- actual housing layout information,
- occupational exposure data,
- vaccination or prior infection status,
- timing of symptom onset,
- behavioural contact patterns,
- and clinical vulnerability indicators not captured in registry data.

The child’s score sits relatively close to the individual threshold compared with a clearly low-risk prediction, so that classification should be considered somewhat uncertain.

5. MODEL CONFIDENCE NOTE

The household-level score was substantially above the model threshold, suggesting the model’s classification at the household level was relatively strong rather than borderline. Two individual

scores were also clearly above threshold, while the child’s score was closer to the cutoff and therefore more uncertain.

However, these outputs remain model-generated statistical assessments rather than verified infection risks or confirmed outcomes. The correctness of the predictions is unknown, and the results should be treated only as a limited decision-support reference for qualified experts interpreting the model’s behaviour.

4.1.3 Grok 4.2

Grok 4.2 — Sample 1

CASE SUMMARY

This is a 4-person household (ID 1215809) with one identified index case. The model assigns a 62.5% probability of household transmission and flags the household as one where additional cases are expected according to its learned patterns. Among the three non-index members, two are flagged as higher susceptibility by the individual-level model (one at 73.8% and one at 63.6%), while the third receives a lower score of 21.6% and is not flagged. The flagged individuals include a middle-aged adult and a very elderly person; the non-flagged member is a young child. These outputs reflect how the model weighs Swedish registry-derived characteristics for this specific household profile.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 62.5%, which exceeds the empirical decision threshold of 0.256 and results in a transmission flag.

The top factors pushing the model’s score higher include:

- The **index-to-household ratio** (0.25), indicating one index case in a four-person home, which the model associates with elevated transmission potential.
- **Household disposable income measures** (particularly the mean family income around 561,300 SEK annually), where higher values in this household contributed positively to the risk score.
- **Household size of 4**, which the model treats as increasing the chance of within-home spread.
- Moderate age spread (standard deviation and range) and the presence of two adults aged 18–64.

Overall, the model’s reasoning combines structural factors (larger household size and some age diversity) with socioeconomic markers (higher family income). This partially aligns with known epidemiological patterns—larger households can facilitate transmission, and age mixing can create bridges between age groups with different behaviors and risks. However, the positive influence of higher income stands out as potentially acting as a proxy for other unmeasured variables (such as urban living patterns, occupation, or testing behavior in 2020) rather than a direct protective or risk factor. The pattern is coherent in structure but merits expert scrutiny regarding what income may represent in this context.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 11039582 (age ~36 in 2020): The model assigns a 73.8% susceptibility score and flags this person. The strongest upward drivers are higher personal and family income figures, adult age in the 18–64 range, and household-level age variation. Housing type (multi-family dwelling) also contributed modestly upward. From a domain view, the model appears to link working-age adults in higher-income or certain housing settings with higher susceptibility scores, possibly reflecting mobility or contact patterns common in 2020.

Member 535501 (age ~103 in 2020): The model assigns a 63.6% susceptibility score and flags this individual. Key upward influences include high family income, advanced age, male gender composition in the household, and the presence of an elderly index or household member. Offsetting factors (some household age averages) pulled slightly downward but not enough to avoid a flag. The model’s emphasis on very advanced age is plausible, as frailty and care needs often correlate with vulnerability, and homecare-related variables also appear in the broader profile.

Member 7242083 (age ~7 in 2020): The model assigns a 21.6% susceptibility score, below the 0.192 threshold, so this child is not flagged. Strong downward drivers include low or zero personal/family income (typical for a child), young age, lower household crowding, and absence of certain background or index characteristics. The model consistently down-weights young children in this configuration, which may reflect lower recorded contact or different exposure patterns captured in the training data.

EXPERT CONSIDERATIONS

The household flag combined with two individual flags directs model attention to this home, particularly the adult and elderly members. Domain-consistent elements include the roles of household size, age diversity, and elderly presence. Income and housing type emerge as influential but likely serve as proxies for occupation, urban density, mobility, or healthcare-seeking behavior rather than direct causal drivers. The medication history variables (antibiotics, respiratory drugs, etc.) had minimal influence here, suggesting the model is not heavily relying on recorded comorbidity proxies for this case.

Experts should consider additional context such as actual testing history, occupation, school/-daycare attendance, mask or ventilation practices, and exact timing of the index case—none of which are available to the model. The child’s score is well below threshold (low uncertainty), while the elderly person’s 63.6% score sits in a more intermediate zone. These outputs can help prioritise households for further investigation but require integration with on-the-ground public health intelligence.

MODEL CONFIDENCE NOTE

The household probability (62.5%) is substantially above threshold, and the two flagged individuals are also clearly above their respective cutoffs, suggesting the model is relatively decisive for this case rather than sitting in an uncertain boundary zone. The third member’s low score further sharpens the model’s differentiation within the home. As with all such outputs, these are statistical pattern recognitions from 2020 registry data, not verified infection risks or clinical predictions. They serve best as one limited reference tool to support, never replace, expert

judgment and targeted public health action.

4.1.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 1

Here is the structured interpretation of the model’s output for the household, written for public health policy experts.

1. CASE SUMMARY

This household consists of four people: one identified “index case” (the initial person of interest) and three other members. The model’s task is to flag households where further COVID-19 transmission might occur, based on statistical patterns in registry data. For this household, the model assigns a **62.5% probability** of within-household transmission, which exceeds its internal flagging threshold. Among the three non-index members, the model flags two as “susceptible” (with scores of 73.8% and 63.6%) and one as not flagged (21.6%). **These are model outputs, not confirmed infections.**

2. HOUSEHOLD MODEL ASSESSMENT

The model flagged this household for potential transmission. The key factors that most influenced this decision were:

- **Index-to-household ratio (strongest influence):** The household has one index case and three others. The model weighted having a single index case relative to household size as a factor pushing toward a higher transmission score.
- **Household disposable income (second strongest):** The household’s annual disposable income is relatively high (approx. 561,300 SEK). The model associated this with a higher risk score — a pattern that may reflect income acting as a proxy for unmeasured factors (e.g., housing layout, mobility, or occupation), rather than income directly causing transmission.
- **Household size (third strongest):** A size of four persons moderately pushed the score higher, which aligns with general expectations that more contacts increase transmission opportunities.

Coherence check: The top factors — index case share, income, and household size — do not form an obvious epidemiological pattern. While household size is plausible, the strong positive influence of higher income is unexpected (lower-income households are often at higher infectious disease risk due to crowding or job type). This suggests the model may be capturing statistical correlations in the registry data that are not obviously causal. Experts should treat this pattern as uncertain rather than clinically intuitive.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 11039582 (age 36) – Flagged (73.8%) The model’s score for this person was most strongly pushed higher by:

- **Household disposable income (high):** Again, the model associated higher household income with a higher susceptibility score.
- **Age 36:** The model weighted middle adult age as increasing susceptibility relative to younger or much older ages in this household.
- **Personal disposable income (high):** The model added further upward weight from the individual’s own income level.

Plausibility: The strong income effect is consistent across household and individual levels but remains surprising from a traditional infectious disease perspective.

Member 535501 (age 103) – Flagged (63.6%) The most influential factors for this very elderly person were:

- **Household disposable income (high):** Same upward pattern.
- **Age 103:** The model weighted advanced age as pushing the score higher.
- **Sex (male):** Being male added upward weight (the model shows a separate “male count” influence).
- **Several features pushed the score lower:** The model’s score was reduced by household mean income (different from total income), younger minimum age in the household, and the index-to-household ratio.

Plausibility: Advanced age as a risk factor is domain-consistent, though the model’s overall score for this 103-year-old (63.6%) is notably lower than for the 36-year-old (73.8%), which may seem counterintuitive. This highlights that the model weighs combinations of features, not age alone.

Member 7242083 (age 7) – Not flagged (21.6%) This child’s low score was primarily driven downward by:

- **Household disposable income (very low value – 0 in raw data, possibly missing or zero income):** The model treated this as a strong downward factor.
- **Age 7:** Young age was associated with a lower susceptibility score.
- **Low personal disposable income (also zero):** Reinforced the downward push.

Plausibility: The model consistently assigning lower susceptibility to a young child is plausible. However, the zero-income values may represent data anomalies (e.g., imputed or missing values) rather than true economic status — a point for expert review.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Household size, advanced age for the 103-year-old, and low susceptibility for a young child are intuitive.
- **Uncertain / surprising patterns:** Higher household disposable income repeatedly pushed the model’s scores higher across multiple levels. This is not a standard epidemiological risk factor for household transmission and may reflect confounding (e.g., income correlated with testing access, mobility, or housing type not captured by floor area).

- **Proxy variables:** Income appears to be acting as a proxy for unmeasured socioeconomic or behavioural factors. Homecare alarms (TRYGG) did not appear in the top influences for this household, despite being available in the model.
- **Additional information needed before action:** Experts would need to know whether the model's income-related pattern replicates in other households and whether zero-income values (as for the 7-year-old) are genuine or data artifacts. No operational decision should be made based solely on these outputs.
- **Uncertain zone:** None of the scores are very close to the thresholds (household threshold 0.256 vs. 0.625; individual thresholds 0.192 vs. scores of 0.738, 0.636, 0.216). The model is relatively confident in its flagging decisions for this household.

5. MODEL CONFIDENCE NOTE

The model's probability estimates are well above (or below) the empirical flagging thresholds, indicating that it makes these classifications with moderate statistical confidence — but **confidence in model output is not the same as correctness**. We do not know whether transmission actually occurred in this household. These results should be treated as one statistical reference for expert review, not as a clinical or public health decision.

4.1.5 Llama 4

Llama 4 — Sample 1

COVID-19 Household Transmission Model Analysis

Household ID: 1215809

Case Summary

This is a 4-person household where the model has identified a 62.5% probability of within-household COVID-19 transmission. The household includes one index case and three exposed members of varying ages (approximately 7, 36, and 103 years old). Based on the household's characteristics, the model flags this as a high-risk transmission scenario. Of the three non-index members, the model identifies two individuals (ages 36 and 103) as susceptible, while the 7-year-old child receives a lower susceptibility score that falls below the model's flagging threshold.

Household Model Assessment

The model assigns a 62% transmission probability to this household, significantly above the 25.6% threshold. Three primary factors drive this assessment:

Index case concentration (0.25 ratio): With one infected person in a 4-person household, the model weighs this as a moderate exposure scenario. A single index case means focused transmission risk rather than multiple simultaneous sources.

Family income level (561,300 SEK annually): The model identifies this mid-to-high income as a risk-increasing factor. This might seem counterintuitive, but could reflect patterns in the

training data around social mobility, workplace exposure, or living arrangements common at this income level.

Household size (4 people): A four-person household creates multiple transmission pathways while maintaining close daily contact typical of family units.

The pattern here is interesting—the model isn’t primarily responding to crowding or healthcare dependency (homecare services show zero influence). Instead, it’s picking up on demographic and socioeconomic structure. This suggests the model learned associations between household composition patterns and transmission risk in the Swedish registry data, though the mechanism isn’t immediately obvious from an epidemiological standpoint.

Individual Susceptibility Profiles

Member 11039582 (Age 36) — 73.8% susceptibility, FLAGGED

The model strongly flags this adult, driven primarily by:

- **Family income (748,400 SEK):** The highest single contributor to this person’s score. This elevated household income appears throughout as a risk marker, potentially proxying for factors like commuting patterns or social networks.
- **Age (36 years):** Working-age adults may have been observed in the training data to have higher exposure or transmission rates compared to elderly or very young household members.
- **Personal income (416,600 SEK):** Individual economic activity reinforces the model’s assessment.

The model’s reasoning here forms a coherent profile of an economically active household member, though it’s worth noting that income as a risk factor may be capturing unmeasured behavioral or structural variables.

Member 535501 (Age 103) — 63.6% susceptibility, FLAGGED

Despite advanced age, this member receives a moderate-high score:

- **Family income** remains the strongest driver, following the same pattern.
- **Age (103 years):** Advanced age contributes to the score, though less dramatically than you might expect if age were purely a biological vulnerability marker.
- **Male household count:** The presence of males in the household increases this person’s score, potentially reflecting contact patterns or caregiving arrangements observed in the data.

Interestingly, the model shows *negative* influence from household mean age and minimum age—suggesting it’s responding to the household’s age diversity rather than treating elderly status as a simple risk factor. This centenarian’s score is high, but the model’s logic seems to emphasize household dynamics over individual frailty.

Member 7242083 (Age 7) — 21.6% susceptibility, NOT FLAGGED

The child receives the lowest score, with all top features pushing *downward*:

- **Zero personal and family income:** Children without independent income received lower scores in the model’s training patterns.
- **Young age (7 years):** The model associates childhood age with lower susceptibility, contrasting with the adult members.
- **Household crowding and composition factors:** Various household structure features consistently reduce this child’s score.

This profile suggests the model learned that young children in this type of household structure had lower observed transmission rates in the 2020 Swedish data—possibly reflecting school closures, reduced social mixing, or different testing patterns during that period.

Expert Considerations

Domain consistency: The model’s age patterns (higher scores for working-age adults and elderly, lower for young children) align with known contact and exposure patterns. However, income as a strong positive predictor is less intuitive from a biological perspective and likely represents unmeasured confounders.

Proxy variables: Family income appears to be standing in for multiple unmeasured factors—possibly housing quality, occupational exposure, social network density, or healthcare access patterns. The model learned statistical associations from registry data but can’t directly tell us *why* these patterns exist.

Missing context: The model has no information about vaccination status, specific occupations, school attendance, mask usage, or actual contact patterns within the household—all factors that significantly affect real-world transmission risk.

Threshold sensitivity: The two flagged adults have scores (73.8% and 63.6%) comfortably above the 30% threshold used here, suggesting reasonably confident flagging. The child’s score (21.6%) is also clearly below threshold. However, if policy thresholds were different, member 535501 would be a borderline case.

Model Confidence Note

The household-level prediction (62.5%) sits well above its threshold, and two of three individual predictions are clearly separated from their thresholds, suggesting the model’s pattern-matching is reasonably confident for this case. However, remember that these probabilities reflect what a statistical model trained on 2020 Swedish registry data assigns to this household’s feature profile—they are not validated predictions of actual infection outcomes. These outputs should be viewed as one limited reference point among many factors that qualified health experts would consider when assessing household transmission risk.

Note: This analysis explains the model’s reasoning process and feature associations, not actual infection outcomes or clinical recommendations.

Evaluation Scores — Sample 1

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet

4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.2 Sample 2

4.2.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 2

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 2432745 | Model Analysis

1. CASE SUMMARY

This household comprises six individuals, including two confirmed COVID-19 index cases and four exposed members. The model assigned a transmission score of 79% to this household—well above the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission likely. Among the four non-index members, the model flagged three as susceptible (members 2470031, 197687, and 12177548, with scores of 51%, 53%, and 62% respectively), while one six-year-old child (member 11108007) received a score of 22% and was not flagged. The model’s assessment appears to distinguish between household members based on age, income patterns, and household composition factors.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 79%, substantially exceeding the 30% threshold, which suggests the model is expressing a clear preference for flagging this case as high-risk. The household’s feature profile appears to have triggered multiple risk signals in the model’s assessment.

Key factors driving the model’s household-level score:

Household size (6 persons) was the single most influential factor, pushing the model’s score higher. Larger households provide more opportunities for person-to-person contact, and the model appears to have learned this pattern from the training data.

Index case characteristics played a prominent role. The mean age of index cases (32.5 years) and maximum age (45 years) both elevated the model’s score, as did having two index cases rather than one. The ratio of index cases to total household size (33%—meaning one-third of the household was already confirmed positive) further contributed to the model’s assessment of transmission risk.

Housing type (Småhus/small house) and moderate living area per person (14.5 m²) both pushed the score higher. This combination suggests the model may be identifying households where spatial separation is challenging despite not being extremely overcrowded.

Household income (mean family income of approximately 426,000 SEK/year) contributed modestly to the elevated score, though its influence was smaller than demographic and structural factors.

The pattern is epidemiologically coherent: larger household size, multiple index cases already present, young-to-middle-aged index cases (who may have higher contact rates), and moderate space constraints align with known transmission risk factors. The model appears to be weighting structural and demographic features in a plausible manner.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 2470031 (Age 29)

The model assigned a susceptibility score of 51%, narrowly above the flagging threshold. This individual's age was the strongest factor pushing the score higher—the model appears to weight adults in their late twenties and thirties as higher risk compared to children or elderly persons. Their individual income (approximately 256,000 SEK/year) also elevated the score slightly. Working against flagging were household crowding metrics and the absence of a member aged 75+, both of which reduced the model's score. A history of antihistamine prescriptions (lmed_R06A) modestly increased the score, possibly serving as a proxy for immune system engagement or allergic conditions. This individual sits in the model's uncertain zone—small changes in their feature profile would likely alter the binary flag.

Member 197687 (Age 42)

The model assigned a susceptibility score of 53%, just above the threshold, flagging this individual as susceptible. The ratio of index cases to household size was the dominant factor elevating the score, followed closely by age (42 years). The model's emphasis on this ratio suggests it has learned that households where a larger proportion is already infected pose higher risk to remaining members. Individual income (approximately 201,000 SEK/year) also contributed positively. Countervailing factors included the number of males in the household (which reduced the score) and the absence of certain medication histories. This individual, like member 2470031, sits very close to the threshold, indicating the model is expressing marginal confidence in the flagging decision.

Member 11108007 (Age 6)

The model assigned a susceptibility score of 22%, well below the threshold, and did not flag this child. Age was by far the most influential factor—the model strongly weights young children as lower risk, with a SHAP value more than twice as large as any other feature for this individual. The complete absence of personal income (both disposable and family income recorded as zero for this child) further drove the score downward substantially. The only factor working against this pattern was the presence of two index cases in the household, which modestly elevated the score. The child's Swedish background (immigration background code 12, indicating born in Sweden to Swedish-born parents) also reduced the score slightly. This is the model's most confident assessment among the four members, with the score sitting far from the threshold.

Member 12177548 (Age 18)

The model assigned a susceptibility score of 62%, the highest among the four exposed members, and flagged this individual as susceptible. Family income (approximately 700,000 SEK/year—the highest in the household) was the dominant driver, pushing the score substantially higher. Age (18 years) and the presence of two index cases also contributed meaningfully. The model's strong weighting of high family income is somewhat puzzling from an epidemiological standpoint, as higher income typically correlates with better health outcomes and housing quality. This may indicate the model is capturing a pattern specific to the training data (2020 Swedish

households with confirmed index cases) that does not generalize straightforwardly. The presence of a child under six in the household (member 11108007) also elevated this individual’s score. Working against flagging were household crowding metrics and the presence of foreign-background members, both of which reduced the score. The model appears reasonably confident in this flagging, as the score sits well above the threshold.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s strong weighting of household size, multiple index cases, and the age profile of index cases aligns well with established epidemiological understanding of household transmission dynamics. The protective effect assigned to very young age (the 6-year-old child) is consistent with lower documented transmission risk for young children in early pandemic data, though this pattern evolved over time and varied by variant.

Uncertain or surprising patterns: The model’s treatment of income is inconsistent. For some members, higher personal or family income elevated susceptibility scores, which contradicts typical health gradient patterns where higher income confers protective advantages through better housing, healthcare access, and ability to isolate. This suggests the model may be capturing correlations specific to the training dataset rather than causal mechanisms. Income may be acting as a proxy for unmeasured factors such as occupation type, mobility patterns, or household density in ways that differ from general population expectations.

Proxy features: The absence of medication histories and homecare service use likely serves as a proxy for overall health status and frailty. The immigration background and country of birth variables may proxy for social network structures, healthcare access patterns, or multigenerational household arrangements not fully captured by the structural features.

Information gaps: Domain experts would benefit from knowing: actual household layout and room usage patterns (not captured by total floor area); index case symptom severity and viral load; exact timing of index case confirmation relative to household exposure; occupational exposure risks; and any pre-existing isolation measures. The model’s threshold of 30% is an empirical machine learning value optimized on a balanced validation set and carries no clinical or policy meaning—experts would need to calibrate this against their intervention capacity and risk tolerance.

Threshold proximity: Members 2470031 (51%) and 197687 (53%) both sit very close to the 30% threshold. For these individuals, the model is expressing minimal confidence, and the binary flag should be treated as unreliable.

5. MODEL CONFIDENCE NOTE

The household-level score of 79% sits well above the 50% threshold, suggesting the model is expressing a reasonably strong preference for flagging this household as high-risk—though this reflects the model’s internal assessment pattern, not a verified transmission probability. Among individuals, the model shows clear confidence only for the 6-year-old child (22%, well below threshold) and moderate confidence for the 18-year-old (62%, clearly above threshold). The two adult members with scores of 51% and 53% occupy the model’s uncertain zone, where small feature variations would flip the binary decision. These outputs provide a limited reference

point for qualified experts to integrate with clinical judgment, epidemiological context, and intervention feasibility—they are not standalone clinical decisions, and the model’s correctness remains unknown.

4.2.2 GPT-5.3

GPT-5.3 — Sample 2

1. CASE SUMMARY

This household consists of six people, including two identified index cases and four non-index household members. The household-level model assigned a transmission score of 78.9%, substantially above the model’s internal threshold of 25.6%, meaning the household was strongly flagged by the model as consistent with patterns associated with secondary household spread. At the individual level, the model assigned elevated susceptibility scores to three non-index members: a 29-year-old adult (50.7%), a 42-year-old adult (53.0%), and an 18-year-old young adult (62.3%). A 6-year-old child received a considerably lower score (22.1%) and was not flagged by the model. These scores represent the model’s learned statistical assessment from registry patterns rather than verified infection probabilities or observed outcomes.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level score of 78.9% was well above the model threshold, indicating a relatively strong model signal for potential within-household transmission according to the training patterns the system learned.

The most influential factor was **household size**, with six residents strongly increasing the model score. This is epidemiologically plausible, as larger households generally create more opportunities for repeated close contact.

Several additional features reinforced the model’s elevated assessment:

- **Two index cases within the household** contributed substantially. The model appears to interpret multiple initial cases as increasing the likelihood that additional household members could also be flagged.
- The **age profile of the index cases**, particularly index cases in adulthood and one older index case aged 45, also increased the score. This suggests the model associates working-age adults with household transmission patterns.
- The household’s **dwelling type** was weighted upward. The recorded housing category (“small house/other dwelling type”) may function as a proxy for family structure or occupancy patterns rather than housing quality itself.
- **Higher household disposable income measures** modestly increased the score. As in many registry-based models, income likely acts indirectly through factors such as occupational exposure, commuting patterns, housing structure, or social interaction networks rather than income itself being causal.
- The model also weighted **limited living area per person** as increasing risk, which aligns more directly with plausible contact-density mechanisms.

Overall, the model’s reasoning forms a fairly coherent structural pattern. Larger households, multiple index cases, and constrained living space are all consistent with known epidemiological drivers of household exposure opportunities. The income-related effects remain more ambiguous and should be interpreted cautiously as probable proxy variables rather than direct risk indicators. Interestingly, elderly composition contributed very little in this household, suggesting the model’s assessment was driven primarily by household scale and exposure structure rather than frailty-related factors.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Non-index member 2470031 (age 29)

The model assigned this individual a susceptibility score of 50.7%, above the flagging threshold. The strongest upward influences were:

- **Adult age (29 years)**, which increased the model score.
- **Presence of two index cases in the household**, suggesting the model interpreted multiple exposure sources as important.
- **Moderate disposable income levels**, which again likely function as indirect socioeconomic markers.

Some factors lowered the score, including indicators suggesting less crowding and greater household floor area. The model also gave a small upward weight to antihistamine medication history, which may reflect respiratory or allergic conditions, though the contribution was modest. Overall, the model appears to view this person as a centrally exposed adult within a larger household structure.

Non-index member 197687 (age 42)

This individual received a susceptibility score of 53.0%, also above threshold.

Key factors increasing the score included:

- **Adult age (42 years)**.
- A relatively high **index-case-to-household ratio**, meaning a large share of household members were already index cases.
- **Disposable income variables**, which again raised the score modestly.

Several household composition variables reduced the score somewhat, including male household composition and lower crowding indicators. The overall pattern nevertheless suggests the model associated this person with sustained exposure within a high-contact household environment. The reasoning appears broadly plausible from a domain perspective because adults in larger households may have extensive within-household interaction patterns.

Non-index member 11108007 (age 6)

The model assigned this child a score of 22.1%, below the threshold, so the child was not flagged. The strongest downward influences were:

- **Young age (6 years)**, which substantially lowered the score.
- **Very low individual and family income values**, likely reflecting the economic profile of a child rather than meaningful deprivation.
- Immigration-background variables also modestly reduced the score.

The strongest upward factor was simply the presence of **multiple index cases**, which pushed the model toward higher susceptibility despite the otherwise low-risk profile.

The pattern suggests the model strongly differentiated young children from adult household members in this case.

Non-index member 12177548 (age 18)

This young adult received the highest individual score in the household at 62.3%.

The strongest upward influences were:

- **Higher household income measures.**
- **Age 18**, which the model treated more similarly to adults than to children.
- **Multiple index cases in the household.**

The model also increased the score because the household included a young child under six, possibly reflecting broader family interaction structure. Lower crowding indicators reduced the score somewhat but not enough to offset the stronger positive influences.

This overall pattern is reasonably coherent: the model appears to place young adults closer to adult exposure patterns than to child-associated patterns.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear domain-consistent, particularly the strong emphasis on:

- larger household size,
- multiple simultaneous index cases,
- limited space per person,
- and adult household members as key transmission-linked profiles.

These align with plausible mechanisms of repeated household exposure.

The recurring influence of income-related variables is more difficult to interpret directly. These features likely serve as proxies for unmeasured structural factors such as occupation, commuting intensity, housing type, or social mixing patterns.

One noteworthy pattern is the model’s consistent separation between younger children and adults/young adults. The 6-year-old child was assigned a markedly lower score despite living in a highly flagged household, suggesting the model learned age-dependent susceptibility patterns from the registry data.

Before using these outputs operationally, experts would likely require additional contextual information including vaccination status, symptom timing, household behavioural patterns, school attendance, occupational exposure, and actual housing layout. The child’s score sits

relatively close to the threshold compared with strongly negative predictions, so uncertainty remains meaningful for that classification.

5. MODEL CONFIDENCE NOTE

The household-level score was far above the threshold, suggesting the model’s household classification was relatively strong rather than borderline. Three adult or near-adult household members also had clearly elevated scores above threshold, while the child’s score remained below but not extremely distant from the cutoff.

However, these outputs remain statistical model assessments rather than verified transmission events or true infection risks. The correctness of the predictions is unknown, and the results should be treated only as a limited interpretive reference for qualified experts rather than as evidence for clinical or public health decisions.

4.2.3 Grok 4.2

Grok 4.2 — Sample 2

CASE SUMMARY

This is a 6-person household (ID 2432745) with two index cases. The model assigns a 78.9% probability of household transmission and flags the household for potential secondary spread. At the individual level, three of the four non-index members are flagged as higher susceptibility (scores 50.7%, 53.0%, and 62.3%), while one young child is not flagged (22.1%). The flagged members are primarily working-age adults and a late teenager. These results reflect the model’s assessment of patterns in this large household’s registry-derived profile from 2020 Swedish data.

HOUSEHOLD MODEL ASSESSMENT

The household-level model gives a transmission probability of 78.9%, well above the empirical threshold of 0.256, resulting in a clear transmission flag.

The strongest factors pushing the score higher are:

- **Household size of 6**, which the model strongly associates with increased transmission potential — larger homes naturally offer more opportunities for close contact.
- Presence of **two index cases** and an index-to-household ratio of about 33%, meaning a substantial share of the household already has recorded cases.
- **Index case ages** (mean around 32–33 years, max 45 years), pointing to adult index cases in the prime working-age group.
- Housing type (small house / detached housing category) and moderately higher family disposable income (roughly 426,000 SEK annually) also contributed positively.

The model’s emphasis on larger household size, multiple index cases, and adult age composition forms a coherent and epidemiologically plausible pattern. These structural elements align with well-established risk factors for within-household spread. The positive influence of income and specific housing type may reflect proxies for urban/suburban location, occupation-related mobility, or testing patterns rather than income itself driving risk.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 2470031 (age ~29 in 2020): The model assigns a 50.7% susceptibility score and flags this person. Key upward drivers include adult working age, personal disposable income, multiple index cases in the home, and antihistamine prescription history. Some downward pressure came from household crowding and age composition. The model links this working-age profile in a multi-index household with moderate susceptibility.

Member 197687 (age ~42 in 2020): The model assigns a 53.0% susceptibility score and flags this individual. The strongest influences pushing higher include the index-to-household ratio, adult age, and personal income level. Offsetting factors include male household composition and crowding measures. The model appears to weigh middle-adult age and income-related factors as increasing the score in this setting.

Member 11108007 (age ~6 in 2020): The model assigns a 22.1% susceptibility score, below the 0.192 threshold, so this young child is not flagged. Dominant downward drivers are young age and zero personal/family income (expected for a child), reinforced by immigration background and proportion of children in the household. Multiple index cases pushed slightly upward but not enough to change the outcome. The model consistently assigns lower scores to young children in this configuration.

Member 12177548 (age ~18 in 2020): The model assigns a 62.3% susceptibility score and flags this person. Major upward factors include high family income, age 18 (young adult), multiple index cases, older index case age, and presence of a child under 6. Crowding and foreign background elements pulled modestly downward. The model associates this late-teen profile with the highest individual susceptibility in the household.

EXPERT CONSIDERATIONS

The model directs attention to this large household as high-priority, with particular focus on the three working-age or near-adult members rather than the young child. Domain-consistent patterns include the strong roles of household size, multiple index cases, and adult age groups. Medication history (e.g., antihistamines or analgesics) played only minor roles here. Income and housing type likely serve as proxies for unmeasured factors such as work-related contacts, transportation use, or residential density common in 2020.

Experts may wish to overlay additional information not available to the model: exact timing between index cases and potential exposures, school or workplace attendance, ventilation and isolation practices within the home, and detailed testing history. The young child's score is firmly in the low range, while the three flagged adults sit in moderate-to-clear territory above threshold. These outputs can support prioritisation but should be integrated with traditional epidemiological investigation.

MODEL CONFIDENCE NOTE

The household probability of 78.9% is substantially above the decision threshold, indicating a confident transmission flag from the model's perspective. Individual scores show good separation: the child is well below threshold, while the three flagged members are above (two in a moderate zone, one more clearly). As always, these are statistical assessments derived from patterns in historical registry data, not verified risks or clinical predictions. They provide a structured

reference to inform — but not determine — expert public health decision-making.

4.2.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 2

Here is the structured interpretation of the model’s output for the second household, written for public health policy experts.

1. CASE SUMMARY

This household contains six people, including two individuals designated as index cases and four other members. The model assigns a **78.9% probability** of within-household COVID-19 transmission, well above its internal flagging threshold. Among the four non-index members, the model flags three as “susceptible” (with scores of 50.7%, 53.0%, and 62.3%) and one young child as not flagged (22.1%). **These are statistical model outputs, not confirmed infections.**

2. HOUSEHOLD MODEL ASSESSMENT

The model strongly flagged this household for potential transmission. The most influential factors driving this decision were:

- **Household size (six persons):** This was by far the strongest factor. The model consistently associates larger households with higher transmission scores, which aligns with the intuitive idea that more people mean more close contacts.
- **Average age of index cases (32.5 years):** The two index cases are relatively young adults. The model weighted this as increasing transmission risk.
- **Housing type (detached/semi-detached house – *Småhus*):** Living in a house rather than an apartment pushed the score higher. This is unexpected, as one might assume apartments (shared stairwells, ventilation) carry higher risk.
- **Number of index cases (two):** Having two initial cases rather than one pushed the score upward.
- **Living area per person (14.5 m²):** This is relatively spacious (above many European crowding thresholds), yet the model treated it as increasing risk – again somewhat counterintuitive.

Coherence check: The pattern here is mixed. Household size is a plausible risk factor. However, living in a detached house and having more living space per person both pushed toward higher risk – the opposite of what crowding-based intuition would suggest. This may indicate the model is picking up socioeconomic correlations (e.g., larger houses with more rooms may enable mixing among more people) rather than a simple crowding effect. Experts should interpret the housing and space findings cautiously.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 2470031 (age 29) – Flagged (50.7%) The model’s score for this person was most influenced by:

- **Age 29 (upward):** Young-adult age pushed the score higher.
- **Household crowding index (downward):** Less crowding pushed the score lower – a rare downward influence from crowding.
- **Number of index cases (upward):** Two index cases in the household increased this person’s susceptibility score.
- **Personal disposable income (upward):** Higher income again pushed upward.

Member 197687 (age 42) – Flagged (53.0%) Key influences:

- **Index-to-household ratio (upward):** Two index cases among six people created a moderate ratio that pushed the score higher.
- **Age 42 (upward):** Middle age contributed positively.
- **Male count in household (downward):** The presence of more males in the household pushed this person’s score lower – an uncertain pattern.
- **Personal disposable income (upward):** Consistent with other members.

Member 11108007 (age 6) – Not flagged (22.1%) This child’s low score was driven primarily downward by:

- **Young age (age 6):** The strongest downward factor.
- **Household disposable income (zero value):** A zero value (possibly missing or imputed data) strongly pushed the score down.
- **Personal disposable income (zero):** Reinforced the downward push.
- **Immigration background code (12):** This also pushed the score lower.

Plausibility: The model consistently assigns very low susceptibility to young children, which is epidemiologically plausible (children often have milder or asymptomatic courses and may be less likely to be flagged as cases in registry data).

Member 12177548 (age 18) – Flagged (62.3%) – highest score among non-index members
Key influences:

- **Household disposable income (high – approx. 699,900 SEK/year, upward):** Strongest positive factor.
- **Age 18 (upward):** Transition to young adulthood increased the score.
- **Number of index cases (upward):** Again, two index cases raised the score.
- **Household crowding index (downward):** Less crowding pushed the score down slightly.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Household size as a strong risk factor, older children and adults flagged more than young children, and the presence of multiple index cases raising household-level risk – all align with reasonable epidemiological expectations.

- **Uncertain / surprising patterns:** Living in a detached house and having more living space per person both pushed the model’s household-level score *higher*, not lower. This contradicts simple crowding-based logic and suggests the model may be capturing correlated but unmeasured factors (e.g., house-dwellers in this dataset may have different testing rates, mobility, or occupational exposure).
- **Proxy variables:** Income appears repeatedly as a strong upward driver, likely acting as a proxy for occupation (e.g., non-remote work), mobility, or testing access – not a direct biological risk factor. Homecare alarms (TRYGG) did not appear in top influences for this household.
- **Additional information needed:** Experts would need to know whether zero-income values (as for the 6-year-old) are genuine or data artifacts. They would also benefit from understanding regional or occupational covariates that might explain why house-dwelling and more space correlate with higher model scores.
- **Uncertain zone:** None of the scores are close to the thresholds (household threshold 0.256 vs. 0.789; individual threshold 0.192 vs. scores of 0.507, 0.530, 0.221, 0.623). The model flags this household and its three members with moderate statistical confidence.

5. MODEL CONFIDENCE NOTE

The model’s probability estimates for this household (78.9% for transmission, individual scores between 22% and 62%) are well clear of the empirical flagging thresholds, indicating that the model makes these classifications with relatively high statistical confidence. However, confidence in model output is not evidence of correctness. These results should be treated as a limited statistical reference for expert review, not as a basis for public health action without further validation.

4.2.5 Llama 4

Llama 4 — Sample 2

COVID-19 Household Transmission Model Analysis

Household ID: 2432745

Case Summary

This is a 6-person household with two index cases, where the model assigns a 78.9% transmission probability—well above the flagging threshold. The household includes adults aged 29, 42, and 18, along with a 6-year-old child. The model identifies three of the four non-index members as susceptible (the two younger adults and the 18-year-old), while the 6-year-old child receives a score below the flagging threshold. This represents a high-concentration exposure scenario with multiple infected members in a larger household setting.

Household Model Assessment

The model assigns a 79% transmission probability, making this one of the stronger predictions

you'll see. Four key factors drive this assessment:

Household size (6 people): This is the dominant factor by far. Larger households create more transmission pathways—more people means more contacts, more shared spaces, and more opportunities for viral spread. The model clearly learned from the training data that six-person households had elevated transmission rates.

Index case demographics (mean age 32.5, max age 45): Two index cases of working age contribute to the model's high score. Multiple infected members, particularly adults in their prime activity years, amplifies the exposure burden throughout the household.

Housing type (Småhusbo - single-family home): Interestingly, the model associates this dwelling type with higher risk. This might seem counterintuitive—you'd expect apartments to be riskier due to shared ventilation and common spaces. However, single-family homes housing six people may indicate multigenerational or extended family arrangements with different interaction patterns than typical nuclear families.

Index case concentration (0.33 ratio): Two index cases out of six people means one-third of the household is infected—a high proportion that the model weighs as a significant risk amplifier. The pattern here is quite coherent: a large household with multiple active infections creates a high-risk transmission environment. Unlike some cases where the model's logic seems to rest on proxy variables, this assessment aligns well with basic transmission dynamics—more people, more infected sources, more risk.

Individual Susceptibility Profiles

Member 2470031 (Age 29) — 50.7% susceptibility, FLAGGED

This young adult receives a moderate susceptibility score driven by:

- **Age (29 years):** Working-age adults consistently score higher in this model, likely reflecting exposure patterns and contact rates observed in the 2020 Swedish data.
- **Two index cases in the household:** The presence of multiple infected members increases this person's score substantially, reflecting heightened exposure risk.
- **Personal income (255,500 SEK):** Economic activity correlates with higher scores, possibly capturing mobility and contact patterns.

Interestingly, lower crowding (more space per person) actually *reduces* this person's score, suggesting the model recognizes that spacious housing provides some protective effect. The antihistamine medication history (lmed_R06A) pushes the score upward, which might reflect underlying health conditions or healthcare engagement patterns.

Member 197687 (Age 42) — 53.0% susceptibility, FLAGGED

The oldest adult receives a similar moderate-high score:

- **Index case concentration (33% of household):** Having one-third of household members infected is the strongest driver for this person's score.
- **Age (42 years):** Another working-age adult, following the pattern we've seen.
- **Personal income (200,500 SEK):** Economic activity again contributes to the susceptibility score.

The model shows negative influence from the high male count in the household and lower crowding, suggesting these factors provide some protective effect. The overall profile suggests a working adult in a large, partially-infected household where spatial factors offer modest mitigation.

Member 11108007 (Age 6) — 22.1% susceptibility, NOT FLAGGED

The youngest child receives by far the lowest score, with nearly all factors pushing downward:

- **Young age (6 years):** Childhood age strongly reduces the model’s score, the single most powerful protective factor.
- **Zero personal and family income:** Children without independent economic profiles score lower, consistent with what we’ve seen in other cases.
- **Immigration background:** This child’s background code (12) is associated with lower scores in the model’s learned patterns.

Notably, having two index cases in the household is the only factor pushing this child’s score *upward*—but it’s overwhelmed by age and other protective factors. This suggests the model learned that young children, despite high household exposure, had lower observed transmission rates in the 2020 data (possibly reflecting different testing patterns, symptom presentation, or actual transmission biology).

Member 12177548 (Age 18) — 62.3% susceptibility, FLAGGED

The oldest child/young adult receives the highest susceptibility score:

- **Family income (699,900 SEK):** Higher household income drives the score upward, continuing the pattern we’ve observed.
- **Age (18 years):** At the threshold of adulthood, this person’s age profile places them in a higher-risk category than younger children.
- **Two index cases present:** Multiple infected household members significantly elevate this person’s exposure risk.
- **Maximum index case age (45):** Having an older adult as one of the index cases contributes to the score.

The model recognizes that lower crowding provides some protection (negative influence), but the combination of age, household composition, and socioeconomic factors outweigh this. The presence of a child under 6 in the household actually *increases* this person’s score, potentially reflecting childcare responsibilities that increase contact.

Expert Considerations

Strong domain consistency: Unlike many cases where income or proxy variables dominate in puzzling ways, this household shows clear structural risk factors—large size, multiple index cases, and working-age composition. The model’s reasoning here aligns well with transmission epidemiology.

Age stratification: The clear differentiation between the 6-year-old (22% score) and the 18-year-old (62% score) suggests the model captured meaningful age-related patterns in the 2020

Swedish data, though whether these reflect biology, behavior, testing patterns, or school policies remains unclear.

Two-index-case dynamics: Having multiple infected members is consistently recognized as a risk amplifier across all individual assessments, which makes epidemiological sense—more sources mean more exposure opportunities.

Housing type puzzle: The single-family home association with higher risk deserves scrutiny. This could reflect unmeasured factors like multigenerational living arrangements, household density, or socioeconomic clustering in the training data.

Threshold clustering: Three of four flagged individuals have scores between 50-62%, relatively close to each other but comfortably above the threshold. This suggests the model sees them as similar risk profiles—all working-age or near-working-age members in a high-exposure setting.

Model Confidence Note

The household-level prediction (79%) is quite confident—far above the threshold and among the higher scores the model assigns. Individual predictions show more variation but are generally decisive, with three members clearly above the flagging threshold and one clearly below. The scores suggest the model has recognized a prototypical high-risk scenario: large household, multiple index cases, predominantly working-age composition. However, these remain pattern-matching outputs from a statistical model trained on 2020 Swedish registry data—not validated infection predictions or clinical assessments. They should be interpreted as one data point among many in expert decision-making.

Note: This analysis explains the model’s decision-making process based on learned patterns from historical data, not actual infection outcomes or medical recommendations.

Evaluation Scores — Sample 2

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.3 Sample 3

4.3.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 3

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 7052823 | Model Analysis

1. CASE SUMMARY

This household comprises five individuals, including one confirmed COVID-19 index case and four exposed members. The model assigned a transmission score of 81% to this household—substantially above the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission highly likely. All four non-index members were flagged as susceptible, with scores ranging from 70% to 85%: a 57-year-old (member 1159155, 85%), a 59-year-old (member 555428, 70%), a 53-year-old (member 11162391, 84%), and a 20-year-old (member 6072977, 80%). This represents the model’s most uniform assessment of high risk across all exposed household members, with no protective age patterns evident.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 81%, far exceeding the 30% threshold, making this one of the model’s strongest flagging decisions. The assessment appears driven primarily by household size, index case age profile, and exceptionally high income levels.

Key factors driving the model’s household-level score:

Household size (5 persons) was the single most influential factor, pushing the model’s score substantially higher. Larger households increase contact opportunities, a pattern the model has clearly learned from the training data.

Index case age characteristics dominated the model’s assessment. The maximum index case age (57 years), minimum age (20 years), and mean age (38.5 years) all elevated the score considerably. This broad age span among index cases (37 years) suggests multiple generations or age groups were already infected, which the model interprets as a strong transmission signal.

Household income was exceptionally influential. Mean family income (approximately 935,000 SEK/year) ranked third among all features, pushing the score dramatically higher. The zero standard deviation in family income indicates all household members report identical family income values, suggesting this may be a single-earner or jointly-reported household. This extremely high income level (nearly double the levels seen in previous cases) continues the puzzling pattern of income as a risk-increasing factor.

Absence of elderly members (zero members aged 65+) paradoxically increased the model’s score. Combined with zero income variance across household members, these features suggest a cohesive adult household structure that the model associates with elevated transmission risk. The pattern is partially coherent: household size and multigenerational index case age structure align with transmission risk factors. However, the model’s strong weighting of very high income

as risk-increasing remains epidemiologically counterintuitive and likely reflects training data artifacts rather than causal mechanisms.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 1159155 (Age 57)

The model assigned a susceptibility score of 85%, the highest in the household and well above the flagging threshold. Family income (approximately 935,000 SEK/year) was the dominant driver, contributing massively to the elevated score. Age (57 years) was the second-strongest factor—the model appears to weight individuals in their late fifties as particularly high risk. The ratio of index cases to household size and the maximum index case age both further elevated the score, as did personal disposable income (approximately 315,000 SEK/year), household crowding patterns, the presence of a child among index cases, household gender diversity, and the proportion with foreign background. Every influential feature pushed the score higher; no protective factors emerged. The model expresses very high confidence in flagging this individual.

Member 555428 (Age 59)

The model assigned a susceptibility score of 70%, above the threshold, flagging this individual as susceptible. Again, family income (approximately 935,000 SEK/year) dominated, pushing the score dramatically higher. Age (59 years) contributed even more substantially than for the previous member—the oldest non-index member receives the strongest age-based risk signal. Personal disposable income (approximately 39,000 SEK/year, notably lower than other household members) also elevated the score. Several factors worked against flagging: the absence of children under 18, lower mean household personal income, lower index-to-household ratio, male sex, and lower proportion of children in the household all reduced the model’s score. Household crowding modestly increased it. This individual shows the most mixed feature profile in the household, with competing risk and protective signals, though the net assessment remains clearly above threshold.

Member 11162391 (Age 53)

The model assigned a susceptibility score of 84%, nearly identical to the first member, and well above the flagging threshold. The familiar pattern emerges: family income (approximately 935,000 SEK/year) was by far the strongest driver, followed by age (53 years). The ratio of index cases to household size, high proportion born in Sweden, number of index cases, generous living area per person (48 m²), mean log-transformed family income, and female count all pushed the score higher. Only mean family income (in non-transformed form) worked against flagging, suggesting the model distinguishes between different income representations. The model’s high confidence in flagging appears driven almost entirely by income and demographic structure rather than health or care utilization factors.

Member 6072977 (Age 20)

The model assigned a susceptibility score of 80%, above the threshold, flagging this youngest household member as susceptible. Notably, this individual shows a different feature influence

pattern than the others. Mean log-transformed family income led the drivers, followed by maximum household age, index-to-household ratio, and the individual's own age (20 years). The three-generation household structure (despite only spanning ages 20-59, the 37-year index case age range triggers this classification), presence of elderly members aged 65+ (this appears to be a data inconsistency, as earlier features showed zero elderly), number of index cases, and small-house dwelling type all elevated the score. Being female and median household income both reduced it slightly. The model's flagging of this 20-year-old contrasts sharply with protective effects assigned to children and young adults in previous cases, likely driven by the household's overall high-income, multi-index-case profile overwhelming age-based protective signals.

4. EXPERT CONSIDERATIONS

Plausible patterns: Household size as a transmission driver and the presence of multiple index cases (or a wide age range among index cases suggesting serial transmission) align with known household transmission dynamics. The model's attention to household demographic composition reflects legitimate epidemiological considerations.

Problematic patterns: The model's overwhelming reliance on extremely high income (approximately 935,000 SEK/year, far above typical household earnings) as the primary risk driver across all four members is epidemiologically implausible and likely represents spurious correlation from the 2020 Swedish training data. High-income households typically have better access to healthcare, larger living spaces enabling isolation, and lower exposure through occupation type. This pattern seriously undermines confidence in the model's reasoning for this case.

Age pattern reversal: Unlike previous cases where young children received strong protective signals, this 20-year-old is flagged with 80% confidence. The model appears to prioritize household-level income and index case features over individual age when the former are sufficiently extreme, creating inconsistent treatment of younger household members across cases.

Proxy features: Income may proxy for unmeasured occupational mobility, testing access patterns, or household composition factors specific to high-earning Swedish households in 2020. The zero variance in family income across members and its extreme value suggest this household may represent an unusual demographic or reporting pattern.

Information gaps: Experts would need: actual household spatial layout (critical for a 5-person household with 240 m² total area, or 48 m² per person); timing and symptom severity of the index case(s); occupational exposures for the four working-age adults; and whether the "multiple index cases" classification reflects true co-primary infections or misclassified secondary cases.

Threshold considerations: All four members sit well above the 30% threshold (70-85% range), with no uncertain cases. However, the uniform flagging and heavy income-weighting raise questions about whether the model is genuinely discriminating risk or simply responding to an outlier feature profile.

5. MODEL CONFIDENCE NOTE

The household score of 81% sits far above the 30% threshold, and all four individual scores (70-85%) are likewise well above threshold, suggesting the model is expressing very strong preferences for flagging across the board. However, this apparent confidence is misleading: the

model’s reasoning is dominated by a single feature—extremely high household income—that lacks epidemiological plausibility as a primary transmission driver. The uniform flagging of all members, including a 20-year-old who might receive protective age signals in other contexts, further suggests the model may be responding to an unusual feature combination rather than performing nuanced risk assessment. These outputs should be treated with particular caution, as the model’s numerical confidence does not reflect genuine epidemiological coherence, and correctness remains unknown.

Sample 3sec:out3

4.3.2 GPT-5.3

GPT-5.3 — Sample 3

1. CASE SUMMARY

This household consists of five people with one identified index case and four non-index household members. The household-level model assigned a transmission score of 81.3%, substantially above the model’s internal threshold of 25.6%, indicating that the household strongly matched patterns the model associated with within-household spread in the Swedish registry data.

At the individual level, all four non-index household members were flagged by the model. The assigned susceptibility scores ranged from 69.5% to 85.2%, with particularly high scores among adults aged 53–59 and one 20-year-old young adult. The model’s outputs reflect statistical associations learned from historical registry patterns rather than verified infection risk or confirmed transmission outcomes.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level score of 81.3% was far above the model threshold, suggesting a relatively strong model signal rather than a borderline classification.

The single strongest factor was **household size (five people)**. Larger households naturally create more opportunities for repeated close contact, so this is epidemiologically plausible.

Several additional features strongly increased the model score:

- The **age profile of the index case** was highly influential. The model weighted both the maximum and mean age of the index case upward, particularly ages in the late 30s to late 50s. This suggests the model associated middle-aged index cases with greater likelihood of additional household members being flagged.
- **High household disposable income values** strongly increased the score. Multiple income-related variables appeared among the most influential features. This pattern is difficult to interpret directly and likely reflects indirect socioeconomic effects rather than income itself being biologically relevant.
- The model also weighted the **uniformity of household income** upward. The absence of income variability may indicate a household composed mainly of economically established adults.

- The household had **no members aged 65 or older**, yet this absence slightly increased the model score. This is somewhat counterintuitive from a traditional vulnerability perspective and may reflect dataset-specific behavioural or demographic patterns learned by the model rather than a clinically meaningful effect.

Overall, the model’s reasoning forms a partially coherent pattern. Household size and adult age structure align with plausible transmission dynamics, particularly if the household contains multiple socially active adults. However, the very strong influence of high-income variables is more difficult to interpret mechanistically and should be treated cautiously as a proxy for unmeasured social or occupational factors.

Interestingly, crowding and homecare-related features played very little role despite often being considered important in household transmission settings. In this case, the model appears to have prioritised demographic and socioeconomic structure over frailty or care-dependency indicators.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Non-index member 1159155 (age 57)

The model assigned this person a susceptibility score of 85.2%, the highest in the household.

The strongest factors increasing the score were:

- **High household/family income measures.**
- **Age 57**, which strongly increased the model’s assessment.
- Household exposure structure, including the **presence of an index case** and the **ratio of index cases to household size**.

The model also gave some weight to household crowding and mixed-gender household structure. Overall, the reasoning is broadly plausible in that older adults in larger households may have sustained close-contact exposure patterns.

Non-index member 555428 (age 59)

This individual received a susceptibility score of 69.5%, also clearly above threshold.

The main upward influences were:

- **High family income values.**
- **Age 59**, which strongly increased the score.
- Moderate contribution from individual disposable income.

Some features slightly reduced the score, including male sex and the absence of children associated directly with this individual profile. The model also gave a small upward contribution to household crowding indicators.

The age-related pattern appears plausible, though the dominant role of income variables again suggests indirect socioeconomic proxy effects.

Non-index member 11162391 (age 53)

The model assigned this person a susceptibility score of 84.4%.

Key upward influences included:

- **High household income.**
- **Age 53.**
- The household’s **index-case structure**, including both the number of index cases and their proportional presence within the household.
- Living-area-per-person measures also increased the score, which is somewhat unexpected because larger personal space might normally be expected to reduce transmission opportunity.

The model also associated being born in Sweden within this household context with a higher score. This likely reflects broader population-level statistical correlations rather than interpretable causal mechanisms.

Non-index member 6072977 (age 20)

This young adult received a susceptibility score of 80.3%, also substantially above threshold.

The most influential factors were:

- Household-level income measures.
- Household age structure, particularly the presence of older adults and broad age diversity.
- The ratio of index cases within the household.
- Age 20, which the model treated similarly to adult exposure profiles rather than child profiles.

The model also weighted the household’s three-generation structure and detached-house dwelling type upward. This suggests the model interpreted the household as a socially interconnected multigenerational environment.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear domain-consistent, particularly:

- the importance of larger household size,
- concentration of adults within the household,
- and the strong role of index-case structure.

These factors align with plausible mechanisms of repeated household exposure.

However, the repeated dominance of income-related variables is difficult to interpret directly. These variables likely act as proxies for unmeasured factors such as occupational exposure, commuting patterns, household mobility, or social interaction structure. The finding that larger living area per person sometimes increased susceptibility scores is also somewhat counterintuitive and may reflect complex correlations within the training data rather than a meaningful epidemiological relationship.

A notable feature of this household is the absence of strong medical-history or healthcare-utilisation influences. The model’s reasoning appears driven primarily by demographic and structural household characteristics rather than frailty or chronic illness indicators.

Before acting on these outputs, experts would likely need additional information including workplace exposure, behavioural contact patterns, vaccination status, symptom chronology, housing layout, and actual interpersonal interaction patterns within the home.

5. MODEL CONFIDENCE NOTE

The household-level score was well above the model threshold, and all four non-index members also received clearly elevated scores above the individual cutoff. This suggests the model's classifications were relatively strong rather than borderline.

However, the correctness of these predictions is unknown. These outputs represent statistical pattern matching learned from registry data and should be interpreted only as limited model-based reference information for expert review, not as evidence of actual transmission or as a basis for clinical or policy decisions on their own.

4.3.3 Grok 4.2

Grok 4.2 — Sample 3

CASE SUMMARY

This is a 5-person household (ID 7052823) with one index case. The model assigns an 81.3% probability of household transmission and flags the household as high-risk for secondary cases. All four non-index members receive high susceptibility scores and are flagged by the individual-level model (ranging from 69.5% to 85.2%). The non-index members consist of several middle-aged adults and one young adult. These outputs reflect the model's assessment of this household's profile based on 2020 Swedish registry patterns.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 81.3%, substantially above the empirical threshold of 0.256, resulting in a clear transmission flag.

The key factors pushing the model's score higher include:

- **Household size of 5**, which the model associates with greater opportunity for within-home spread.
- **Index case age profile** (maximum age 57, minimum 20, mean around 38.5), indicating a mix of adult ages spanning working-age and older working-age groups.
- **Very high family disposable income** (mean around 935,000 SEK annually), which strongly contributed to elevating the score.
- Absence of members aged 65+ and one index case.

The model's reasoning combines larger household size and multi-age adult composition—factors that align with established epidemiological understanding of transmission dynamics in shared living spaces. The prominent role of high family income is notable and may function as a proxy for other unmeasured characteristics, such as residential location, occupational exposure, or testing/access patterns in 2020, rather than income directly influencing biological susceptibility.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 1159155 (age ~57 in 2020): The model assigns an 85.2% susceptibility score and flags this person. The strongest upward drivers are high family income, adult age in the late 50s, presence of an index case, and the index-to-household ratio. Additional modest contributions came from crowding measures and index case characteristics. The model links this middle-aged profile in a higher-income household with very high susceptibility.

Member 555428 (age ~59 in 2020): The model assigns a 69.5% susceptibility score and flags this individual. Dominant positive influences are high family income and age in the late 50s. Some downward pressure came from the number of children under 18 and certain household income averages, but not enough to avoid a flag. The model weighs advanced working-age combined with high income as increasing the score.

Member 11162391 (age ~53 in 2020): The model assigns an 84.4% susceptibility score and flags this person. The top factors pushing higher are high family income, middle-adult age, index-to-household ratio, and proportion born in Sweden. Living area per person and female household composition also contributed positively. The model assigns one of the highest individual scores to this profile.

Member 6072977 (age ~20 in 2020): The model assigns an 80.3% susceptibility score and flags this young adult. Major upward drivers include high household family income, household maximum age, index-to-household ratio, three-generation household structure, presence of elderly members, and housing type (detached house/agricultural). Young adult age also pushed the score upward. The model associates this combination with high susceptibility.

EXPERT CONSIDERATIONS

The model strongly flags the entire household, directing attention to all non-index members, particularly the middle-aged adults. Consistent patterns include the strong influence of household size and adult age mixing, which align with known transmission risks. High family income emerges as one of the most influential factors across both household and individual levels—likely serving as a proxy for unmeasured variables such as housing quality, geographic area, work-related contacts, or healthcare engagement rather than a direct risk factor. Medication and homecare variables had negligible influence here.

Domain experts should incorporate information unavailable to the model, including precise timing of the index case, specific occupations, school/workplace attendance, isolation measures within the home, and actual testing or symptom data. All individual scores are well above the threshold, with no members in the uncertain boundary zone. These model outputs can help identify households meriting closer review but must be interpreted alongside traditional epidemiological and clinical context.

MODEL CONFIDENCE NOTE

The household transmission probability of 81.3% is markedly above the decision threshold, and all four individual susceptibility scores are clearly elevated well above their cutoff, indicating decisive flagging by the model for this case. These are statistical pattern-based assessments derived from historical Swedish registry data, not confirmed infection risks or clinical predictions. They should be used only as a supplementary reference to support expert judgment in public health decision-making.

4.3.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 3

Here is the structured interpretation of the model’s output for the third household, written for public health policy experts.

1. CASE SUMMARY

This household contains five people, including one designated index case and four other members. The model assigns an **81.3% probability** of within-household COVID-19 transmission, well above its internal flagging threshold. All four non-index members are flagged as “susceptible” by the model, with scores ranging from 69.5% to 85.2% — an unusually high and uniform set of individual risk assignments. **These are statistical model outputs, not confirmed infections.**

2. HOUSEHOLD MODEL ASSESSMENT

The model very strongly flagged this household for potential transmission. The most influential factors were:

- **Household size (five persons):** The strongest positive factor. Larger households consistently drive the model’s transmission score higher.
- **Oldest age among index cases (57 years):** The single index case is 57 years old. The model weighted this as a major upward driver.
- **Household disposable income (very high – approx. 935,000 SEK/year):** This is the highest income seen across the three example households. The model treated it as a strong positive factor.
- **Youngest age among index cases (20 years):** The index case is the only person in that role, so both “min” and “max” refer to the same 57-year-old? *Note: The report shows `index_min_age_2020_std = 20.0` — this appears inconsistent (57 vs 20). This may indicate a data artifact or a second, younger index case not clearly listed. Experts should verify.*
- **No elderly members (age 65+ count = 0):** Surprisingly, the absence of elderly persons pushed the score higher — counter to the intuition that older adults are more vulnerable.

Coherence check: The pattern is mixed. Household size and high income are consistent with earlier households. However, the absence of elderly members increasing risk, and the apparent age discrepancy in index case data, are unexpected. The model appears to be picking up a specific income- and size-related signal that overrides age-based intuition. This household’s very high income (nearly double the previous highest) may be a dominant proxy for unmeasured factors such as occupation-related exposure or mobility.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Note: All four non-index members were flagged. Their scores are unusually high and tightly clustered.

Member 1159155 (age 57) – Flagged (85.2%) – highest score Key upward drivers:

- **Household disposable income (very high):** Strongest factor by far.
- **Age 57:** Middle-older age pushed the score higher.
- **Number of index cases (one) and index-to-household ratio:** Both added upward weight.

Member 555428 (age 59) – Flagged (69.5%) – lowest among the four Key influences:

- **Household disposable income (very high, upward):** Again dominant.
- **Age 59 (upward):** Similar to above.
- **Number of children under 18 in household (zero, downward):** No children present pushed the score lower — the only notable downward factor for this person.
- **Sex (female, downward):** Being female slightly reduced the score relative to males in this household.

Member 11162391 (age 53) – Flagged (84.4%) Key upward drivers:

- **Household disposable income (very high):** Strongest factor.
- **Age 53 (upward):** Consistent with other adults.
- **Proportion born in Sweden (upward):** The model weighted a higher share of Swedish-born members as increasing susceptibility.
- **Living area per person (48 m², upward):** More space per person paradoxically pushed the score higher — similar to the previous household’s detached house pattern.

Member 6072977 (age 20) – Flagged (80.3%) Key upward drivers:

- **Household mean disposable income (very high, upward):** Consistent across all members.
- **Oldest age in household (57, upward):** The presence of an older adult in the household increased this young adult’s score.
- **Index-to-household ratio (upward):** One index case among five.
- **Three-generation household present (upward):** Despite no official elderly (65+), the age range likely spans child/adult/older adult.
- **Housing type (detached house with agricultural land – *SmahusAg*, upward):** Again, house type pushed the score higher.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Larger household size and the presence of an index case of moderate age (57) are plausible drivers. All four non-index members being flagged is at least internally consistent given the high household-level score.
- **Uncertain / surprising patterns:**

- Very high household income (935,000 SEK/year) is the dominant factor across both household and individual levels. This is not a standard epidemiological risk factor and likely acts as a proxy for unmeasured variables (e.g., international travel, occupation unable to work from home, larger social networks).
 - More living space per person and living in a detached house *increased* risk — the opposite of crowding-based expectations.
 - The absence of elderly members (age 65+) pushed the household score *higher*, which is counterintuitive.
 - **Data anomaly:** The report shows `index_min_age_2020_std` = 20.0 but the only listed index case has age 57. This inconsistency should be investigated before any operational use.
- **Proxy variables:** Income is the clearest proxy here — likely standing in for occupation, mobility, or testing access. Housing type may proxy for geographic location (rural vs. urban) or household layout (more rooms, more mixing opportunities).
 - **Additional information needed:** Experts would need to resolve the age inconsistency, understand why very high income correlates with higher model scores in this dataset, and examine whether this household’s characteristics (e.g., occupation, travel history) align with known exposure risks.
 - **Uncertain zone:** None. All scores are far from their respective thresholds (household 0.813 vs. 0.256; individual scores 0.695–0.852 vs. 0.192). The model is statistically confident in its flagging, but that does not imply correctness.

5. MODEL CONFIDENCE NOTE

The model’s probability estimates for this household (81.3% for transmission, individual scores between 69.5% and 85.2%) are very far from the empirical flagging thresholds, indicating high statistical confidence in its classifications. However, statistical confidence is not evidence of real-world correctness. These outputs should be treated as a limited reference — and the apparent data inconsistency regarding index case age warrants expert review before any interpretation is acted upon.

4.3.5 Llama 4

Llama 4 — Sample 3

COVID-19 Household Transmission Model Analysis

Household ID: 7052823

Case Summary

This is a 5-person adult household where the model assigns an 81.3% transmission probability—among the highest scores the model generates. The household consists entirely of working-age and near-retirement adults (ages 20, 53, 57, and 59), with one index case. Re-

markably, the model flags all four non-index members as susceptible, with scores ranging from 69.5% to 85.2%—indicating the model sees this as a universally high-risk exposure scenario. The absence of young children or elderly members, combined with high household income, creates an unusual demographic profile where every exposed person receives a strong susceptibility score.

Household Model Assessment

The model assigns an 81% transmission probability, driven by several converging factors:

Household size (5 people): A five-person household provides the structural foundation for elevated transmission risk—enough people to create multiple transmission pathways without being so large as to dilute contact patterns.

Index case age characteristics (max 57, min 20, mean 38.5): The model responds strongly to the index case’s demographic profile. The relatively wide age span (20-57) among what appears to be index cases or a single index case with these characteristics suggests a multigenerational household structure.

Family income (935,000 SEK annually): This is the third-strongest driver and represents a notably high household income. At nearly one million kronor per year, this places the household well above typical Swedish income levels. The model consistently associates higher income with increased transmission risk, possibly capturing patterns around occupational exposure, social networks, or mobility that were present in the 2020 training data.

Zero elderly members (65+): Interestingly, the *absence* of elderly household members pushes the score upward. This might reflect that all-adult working-age households had higher observed transmission rates in the training data, possibly due to more external contacts and less ability to isolate within the home.

Zero income variance: All household members appear to share the same family income level (zero standard deviation), suggesting a tightly coupled economic unit. The model treats this uniformity as a risk factor, potentially indicating shared financial responsibilities that require continued external activity.

The pattern is coherent but demographically unusual—this is an all-adult household with very high collective income, which the model interprets as a high-transmission scenario. This aligns with patterns where working-age adults have high contact rates and mobility.

Individual Susceptibility Profiles

Member 1159155 (Age 57) — 85.2% susceptibility, FLAGGED

The oldest household member receives the highest susceptibility score:

- **Family income (935,000 SEK):** The dominant driver across all members, this exceptionally high household income appears to proxy for exposure patterns or lifestyle factors the model learned from the training data.
- **Age (57 years):** Late working age, approaching retirement but still in high-activity years. The model strongly weighs this age bracket as susceptible.
- **Index case count and ratio:** Being one of four non-index members in a five-person household (20% index concentration) elevates the score.

- **Personal income (315,200 SEK):** Individual economic activity reinforces the household pattern.

The model sees a near-retirement-age adult in a high-income, all-adult household as extremely susceptible—the highest score in this household.

Member 555428 (Age 59) — 69.5% susceptibility, FLAGGED

The eldest member actually receives a *lower* score than the 57-year-old:

- **Family income (935,000 SEK)** remains the primary driver.
- **Age (59 years):** Just two years older than member 1159155, but the model treats this differently, suggesting the scoring isn't simply linear with age.
- **Personal income (39,200 SEK):** Notably lower personal income than other adults, which the model treats as a risk factor—perhaps indicating different activity patterns or dependency status.

Interestingly, the absence of children in the household ($\text{AntalBarnUnder18} = 0$) actually *reduces* this person's score, and being male ($\text{Kon} = 1$) also provides a slight protective effect. Despite being the oldest, this member receives the lowest susceptibility score in the household, though still strongly flagged.

Member 11162391 (Age 53) — 84.4% susceptibility, FLAGGED

The middle-aged member receives the second-highest score:

- **Family income (935,000 SEK)** continues as the overwhelming driver.
- **Age (53 years):** Mid-career age, solidly in the working-age band the model associates with high susceptibility.
- **Index case ratio (20%):** The proportion of infected members in the household contributes substantially.
- **Swedish birth country proportion:** The household's Swedish-born composition increases this person's score, possibly reflecting social network or contact patterns observed in the training data.
- **Living area per person (48 m²):** More personal space than typical, yet the model treats this as a risk-increasing factor here—perhaps indicating larger homes correlate with higher-income lifestyles that involve more external contact.

Member 6072977 (Age 20) — 80.3% susceptibility, FLAGGED

The youngest adult receives a high but slightly lower score:

- **Household income structure:** Rather than personal income, various household income metrics dominate—mean family income, household age composition.
- **Age (20 years):** Young adult age, likely a student or early-career individual, which the model treats as moderately high-risk.
- **Three-generation household indicator:** Despite all members being adults, the age span (20 to 59) triggers the model's three-generation flag, which *increases* the score—suggesting households with wide age diversity had higher transmission in the training data.

- **Presence of near-elderly members (approaching 65):** Having household members close to elderly status increases this young person’s score, possibly reflecting caregiving or close-contact dynamics.
- **Housing type (SmahusAg - detached house):** Similar to earlier cases, single-family homes are associated with higher risk in the model’s logic.

Expert Considerations

Highly unusual pattern: An all-adult household where *every* exposed member scores 69-85% susceptibility is rare. The model has identified a demographic profile—high-income, all working-age, no children, no elderly—that it strongly associates with transmission risk.

Income as a dominant proxy: The 935,000 SEK family income appears in every individual assessment as the primary driver. At this income level, the model may be capturing unmeasured factors like: white-collar occupations requiring in-person work, international travel, social networks involving higher-contact activities, or residential patterns (large single-family homes in suburban areas requiring commuting).

Age paradox: The 59-year-old receives a *lower* score than the 57-year-old, and the 20-year-old scores higher than expected. This suggests the model isn’t treating age as a simple linear risk factor but has learned specific patterns about different age cohorts’ transmission risks in the 2020 Swedish context.

Absence as a risk factor: Having *no* elderly members increases the household score, suggesting the model learned that all-working-age households had higher transmission rates—possibly because they couldn’t isolate as effectively or had more external obligations.

Missing behavioral context: The model has no information about remote work capability, vaccination status, social activities, or actual contact patterns—all crucial for understanding why this high-income adult household might be high-risk.

Threshold clustering: All four scores are tightly clustered (69-85%), well above the 30% threshold, indicating the model sees them as a uniform high-risk group rather than differentiating meaningfully between individuals.

Model Confidence Note

The model expresses very high confidence in this case, with a household score of 81% (far above threshold) and all four individuals scoring between 69-85% (decisively above the individual threshold). The tight clustering of individual scores suggests the model is responding primarily to shared household characteristics rather than individual-level differentiation. However, these scores reflect patterns the model learned from 2020 Swedish registry data about high-income, all-adult households—not validated infection predictions. The unusually uniform high-risk assessment should prompt experts to examine what specific behavioral or structural factors in this demographic profile drove the patterns the model learned, and whether those patterns would hold in different contexts or time periods.

Note: This analysis explains the model’s statistical pattern recognition based on 2020 training

data, not actual infection outcomes or clinical recommendations.

Evaluation Scores — Sample 3

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 IND-R@2	B6 IND-R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.4 Sample 4

4.4.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 4

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 9878768 | Model Analysis

1. CASE SUMMARY

This household comprises two individuals: one confirmed COVID-19 index case and one exposed member. The model assigned a transmission score of 7% to this household—far below the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission unlikely. The single non-index member, a 3-year-old child (member 6775956), received a susceptibility score of 5%, well below the flagging threshold, and was not flagged by the model. This represents the model’s strongest protective assessment among all cases examined, with both household and individual scores sitting at the opposite end of the risk spectrum from the high-flagging cases.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 7%, substantially below the 30% threshold, marking this as a case the model considers very low risk for within-household transmission. Nearly every influential feature pushed the score downward, creating a uniformly protective feature profile.

Key factors driving the model’s low household-level score:

Low household income (mean family income of approximately 108,000 SEK/year) was by far the most influential factor, dramatically reducing the model’s score. This continues the counterintuitive pattern observed across cases where income correlates with risk in the model’s learned associations—here, low income acts protectively. This household’s income is roughly one-tenth that of the highest-income case, suggesting the model has learned a strong income-risk gradient from the training data.

High index-to-household ratio (50%—one of two members is the index case) substantially reduced the score. This inverts the typical expectation: having half the household already infected might seem to increase exposure risk for the remaining member, but the model appears to have learned that small households with high index proportions show different transmission patterns.

Young index case age (28 years for both mean and maximum, indicating a single young adult index case) strongly reduced the score. The model weights young adult index cases as less likely to transmit compared to older or middle-aged index cases.

Small household size (2 persons) reduced the score considerably, consistent with fewer contact opportunities in smaller households—this pattern aligns with epidemiological expectations.

Female index case (100% of index cases are female) further reduced the score. The model may have learned gender-associated behavioral or biological transmission patterns from the training

data.

Young minimum household age (3 years—the non-index child) and presence of a child under 18 both contributed to the protective assessment.

The pattern is partially coherent: small household size as protective aligns with transmission mechanics. However, the dominant role of low income as a protective factor inverts typical health gradients and likely reflects training data artifacts. The protective effect of a young index case and high index-to-household ratio are less intuitive from a transmission standpoint.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 6775956 (Age 3)

The model assigned a susceptibility score of 5%, far below the flagging threshold, and did not flag this child. Age was overwhelmingly the most influential factor—the model assigns extremely strong protective weight to very young children, with this 3-year-old receiving the largest protective signal observed across all cases. The absence of personal income (both family income and disposable income recorded as zero for the child) further drove the score dramatically downward, contributing almost as much as age itself. Being born in Sweden (Födelseland = SVERIGE), young minimum household age, multi-family dwelling residence (apartment building), low mean household income, Swedish background (immigration code 21, indicating Swedish-born to Swedish-born parents), absence of respiratory medication history, and no homecare service utilization all contributed modest additional protective effects. Every single influential feature pushed the score lower—no risk-increasing factors emerged. This represents the model’s most confident protective assessment, with the score sitting further from the threshold than any other individual examined.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s strong protective weighting of very young age (3 years) aligns with documented patterns from early pandemic data showing lower documented transmission risk among young children. Small household size (2 persons) as a protective factor is epidemiologically coherent, as fewer household members reduce contact opportunities and transmission pathways.

Counterintuitive patterns: Low income as the dominant protective factor contradicts established health gradients where lower socioeconomic status typically correlates with worse health outcomes and higher infectious disease risk. This suggests the model learned spurious correlations from the 2020 Swedish training data—potentially related to differential testing access, reporting patterns, or unmeasured confounders that varied systematically with income during that specific period.

High index ratio paradox: The protective effect of having 50% of the household already infected (high index-to-household ratio) seems counterintuitive, as this represents maximum potential exposure for the remaining member. This may indicate the model learned that two-person households with one index case differ systematically from larger households—perhaps through selection effects in who gets tested or reported in registry data.

Proxy features: Income likely proxies for factors beyond socioeconomic status—perhaps occupation type, mobility patterns, or household contact structures that differed in ways specific

to the 2020 Swedish context. The absence of medication histories and homecare services indicates this is a generally healthy household without chronic care needs.

Information gaps: Domain experts would benefit from knowing: the relationship between the two household members (parent-child, siblings with guardian elsewhere, other arrangement); timing of index case confirmation relative to the child’s exposure period; childcare or preschool attendance patterns; presence of additional caregivers outside the household; and actual isolation measures implemented. For a 3-year-old, caregiver contact intensity during the index case’s infectious period would be particularly relevant.

Threshold distance: Both the household score (7%) and individual score (5%) sit extremely far below the 30% threshold, with no uncertain cases. The model expresses maximal confidence in not flagging this household.

5. MODEL CONFIDENCE NOTE

The household score of 7% and individual score of 5% both sit far below the 30% threshold, suggesting the model is expressing very strong confidence that this household represents low transmission risk—though this reflects the model’s learned pattern associations, not verified infection probability. The model’s assessment is driven overwhelmingly by features that align with protective epidemiological patterns (very young child age, small household size) alongside features with questionable causal interpretation (low income as protective, high index ratio as protective). While the numerical confidence is unambiguous, the reliance on income patterns that contradict typical health gradients introduces interpretive uncertainty about whether the model’s reasoning generalizes beyond the specific 2020 Swedish training population. These outputs provide a reference point for qualified experts but should not substitute for clinical judgment, and the model’s actual predictive accuracy for this household remains unknown.

Sample 4sec:out4

4.4.2 GPT-5.3

GPT-5.3 — Sample 4

1. CASE SUMMARY

This household consists of two people: one identified index case and one non-index household member, a 3-year-old child. The household-level model assigned a transmission score of 6.7%, well below the model’s internal threshold of 25.6%, meaning the household was not flagged by the model as high-risk for within-household transmission.

At the individual level, the non-index child received a susceptibility score of 4.9%, also substantially below the individual threshold. Nearly all influential factors pushed the model toward a lower-risk assessment rather than a higher one. These outputs describe the model’s statistical interpretation of this household profile and should not be interpreted as verified infection outcomes or true biological probabilities.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level score of 6.7% was far below the threshold of 25.6%, indicating that the model viewed this household as relatively inconsistent with patterns associated with secondary household spread in the registry training data.

The strongest factor lowering the score was the household’s **family disposable income profile**. Several income-related variables strongly pushed the model toward a lower-risk classification. In this case, the model appears to associate this specific household income structure with lower transmission likelihood. However, these effects should be interpreted cautiously because income variables often act as indirect proxies for broader social or environmental conditions rather than direct determinants of transmission.

Additional features that reduced the model score included:

- **Small household size (2 persons)**, which strongly lowered the score. This is epidemiologically plausible because fewer household members create fewer opportunities for repeated close-contact exposure.
- **Young age of the index case (28 years)**, which reduced the model score. The model appears to associate younger adult index cases with lower likelihood of additional household members being flagged in this household context.
- **Presence of a child aged 3** also reduced the score. The household minimum age and the count of children under 18 both contributed toward a lower-risk assessment.
- The model additionally weighted the fact that the **index case was female** slightly toward lower risk.

Notably, almost all influential features moved the prediction downward. Very few features meaningfully increased the score, suggesting the model’s assessment was internally consistent and relatively unambiguous.

The overall pattern is broadly coherent from a domain perspective. Small households with limited contact networks inside the home would reasonably be expected to generate fewer opportunities for household spread. The model also did not identify crowding, elderly household structure, healthcare dependency, or medical burden as important concerns in this case.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Non-index member 6775956 (age 3)

The model assigned this child a susceptibility score of 4.9%, substantially below the individual threshold, so the child was not flagged.

The strongest factor lowering the score was **very young age (3 years)**. This was by far the most influential feature in the model’s assessment and strongly reduced the assigned susceptibility score. The model therefore appears to distinguish very young children from older children and adults in its learned susceptibility patterns.

Additional downward influences included:

- **Very low disposable income and family income values at the individual level**, which likely reflect the expected economic profile of a young child rather than deprivation

itself. In registry-based systems, children naturally appear with little or no personal income.

- **Swedish birth-country coding** modestly lowered the score. This likely reflects statistical population patterns in the training data rather than a directly interpretable mechanism.
- The household’s dwelling type (“apartment/multi-family housing”) slightly lowered the score in the model’s assessment.
- The absence of healthcare-service utilisation and respiratory medication history also contributed weakly toward a lower score.

Importantly, essentially all meaningful feature contributions pushed the model downward rather than upward. There were no substantial positive influences suggesting elevated susceptibility according to the model.

From a domain perspective, the overall reasoning appears fairly coherent. The model associated this profile with low exposure complexity: a very small household, absence of elderly or medically vulnerable members, no indicators of frailty or healthcare dependency, and a very young non-index member.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear epidemiologically plausible in this case:

- the strong influence of small household size,
- the absence of crowding or frailty indicators,
- and the low model-assigned susceptibility for a very young child.

These factors align with a structurally lower-contact household environment.

The repeated importance of income-related variables remains more difficult to interpret directly. In this case, the model associated the household’s income structure with lower transmission likelihood, but these variables likely act as proxies for broader contextual factors such as occupation patterns, housing conditions, social mobility, or care access.

An important observation is that the model showed unusually consistent directional behaviour: almost every influential feature pushed toward lower risk. This differs from more ambiguous cases where competing factors raise and lower the score simultaneously.

Before acting on these outputs, a domain expert would still require contextual information not available in registry data, including actual contact intensity within the household, childcare arrangements, workplace exposure of the index case, symptom timing, and behavioural mitigation measures.

There are no major borderline classifications in this case. Both household and individual scores sit well below their respective thresholds.

5. MODEL CONFIDENCE NOTE

Both the household-level score (6.7%) and the child’s individual score (4.9%) were far below the model thresholds, suggesting the model’s classifications were relatively strong and not borderline. In addition, nearly all influential factors pointed in the same downward direction, indicating a comparatively stable internal model assessment.

However, the correctness of the predictions remains unknown. These outputs reflect statistical pattern matching learned from Swedish registry data and should be treated only as a limited interpretive reference for expert review, not as evidence of actual transmission outcomes or clinical risk.

4.4.3 Grok 4.2

Grok 4.2 — Sample 4

CASE SUMMARY

This is a small 2-person household (ID 9878768) consisting of one index case and one non-index member. The model assigns a low 6.7% probability of household transmission and does not flag the household. The single non-index member, a young child, receives a very low susceptibility score of 4.9% and is not flagged by the individual-level model. These outputs reflect the model's assessment of this minimal household configuration based on patterns observed in 2020 Swedish registry data.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of only 6.7%, well below the empirical decision threshold of 0.256, so no transmission flag is raised.

The factors most strongly pulling the model's score downward include:

- **Low family disposable income** (mean around 108,000 SEK annually), which the model weighted as reducing the predicted risk.
- **Index-to-household ratio of 0.5** (one index case out of two people), indicating a relatively high proportion of known cases in a tiny household.
- **Young index case age** (28 years), along with the presence of a young child (age 0–17) in the home.
- **Small household size of 2**, which the model associates with lower transmission potential compared to larger homes.

The model's reasoning here forms a coherent pattern driven by small household size, young ages, and lower income. Smaller households generally offer fewer opportunities for secondary spread, and the age profile (young adult with a toddler) aligns with epidemiological expectations of potentially different contact patterns or behaviors. The negative influence of lower income may reflect proxy effects related to living situation, occupation, or mobility patterns captured in the training data.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 6775956 (age ~3 in 2020): The model assigns a 4.9% susceptibility score, far below the 0.192 threshold, and does not flag this young child. The strongest factors pushing the score downward are the child's very young age and zero recorded personal or family income (typical for a young child). Additional modest downward contributions came from birth country (Sweden), housing type, and certain household income and homecare measures.

The model's strong down-weighting of a very young child in a small household with a young

index case appears consistent with domain expectations. Young children may exhibit different exposure or recording patterns in registry data, and the absence of income indicators reinforces the low score in this context.

EXPERT CONSIDERATIONS

In this case, the model does not prioritise the household for intervention, showing clear differentiation from the higher-risk households seen in other reports. Domain-consistent elements include the protective effects (in model terms) of very small household size and young child presence, which match known patterns where transmission risk scales with household crowding and contact opportunities. Lower income acting to reduce the score is interesting and may serve as a proxy for specific residential or social circumstances rather than a direct protective factor. Features such as income and housing type likely capture broader unmeasured variables like geographic location, work-from-home potential, or social networks in 2020. Medication history and homecare variables played minimal roles. Experts should consider additional real-world details unavailable to the model, including the exact timing of the index case, caregiving arrangements for the child, any isolation measures taken, and symptom or testing history. There are no scores near the decision thresholds, reducing ambiguity in the model's output for this household.

MODEL CONFIDENCE NOTE

Both the household transmission probability (6.7%) and the individual susceptibility score (4.9%) are substantially below their respective thresholds, indicating the model is quite decisive in assigning low risk to this small household and young child. As with all outputs from this system, these reflect statistical patterns learned from 2020 registry data rather than verified infection risks or clinical assessments. They should serve only as one limited reference tool to support, never replace, expert public health judgment and contextual knowledge.

4.4.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 4

Here is the structured interpretation of the model's output for the fourth household, written for public health policy experts.

1. CASE SUMMARY

This household consists of two people: one designated index case and one other member. The model assigns a **6.7% probability** of within-household COVID-19 transmission, which is well below its internal flagging threshold. The single non-index member receives a susceptibility score of only **4.9%** and is not flagged by the model. **These are statistical model outputs, not confirmed absence of infection.**

2. HOUSEHOLD MODEL ASSESSMENT

The model did **not** flag this household for transmission. The probability (6.7%) is far below the threshold of 25.6%. Every influential feature pushed the score downward:

- **Household disposable income (108,000 SEK/year – lowest among example households):** This was the strongest downward driver. The model consistently associates lower income with lower transmission risk — the inverse of the pattern seen in higher-income households.
- **Index-to-household ratio (0.5):** With one index case in a two-person household, this ratio is high. Surprisingly, the model treated it as a strong downward factor (negative influence). This contradicts the pattern in larger households where a single index case often pushed risk up.
- **Age of index case (28 years):** The index case is young. The model weighted this as reducing transmission risk.
- **Household size (2 persons):** Very small household size pushed the score downward, which is epidemiologically plausible (fewer contacts).
- **Presence of a child under 18 (one child):** The non-index member is a young child. The model treated the presence of a child as reducing household-level transmission risk.

Coherence check: The pattern is internally consistent but surprising in places. Low income and small household size are plausible protective factors. However, a high index-to-household ratio (1 of 2 people) normally might suggest higher risk, yet the model treated it as risk-reducing — possibly because the single non-index member is a very young child (age 3), and the model may be learning that child–adult transmission is less frequently recorded in registry data. Overall, the model is confidently predicting low risk, but this is a statistical pattern, not a guarantee.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 6775956 (age 3) – Not flagged (4.9%) This very low score was driven entirely downward by:

- **Age 3 (strongest factor):** The model strongly associates very young childhood with low susceptibility scores. This is consistent across all example households (the 6-year-old and 7-year-old also had low scores, though not this low).
- **Household disposable income (zero value):** The family income is recorded as zero — likely a data artifact (e.g., missing or imputed value). The model treated this as a strong downward factor.
- **Personal disposable income (zero):** Same issue — the child’s individual income is zero, which the model interpreted as reducing susceptibility.
- **Country of birth (Sweden, downward):** Being born in Sweden slightly reduced the score relative to other backgrounds in the model’s framework.
- **Housing type (row house / *FlerHy*, downward):** The specific dwelling form pushed the score lower.

Plausibility: The model’s assignment of very low susceptibility to a 3-year-old is broadly consistent with the idea that young children may experience milder or asymptomatic illness and may be less likely to appear as recorded cases in registry data. However, the zero-income values

are a clear data-quality concern. Experts should verify whether these zeros are genuine (e.g., unreported income) or imputation artifacts before interpreting the income-related drivers.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Very small household size, young index case age, and very young child as the only contact — all align with lower transmission risk. The model’s confident low-risk assignment is not unreasonable.
- **Uncertain / surprising patterns:**
 - The model treated a high index-to-household ratio (0.5) as risk-reducing, which is the opposite of what one might expect. This may be because the non-index member is a very young child, and the model prioritises age over ratio.
 - Income again appears as a dominant factor, but here low income pushes risk down — the mirror image of the high-income households. This suggests the model has learned a strong linear relationship between income and risk in this dataset, which may not reflect biological reality.
 - **Zero-income values** appear repeatedly (child’s individual income, family income). These are unlikely to be true economic zeros and may represent missing data. The model treats them as real, which is a concern.
- **Proxy variables:** Income is once again acting as a proxy — here for low-resource households. Homecare alarms and medication histories did not appear in top factors for this household.
- **Additional information needed:** Experts would need to verify the accuracy of income data, particularly zero values. They would also benefit from understanding whether the model has been tested on very small households with young children to validate its low-risk predictions against real outbreak data.
- **Uncertain zone:** Not applicable. The scores (6.7% household, 4.9% individual) are extremely far from their respective thresholds. The model is highly confident in its “no flag” decision.

5. MODEL CONFIDENCE NOTE

The model’s probability estimates for this household (6.7% for transmission, 4.9% for the individual) are very far below the empirical thresholds, indicating high statistical confidence in its low-risk classification. However, statistical confidence does not mean the model is correct — especially given the zero-income values that may be data artifacts. These outputs should be treated as a limited reference, and experts should prioritise data quality review before considering the model’s low-risk assessment as actionable.

4.4.5 Llama 4

Llama 4 — Sample 4

COVID-19 Household Transmission Model Analysis

Household ID: 9878768

Case Summary

This is a 2-person household—likely a young mother (age 28) and her 3-year-old child—where the model assigns just a 6.7% transmission probability, well below the flagging threshold. The single non-index member, the 3-year-old child, receives an exceptionally low susceptibility score of 4.9%, also far below the flagging threshold. This represents the opposite end of the risk spectrum from the previous cases: a small, low-income household with a young child where nearly every model factor pushes toward *lower* transmission risk rather than higher.

Household Model Assessment

The model assigns only a 7% transmission probability—among the lowest scores possible. Every top feature pulls the score *downward*:

Low family income (108,000 SEK annually): This is the single strongest factor and contrasts sharply with previous cases where high income drove scores upward. At roughly 108,000 kronor per year, this household sits well below median Swedish income levels. The model learned from training data that lower-income households had reduced observed transmission rates in 2020—potentially reflecting reduced mobility, different occupational patterns, or simpler social networks.

High index-to-household ratio (50%): With one infected person in a 2-person household, half the household is already infected. Paradoxically, this *reduces* rather than increases the model’s transmission score. This likely reflects the mathematical reality that when 50% of a household is already infected, there are fewer people left to infect—the model is predicting whether *additional* members will be flagged, and here only one person remains.

Young index case (age 28): A 28-year-old index case pushes the score downward. Combined with the household composition (one adult, one young child), this suggests a young single parent scenario. The model may have learned that younger single parents with toddlers had lower observed transmission rates in the 2020 data.

Small household size (2 people): Two-person households have fewer transmission pathways than larger families. Less interpersonal contact, simpler household dynamics, and potentially more ability to isolate within the home.

Female index case: The index case being female (proportion = 1.0) reduces the transmission score. Whether this reflects biological factors, behavioral differences, or social patterns observed in the training data isn’t clear from the model alone.

Young child present (age 0-17 count = 1): Having a child under 18, specifically the minimum age of 3 years, consistently pushes the score downward. This aligns with patterns we’ve seen: young children receive protective scores throughout this model.

The pattern is remarkably coherent and represents the inverse of high-risk households: small

size, low income, young child, young single parent, female index case. Every structural factor points toward lower transmission risk based on the model’s learned patterns.

Individual Susceptibility Profiles

Member 6775956 (Age 3) — 4.9% susceptibility, NOT FLAGGED

This toddler receives one of the lowest susceptibility scores possible, with all factors pulling downward:

- **Young age (3 years):** This is by far the dominant protective factor. The model strongly associates early childhood with low susceptibility scores—a pattern consistent across all cases we’ve examined. Whether this reflects biological susceptibility, different symptom presentation, testing patterns, reduced social contacts, or other factors from the 2020 Swedish context isn’t directly observable from the model.
- **Zero personal and family income:** Children without independent economic profiles consistently receive lower scores. Here, zero family income (rather than the household’s 108,000 SEK reported at the household level, this specific child has no income attributed) reinforces the protective pattern.
- **Swedish birth country:** Being born in Sweden (Fodelseland = SVERIGE) slightly reduces the score, suggesting Swedish-born children had lower observed rates in the training data.
- **Immigration background code 21:** This classification reduces the score, indicating the model learned lower transmission associations with this demographic category.
- **Apartment dwelling (FlerHy - multi-family housing):** Contrary to single-family homes associated with higher risk in previous cases, this apartment-dwelling child receives a protective effect, possibly reflecting that smaller, lower-income housing correlated with lower transmission in the data.

The model sees virtually no risk factors for this child—age, household structure, socioeconomic profile, and demographic characteristics all align toward minimal susceptibility. The 5% score suggests the model views this as close to its lowest-risk profile.

Expert Considerations

Complete risk reversal: This case demonstrates the model’s opposite pole—where the high-risk cases showed large households, high income, working-age adults, and all-adult composition, this case shows small household, low income, young single parent, and a toddler. The model has clearly learned distinct patterns at both ends of the risk spectrum.

Income as a bidirectional proxy: While high income (700-900K SEK) drove scores upward in previous cases, low income (108K SEK) drives scores downward here. This suggests income is proxying for a bundle of behavioral, occupational, and mobility factors that cut both ways—high-income households may have had more external contacts and exposure opportunities, while low-income households may have had more constrained mobility and simpler social networks.

Young children as a protective factor: Across all cases examined, young children (ages 3-7) consistently receive the lowest susceptibility scores. This pattern appears robust in the model’s

learned associations, though the underlying mechanism—biological, behavioral, testing-related, or policy-related (school closures in 2020)—remains unmeasured.

Female index case effect: The model treats female index cases as lowering transmission risk. This could reflect contact patterns, caregiving behaviors, symptom recognition and isolation timing, or other factors present in the 2020 training data.

Two-person household dynamics: Small households present a mathematical edge case—when one person is already infected, the "additional transmission" question applies to fewer people. The model's logic appears to recognize this structural constraint.

Missing context: The model has no information about actual living conditions, childcare arrangements, work-from-home capability, or social support networks—factors that might substantially affect actual transmission risk for a young single parent.

Model Confidence Note

The model expresses very high confidence in the low-risk assessment: the household score (7%) sits far below the 25.6% threshold, and the individual score (5%) is even more decisively below the 30% threshold. The overwhelming alignment of protective factors—small household, low income, young child, young mother—creates a clear low-risk profile in the model's learned patterns. However, as with all model outputs, this represents statistical associations from 2020 Swedish registry data, not a validated prediction of actual infection outcomes. A low model score does not guarantee no transmission occurred, just as a high score does not prove transmission did occur. These outputs provide one reference point for expert assessment, not a definitive risk determination.

Note: This analysis explains why the model assigned a low-risk assessment based on patterns in historical training data, not actual infection outcomes or epidemiological validation.

Evaluation Scores — Sample 4

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 IND-R@2	B6 IND-R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.5 Sample 5

4.5.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 5

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 13054736 | Model Analysis

1. CASE SUMMARY

This household comprises two individuals: one confirmed COVID-19 index case and one exposed member. The model assigned a transmission score of 23% to this household—below the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model does not flag this household as high-risk for transmission. However, the single non-index member (member 12361865) received a susceptibility score of 35%, just above the individual-level flagging threshold of 30%, creating a discordant assessment where the household is not flagged but the individual within it is. This represents a borderline case where the model expresses uncertainty at both levels, with scores hovering near their respective thresholds.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 23%, falling just below the 30% threshold by a narrow margin. This positions the household in the model’s uncertain zone—small variations in the feature profile would likely flip the binary decision. Nearly every influential feature pushed the score downward, creating a predominantly protective profile.

Key factors driving the model’s household-level assessment:

Moderate household income (mean family income of approximately 161,000 SEK/year) was the most influential factor, substantially reducing the model’s score. This income level falls in the lower-middle range compared to other cases examined, and the model treats it as protective—continuing the counterintuitive income-risk gradient pattern.

Small household size (2 persons) reduced the score considerably, consistent with the model’s learned association between smaller households and lower transmission risk. This pattern aligns with epidemiological expectations about reduced contact opportunities.

Young index case age (maximum and mean both 26 years, indicating a single young adult index case) strongly reduced the score. The model consistently weights younger index cases as less likely to generate household transmission.

Spacious living conditions (the household meets the "spacious" classification, suggesting low crowding) reduced the score. Combined with total floor area of 58 m² for two people (29 m² per person), this indicates adequate physical space for potential isolation.

Multi-family dwelling (apartment building, FlerHy) and all-male household composition (0% female index case) both contributed modest protective effects.

The pattern is relatively coherent: small household size, spacious living conditions, and young index case age all align with factors that could plausibly reduce transmission risk. The protective effect of moderate-to-low income remains the primary interpretive puzzle, likely reflecting training

data artifacts rather than causal mechanisms.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12361865 (Age and other demographics not directly stated)

The model assigned a susceptibility score of 35%, narrowly above the 30% flagging threshold, creating a discordant situation where the individual is flagged despite the household not being flagged. This individual sits squarely in the model’s uncertain zone. Interestingly, every single influential feature pushed the score *downward*—there are no visible risk-increasing factors—yet the baseline score before feature adjustments must have been high enough that even after all protective adjustments, the score remained marginally above threshold.

Family income (approximately 65,000 SEK/year, notably lower than the household mean of 161,000 SEK/year, suggesting this may be a dependent or low-earning member) was the strongest protective factor, reducing the score substantially. Mean household family income, absence of gender diversity in the household (all-male composition), young minimum household age, Swedish background (immigration code 12, indicating born in Sweden to Swedish-born parents), male sex, low male count, absence of children under 18, absence of children aged 6-17, and non-multigenerational household structure all further reduced the score.

The pattern is paradoxical: the model applies numerous protective adjustments yet still flags this individual. This suggests the individual’s baseline risk profile (before feature-specific adjustments) positions them above threshold, and the protective features only partially offset this baseline. Age information is not directly visible in the top features, but given the young index case (26 years) and two-person household, this member is likely also a young adult, which would typically confer protection but apparently does not dominate the model’s assessment here.

4. EXPERT CONSIDERATIONS

Threshold ambiguity: This case exemplifies the challenge of binary thresholds applied to continuous risk scores. The household score of 23% sits 7 percentage points below threshold, while the individual score of 35% sits 5 points above—both occupy uncertain territory where the model’s preference is weak and unstable.

Household-individual discordance: The model flags the individual as susceptible while not flagging the household for transmission risk. This logical tension arises from the two-level architecture operating independently: one model assesses "will transmission occur in this household" while the other assesses "among exposed members, who is susceptible." In a two-person household where the individual is flagged, the discordance seems particularly notable—if the one exposed member is susceptible, shouldn’t household transmission be likely?

Protective factors without clear risk drivers: The unusual feature of this case is that all visible influential factors reduce risk, yet the individual remains flagged. This suggests either: (a) the model’s baseline prediction for this feature combination starts high, with protective features only partially offsetting it, or (b) influential risk-increasing factors exist but fall outside the top features shown, or (c) complex feature interactions create risk signals not visible in univariate feature importance.

Income pattern consistency: Low income again acts protectively, reinforcing the pattern across cases that this likely reflects 2020 Swedish training data artifacts—perhaps related to differential testing access, workplace exposure patterns, or selection effects in who entered the registry dataset.

Information gaps: Experts would need: the age of the non-index member (not directly shown); relationship between household members (roommates, partners, family); actual living arrangement within the 58 m² space; occupational exposures; and timing of potential isolation measures. The discordance between household and individual assessments would particularly benefit from understanding whether they share sleeping quarters, meal preparation, or bathroom facilities.

5. MODEL CONFIDENCE NOTE

The household score of 23% and individual score of 35% both sit very close to the 30% threshold, placing this case firmly in the model’s uncertain zone. The household assessment narrowly misses flagging, while the individual assessment narrowly achieves it—small changes in either feature profile would likely reverse these binary decisions. This dual uncertainty, combined with the household-individual discordance, indicates the model is expressing minimal confidence in either assessment. The predominance of protective features for the individual, despite the flagging outcome, further suggests the model’s reasoning here is marginal and potentially unstable. These outputs provide limited actionable information and should be treated as highly uncertain references rather than confident risk assignments—and as always, the model’s actual correctness remains unknown.

Sample 5sec:out5

4.5.2 GPT-5.3

GPT-5.3 — Sample 5

1. CASE SUMMARY

This case involves a two-person household with one identified index case and one exposed non-index household member. At the household level, the model assigned a 23.4% transmission score, which falls below the model’s operational threshold of 25.6%, meaning the household was not flagged as high-risk by the model.

At the individual level, however, the non-index household member received a susceptibility score of 35.3%, above the individual-level threshold, and was therefore flagged by the model as susceptible. The model’s reasoning appears to rely primarily on structural household characteristics such as household size, income profile, age structure, and housing conditions rather than on strong medical-risk indicators.

Importantly, these outputs describe how the model classified this case based on learned statistical patterns in registry data. They do not indicate whether transmission actually occurred or whether the individual was truly infected.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level model assigned a transmission score of 23.4%, slightly below the operational threshold of 25.6%. This means the model did not classify the household as one in which secondary transmission was likely according to its internal criteria.

The most influential factors all pushed the model toward a *lower* transmission assessment:

- **Higher household disposable income** was the strongest factor lowering the score. The model appears to associate higher household income with lower transmission likelihood, potentially reflecting indirect factors such as housing quality, reduced crowding, greater ability to isolate, or different occupational exposure patterns.
- **Small household size (2 persons)** substantially lowered the score. This is epidemiologically coherent in the sense that fewer household members reduce the number of potential close-contact transmission pathways available within the home.
- **Young household age profile** also reduced the score. Both the average household age and the age of the index case were approximately 26 years, and the household contained no elderly members or children. The model therefore viewed this as a relatively simple adult-only household structure rather than a multigenerational or dependency-heavy environment.
- **Spacious housing conditions** contributed further to the lower assessment. The model weighted indicators suggesting relatively adequate living space per person and a total floor area of approximately 58 m². This aligns with established epidemiological reasoning that reduced crowding may lower prolonged close-contact exposure opportunities.
- The dwelling type (“FlerHy”, multi-dwelling/apartment-style housing) modestly reduced the score in this case, although this effect is less intuitive and may reflect patterns specific to the training data rather than a direct housing effect.

Overall, the household-level reasoning forms a relatively coherent pattern. The model consistently weighted lower crowding, smaller household size, younger adults, and higher income toward reduced transmission risk. These are structurally plausible proxies for lower within-home exposure intensity, although the model cannot establish causality.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Non-index member: 12361865

- **Model-assigned susceptibility score:** 35.3%
- **Threshold:** 19.2%
- **Model classification:** Flagged as susceptible

Interestingly, the strongest feature influences for this individual mostly pushed the score downward rather than upward, suggesting the final flagged score may reflect broader baseline patterns learned by the model rather than one dominant risk factor.

Key downward influences included:

- **Higher family income** was the strongest lowering factor. As at the household level, the model appears to associate greater financial resources with conditions that reduce susceptibility scores, potentially reflecting better housing, reduced occupational exposure, or other unmeasured socioeconomic advantages.
- **Young household age structure** lowered the score. The model weighted the absence of elderly members and the relatively young minimum household age toward lower susceptibility.
- **Immigration background encoding** also modestly lowered the score in this specific case. Because this is an encoded registry variable rather than a direct social or biological measure, it should be interpreted cautiously. It may act as a proxy for complex social-network, demographic, or healthcare-access patterns present in the training data.
- **Male sex** slightly lowered the score according to the model’s internal weighting.
- **No children in the household** also reduced the score, suggesting the model may associate households with children — especially school-age children — with more extensive social contact networks.

Notably, the model identified essentially no strong medical-history drivers. Medication-history variables and homecare indicators had near-zero influence, suggesting the model did not view this person as medically frail or heavily burdened by chronic illness.

From a domain perspective, the pattern is somewhat mixed. The household-level characteristics mostly point toward a lower-risk environment, yet the individual model still produced a score above threshold. This may indicate that the individual-level model has a comparatively low operational threshold or is sensitive to subtle combinations of demographic factors not immediately obvious from the top-ranked features alone.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear domain-consistent. Smaller households, higher income, younger adults, and more spacious living conditions are all plausible structural correlates of reduced within-household exposure intensity. The absence of elderly members, homecare dependency, or substantial medication burden also contributed to a generally lower-risk profile. At the same time, some elements remain uncertain. The fact that the individual-level model flagged susceptibility despite nearly all prominent feature influences lowering the score suggests that the prediction may depend on weaker background interactions distributed across many variables rather than a clearly interpretable driver. This reduces interpretability and should encourage caution.

A number of variables likely function as proxies rather than direct mechanisms. Income may stand in for housing quality, occupation type, or capacity to isolate. Homecare utilisation likely serves as a proxy for frailty or dependency. Immigration-background variables may capture broader social or structural patterns that are not explicitly measured elsewhere in the dataset. Before acting on these outputs, a domain expert would likely need additional contextual information, including occupational exposure, timing and intensity of contact within the home,

behavioural isolation measures, ventilation conditions, symptom history, and vaccination status or prior immunity — none of which are represented here.

The household score sits relatively close to the threshold (23.4% vs 25.6%), suggesting a borderline classification rather than a strongly confident “low-risk” assessment.

5. MODEL CONFIDENCE NOTE

The household-level prediction is relatively close to the operational threshold, which suggests moderate uncertainty rather than a strongly separated classification. By contrast, the individual-level score is clearly above the individual threshold, although the underlying feature pattern is not especially strong or clinically intuitive.

These outputs should therefore be interpreted as a limited statistical reference describing the model’s internal reasoning on this case. They do not establish actual transmission, infection status, or causal risk, and their correctness is unknown at inference time.

4.5.3 Grok 4.2

Grok 4.2 — Sample 5

CASE SUMMARY

This is a small 2-person household (ID 13054736) with one index case. The household-level model assigns a 23.4% probability of transmission, which falls just below the empirical decision threshold of 0.256, so the household is not flagged. However, at the individual level, the single non-index member receives a susceptibility score of 35.3%, which exceeds the individual threshold and results in a susceptible flag. This creates a nuanced situation where the model sees limited overall household risk but identifies the exposed member as warranting attention based on their feature profile.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 23.4%, narrowly below the threshold, resulting in no transmission flag.

The strongest factors pulling the score downward (reducing predicted risk) include:

- **Low family disposable income** (mean around 161,000 SEK annually), which the model weighted heavily as lowering transmission potential.
- **Small household size of 2**, limiting opportunities for spread.
- **Young index case age** (maximum and mean age around 26 years).
- Spacious living conditions and multi-family housing type, both contributing to lower risk in the model’s assessment.

The model’s emphasis on small household size and young adult index case aligns well with epidemiological expectations — smaller homes generally have fewer contacts, and younger adults may have different behavioral or exposure patterns. Lower income and spacious housing appear to act as proxies that the model associates with reduced transmission likelihood in this context,

though the exact mechanisms (e.g., location, occupation, or lifestyle factors) would require domain expertise to interpret.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12361865: The model assigns a 35.3% susceptibility score and flags this person as susceptible, despite the household-level assessment. The top factors pulling the score downward include moderate-to-low family income, household income measures, gender diversity (or lack thereof), young minimum household age, and immigration background. Several other demographic and household composition factors also contributed modestly downward.

The model still crosses the individual threshold, suggesting that certain personal or household characteristics specific to this member pushed the susceptibility score higher relative to the overall household picture. This divergence between levels highlights how the individual model can identify targeted risks even when aggregate household transmission potential remains below threshold. From a domain perspective, the combination of low income indicators and background variables may reflect specific social or environmental circumstances captured in the 2020 data.

EXPERT CONSIDERATIONS

This case illustrates the value of the dual-level approach: the household is not prioritised overall, yet the model singles out the non-index member for closer attention. Domain-consistent patterns include the risk-reducing role of small household size and young age. However, the contrast between the low household probability and the flagged individual score is noteworthy and may reflect how the model balances aggregate structural factors against individual-specific ones.

Income, housing type, and immigration background variables likely serve as proxies for unmeasured factors such as occupational exposure, residential density, social networks, or testing behaviours during 2020. Medication and homecare variables had negligible influence. Experts should review additional contextual information unavailable to the model, including the precise timing of the index case relative to potential exposure, living arrangements, any protective measures taken, occupation or daily routines of the non-index member, and actual testing or symptom records. The household probability sits close to the threshold (23.4% vs 0.256), placing it in a relatively uncertain zone compared to clearer cases.

MODEL CONFIDENCE NOTE

The household probability (23.4%) is close to but below the decision threshold, introducing some boundary uncertainty at the aggregate level. The individual score (35.3%) is above its threshold but not extremely high, suggesting moderate rather than decisive flagging for the non-index member. These outputs represent statistical patterns learned from 2020 Swedish registry data and are not verified infection risks or clinical predictions. They are best used as a limited reference to help guide, but never replace, expert public health assessment and further investigation.

4.5.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 5

Here is the structured interpretation of the model’s output for the fifth household, written for public health policy experts.

1. CASE SUMMARY

This household consists of two people: one designated index case and one other member. The model assigns a **23.4% probability** of within-household COVID-19 transmission, which falls just below its internal flagging threshold of 25.6%. Although the household is **not flagged** by the model, the single non-index member receives a susceptibility score of **35.3%** and is flagged as susceptible at the individual level. **These are statistical model outputs, not confirmed infection statuses.**

2. HOUSEHOLD MODEL ASSESSMENT

The model did **not** flag this household for transmission, but the probability (23.4%) is very close to the threshold (25.6%) — a borderline case. All of the top influential features pushed the score downward:

- **Household disposable income (161,000 SEK/year – low):** The strongest downward driver. Consistent with the previous low-income household (report 4), the model associates lower income with lower transmission risk.
- **Household size (2 persons):** Very small household size pushed the score down, which is epidemiologically plausible.
- **Age of index case (26 years):** The young age of the index case reduced the model’s transmission estimate.
- **Spacious living condition (is_spacious flagged):** The household is classified as spacious (low crowding). The model treated this as reducing risk — unlike the previous households where more space sometimes increased risk. This is the first example where spaciousness aligns with intuitive expectations.
- **Housing type (row house – *FlerHy*):** This dwelling form also pushed the score downward.

Coherence check: The pattern here is more internally consistent than in some previous households. Low income, small household size, young index case, and spacious living all point in the same direction (lower risk). The only notable feature is that the model’s household-level probability (23.4%) is just below the threshold, while the individual non-index member is flagged (35.3%). This discordance between the two levels is worth attention.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12361865 – Flagged (35.3%) *Note: This individual’s age is not shown in the top features list, but the household composition suggests they are likely the other adult.*

All top influential features pushed the score **downward**, yet the final score (35.3%) still crossed the individual threshold (19.2%). This indicates that while every listed factor reduced the score, the model’s baseline or other unshown factors kept it above the line.

The strongest downward influences were:

- **Household disposable income (low – approx. 65,100 SEK/year):** The model treated low family income as reducing susceptibility — the same pattern as at household level.
- **Household mean income (also low):** Reinforced the downward push.
- **Gender diversity (both sexes present in household):** This pushed the score down, though the direction is uncertain from a domain perspective.
- **Youngest age in household (likely the index case at 26 years):** Having a young minimum age reduced the non-index member’s score.
- **Immigration background (code 12 – not the foreign-background code 11):** This slightly reduced the score.
- **Sex (female, assuming Kon=1 is male? The report shows Kon_std=1.0, which may indicate male; experts should verify coding):** Pushed the score down.

Plausibility: The model flags this individual despite every top factor pushing the score downward — a borderline case. The fact that the individual score (35.3%) is above threshold while the household score (23.4%) is below threshold suggests the model sees some individual-specific signal (perhaps age or other unlisted features) that increases susceptibility, but not enough to push the household-level probability over the line.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Low income, small household size, young index case age, and spacious living all align with lower transmission risk. This household is the most intuitively coherent of the five examples.
- **Uncertain / surprising patterns:**
 - The **discordance between household and individual flags** is noteworthy: household not flagged (23.4% vs. 25.6% threshold), but the single non-index member is flagged (35.3% vs. 19.2% threshold). This suggests the model’s two levels are not perfectly aligned — the individual model may be more sensitive, or the household model may require a stronger signal to cross its higher threshold.
 - All top individual-level features show negative influences, yet the score still exceeds the threshold. This implies the model’s baseline prediction or other unshown features (possibly the non-index member’s own age or health factors) are positive. Experts would need to see the full feature set to understand what is driving the individual flag.
- **Proxy variables:** Income again acts as a proxy, here for lower socioeconomic status. The

“spacious” flag (`is_spacious`) is a direct crowding measure that behaved intuitively in this case — unlike earlier households where more space paradoxically increased risk.

- **Additional information needed:** Experts would need to know the age and health characteristics of the non-index member to understand why they were flagged individually while the household was not. They would also benefit from understanding the model’s baseline probability (the score when all features are neutral) to interpret the net effect of downward pushes.
- **Uncertain zone:** The household score (23.4%) is very close to the threshold (25.6%) — a difference of only 2.2 percentage points. This is a true uncertain zone. The model is not confident in either direction at the household level, and small changes in features could flip the prediction.

5. MODEL CONFIDENCE NOTE

The household-level prediction (23.4%) is very close to the empirical threshold (25.6%), placing this case in a low-confidence uncertain zone for the household model. In contrast, the individual-level prediction (35.3%) is more comfortably above its lower threshold (19.2%). Correctness is unknown for both levels. These outputs should be treated as a borderline reference — the model is essentially saying this household could go either way, while the individual member is more clearly flagged by the model’s internal logic.

4.5.5 Llama 4

Llama 4 — Sample 5

COVID-19 Household Transmission Model Analysis

Household ID: 13054736

Case Summary

This is a 2-person household—likely two young men (both age 26) living together—where the model assigns a 23.4% household transmission probability, just below the 25.6% flagging threshold. However, the individual-level model tells a different story: the single non-index member receives a 35.3% susceptibility score, placing them above the individual flagging threshold. This creates an interesting divergence where the household assessment suggests low risk, but the individual assessment flags the exposed person as susceptible—a reminder that the two model levels can reach different conclusions on the same case.

Household Model Assessment

The model assigns a 23% household transmission probability—just shy of the flagging threshold. All major factors push *downward*:

Low family income (161,000 SEK annually): This modest income level is the strongest protective factor, consistent with patterns where lower-income households received reduced transmission scores. At roughly 161,000 kronor per year for a two-person household, this sits

below typical Swedish household income levels.

Small household size (2 people): Two-person households minimize transmission pathways—only one exposed person remains after the index case.

Young index case (age 26): A 26-year-old index case drives the score downward, suggesting young adult households had lower observed transmission rates in the training data.

Spacious housing: The household is flagged as "spacious" (low crowding index), meaning high living area per person. More personal space provides physical separation opportunities that reduce transmission risk.

Apartment dwelling (FlerHy - multi-family housing): Contrary to single-family homes associated with higher risk in some cases, this apartment residence provides a protective effect, likely correlating with the lower-income and smaller-space profile.

Male index case (proportion female = 0.0): Having a male rather than female index case provides a slight protective effect—the opposite of what we saw with female index cases reducing scores in other cases.

The pattern is coherent: small, lower-income, spacious apartment with young adult males. All structural factors point toward lower transmission risk. The 23.4% score sits just 2 percentage points below the threshold—this is a borderline case where a small shift in any factor could change the household-level flagging decision.

Individual Susceptibility Profiles

Member 12361865 (Age unknown, likely ~26) — 35.3% susceptibility, FLAGGED

Here's where it gets interesting: despite the household-level "no flag" decision, this individual receives a 35.3% score—above the 30% individual threshold. Every top feature actually *reduces* the score, yet the baseline is high enough that the person is still flagged:

- **Low family income (65,100 SEK reported at individual level):** The strongest protective factor, yet not strong enough to push the score below the individual threshold.
- **Household income structure:** Mean family income metrics continue pulling downward.
- **No gender diversity:** An all-male household (gender_diversity = 0) reduces the score, possibly reflecting different contact or caregiving patterns observed in the training data.
- **Immigration background (code 12):** This classification provides protection, consistent with what we've seen for certain demographic categories.
- **Male (Kon = 1):** Being male reduces the score slightly.
- **No children under 18:** The absence of children in the household provides a protective effect.

What's striking is that **every** identifiable factor pulls the score *down*, yet this person is still flagged as susceptible. This suggests the model's baseline score for this demographic profile starts relatively high, and while protective factors lower it, they don't lower it enough to cross below the 30% threshold. The 35.3% score is barely above threshold—a 5-percentage-point margin.

Expert Considerations

Model-level divergence: This case highlights that household and individual models operate independently and can disagree. The household model says "low risk for any additional transmission" (23.4%), while the individual model says "this specific person is susceptible" (35.3%). Both can be true from a model perspective—the household structure is generally protective, but the one exposed person still has risk factors the individual model weighs.

Threshold sensitivity: Both scores hover near their respective thresholds. The household score (23.4%) sits just 2 points below its cutoff, while the individual score (35.3%) sits just 5 points above its cutoff. Small perturbations in features or threshold calibration would flip these decisions. This is a classic borderline case where model confidence should be considered low.

Baseline susceptibility puzzle: The most unusual pattern is that every measured factor *reduces* risk for the individual member, yet they're still flagged. This suggests unmeasured factors or baseline demographic characteristics (age, household type, geographic location, or other variables not in the top feature list) create an elevated starting point that the protective factors can't fully overcome.

Young male household dynamics: The model appears to have learned specific patterns about young adult male cohabiting households that don't fully mirror other demographic groups. Whether this reflects actual transmission biology, testing behaviors, isolation compliance, or occupational patterns from 2020 isn't observable from the model.

Income as partial protection: At 161,000 SEK household income (or 65,100 SEK individual), we're seeing income provide protection but not as strongly as in the extremely low-income case (108,000 SEK). This suggests the income-risk relationship may be nonlinear—very low income provides strong protection, moderate-low income provides moderate protection, and high income increases risk.

Missing behavioral context: The model has no information about actual living arrangements (roommates vs. partners), work situations, social activities, or isolation capability—factors that could substantially differ between theory and reality for two young adults living together.

Model Confidence Note

This case demonstrates model uncertainty. The household score (23%) barely misses the flagging threshold, while the individual score (35%) barely exceeds it. Both predictions sit in the gray zone where small changes would reverse the decision. The contradiction between household and individual assessments further indicates this is not a clear-cut case. When household and individual models disagree, and when scores cluster near thresholds, experts should treat these outputs as highly uncertain references rather than confident predictions. The model is essentially saying "this case could go either way"—which is honest but provides limited actionable guidance.

Note: This analysis explains the model's reasoning and highlights where its confidence is low. These borderline scores reflect statistical uncertainty in pattern-matching, not validated infection predictions.

Evaluation Scores — Sample 5

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.6 Sample 6

4.6.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 6

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 14260863 | Model Analysis

1. CASE SUMMARY

This household comprises four individuals, including one confirmed COVID-19 index case and three exposed members. The model assigned a transmission score of 38% to this household—modestly above the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission moderately likely, though with limited confidence. Among the three non-index members, the model flagged one as susceptible: a 26-year-old adult (member 13538493, score 55%). The other two members were not flagged: one individual (member 9105013, score 27%) falls just below threshold, while a 5-year-old child (member 12108918, score 5%) receives strong protective assessment. This represents a mixed-confidence case with one clear flagging, one borderline non-flagging, and one confident protective assessment.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 38%, sitting 8 percentage points above the 30% threshold—a modest margin that places the household in a relatively uncertain zone. Notably, every influential feature pushed the score *downward*, suggesting the baseline risk before feature-specific adjustments must have started substantially higher, with protective factors only partially offsetting it.

Key factors driving the model’s household-level assessment:

Moderate-low household income (mean family income of approximately 138,000 SEK/year) was by far the most influential factor, substantially reducing the model’s score. This continues the persistent pattern where lower income acts protectively in the model’s learned associations, contrary to typical health gradient expectations.

Young index case age (both maximum and minimum of 32 years, indicating a single young adult index case) reduced the score considerably. The model consistently weights younger index cases as less likely to generate transmission.

Young mean household age (18.75 years, indicating a household with young children bringing down the average) further reduced the score.

Absence of elderly members (no members aged 65+) and **presence of a young child** (child under 6 in the household) both contributed protective effects, as did moderate living space (64 m² total, or 16 m² per person), median personal income (approximately 134,000 SEK/year), apartment dwelling, and female index case (100% female).

The pattern is partially coherent: absence of vulnerable elderly, presence of young children who may receive protective weighting, and a young index case all align with lower transmission

expectations from the model’s learned patterns. However, the dominant protective role of low-moderate income remains epidemiologically questionable and likely reflects training data artifacts.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 13538493 (Age 26)

The model assigned a susceptibility score of 55%, well above the flagging threshold, identifying this young adult as the primary at-risk individual in the household. Age (26 years) was the strongest driver elevating the score—the model appears to weight young-to-middle-aged adults as higher risk compared to children or elderly. Personal disposable income (approximately 286,000 SEK/year, the highest in the household) also pushed the score substantially higher, reinforcing the income-as-risk pattern. Maximum household age and having two children under 18 in the household further elevated the score modestly, as did the proportion of males and mean household age. Working against flagging were household age variance (suggesting age diversity reduces risk), number of index cases, multigenerational household structure, and proportion with foreign background—all of which reduced the score. The model shows moderate-to-strong confidence in flagging this individual, with the score sitting comfortably above threshold.

Member 9105013 (Age not directly shown)

The model assigned a susceptibility score of 27%, just below the 30% flagging threshold, placing this individual in the uncertain zone where small feature changes could flip the binary decision. Absence of personal income (both family income and disposable income recorded as zero, suggesting this is likely a dependent or non-earning member) was overwhelmingly protective, driving the score dramatically downward. Mean household family income, apartment dwelling, index case maximum age, household crowding index, total household floor area, and the index case being a child all further reduced the score. The sum of homecare alarms across the household also contributed modest protective effect. Only the presence of a child under 6 modestly increased the score. This individual sits at the margin—the model is essentially expressing no clear preference about whether to flag.

Member 12108918 (Age 5)

The model assigned a susceptibility score of 5%, far below the flagging threshold, and did not flag this young child. Age was overwhelmingly the most influential factor—the model assigns extremely strong protective weight to young children, with this 5-year-old receiving the largest protective signal in the household. Absence of personal income (both family and disposable income recorded as zero) further drove the score dramatically downward, as did index case maximum age, presence of a child under 6 (this child), mean household family income, proportion with foreign background, three-generation household structure, homecare services metrics, and total homecare contacts. Every influential feature pushed the score lower. The model expresses very high confidence in not flagging this child, consistent with protective patterns observed for young children across all cases.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s strong protective weighting of very young age (5-year-old child) and modest protective effect for the borderline member aligns with documented early-pandemic transmission patterns. The absence of elderly household members as a protective factor is epidemiologically reasonable, as it reduces the presence of highly vulnerable individuals who might amplify transmission dynamics.

Age-income interaction: The model flags the 26-year-old with highest personal income while not flagging the zero-income member who is likely also an adult (given the household composition). This suggests the model may be capturing correlations between earning status and exposure patterns—perhaps employed adults with income had systematically different contact patterns in 2020 Sweden compared to unemployed or dependent household members.

Threshold proximity: The household score (38%) and one individual score (27%) both sit relatively close to the 30% threshold, indicating modest confidence. Only the 26-year-old (55%) and 5-year-old (5%) receive assessments far from threshold.

Counterintuitive income effects: Low household income acts as the dominant protective factor at the household level, while high personal income drives individual flagging—creating an apparent contradiction. This likely reflects the model learning different patterns at household versus individual scales from the training data, though the causal interpretation remains unclear.

Information gaps: Experts would benefit from knowing: the exact age and relationship status of member 9105013 (the borderline case); occupational exposures for the 26-year-old flagged member; childcare arrangements for the two young children; actual room configuration within the 64 m² space; and timing of index case confirmation relative to household exposure periods.

5. MODEL CONFIDENCE NOTE

The household score of 38% sits modestly above the 30% threshold, suggesting weak-to-moderate confidence in flagging—small variations in the feature profile could reverse the decision. Among individuals, the model shows clear confidence for the 5-year-old (5%, strongly protective) and reasonable confidence for the 26-year-old (55%, clearly flagged). However, the borderline member at 27% occupies the uncertain zone where the binary flag is unreliable. The counterintuitive protective role of low income at the household level, combined with income-as-risk at the individual level, introduces interpretive challenges that cannot be resolved from these outputs alone. These predictions provide a limited reference point for qualified experts to integrate with domain knowledge and contextual factors—they are not standalone clinical decisions, and the model’s actual predictive accuracy for this household remains unknown.

Sample 6sec:out6

4.6.2 GPT-5.3

GPT-5.3 — Sample 6

1. Case Summary

This case involves a four-person household with one identified index case and three non-index

household members. The household-level model assigned a 37.6% probability of within-household transmission, which is above the model’s operational threshold of 25.6%, so the household was flagged by the model as higher priority for potential transmission concern.

At the individual level, the model differentiated substantially between household members. One adult non-index member received a comparatively high susceptibility score of 54.9% and was flagged by the model, while the other two members received lower scores of 26.8% and 5.1% respectively and were not flagged. The model’s reasoning appears to rely mainly on demographic structure, age composition, and income-related variables rather than healthcare utilisation or medication history in this household.

Importantly, these outputs describe how the statistical model behaved on this case. They do not establish whether transmission actually occurred or whether any individual truly became infected.

2. Household Model Assessment

The household-level model assigned a transmission score of 37.6%, above the empirical threshold of 25.6%. Although this is clearly above the cutoff, it is not an extremely high score, suggesting a moderate model indication rather than a strongly decisive classification.

The strongest factor lowering the household score was household disposable income. The model associated higher average family disposable income with a lower transmission score. In practical terms, the household appears relatively economically stable compared with other households in the training data, and the model weighted this as protective. Several other features also pushed the score downward:

- The index case was approximately 32 years old, and the model associated this age profile with lower transmission risk in this case.
- The household’s mean age was relatively young (approximately 19 years), which also reduced the score.
- The household did not include elderly members aged 65+, which the model interpreted as lowering risk.
- Larger total living area also modestly reduced the score, suggesting the model may associate more available space with fewer opportunities for close-contact exposure.
- The dwelling type was classified as a multi-unit residential setting (“FlerBo”), but in this specific model assessment it slightly lowered the score rather than increasing it.

Notably, many household features had near-zero influence, including healthcare utilisation indicators and measures of elderly care dependency. This suggests the model’s decision was driven primarily by demographic and socioeconomic structure rather than medical vulnerability. Overall, the pattern is reasonably coherent from a public health perspective. The model appears to associate higher income, greater living space, and absence of elderly residents with lower household transmission propensity. These are structurally plausible proxy indicators for housing conditions, crowding, and social vulnerability. However, despite several downward influences, the household still crossed the threshold, implying that the model identified enough residual risk

from the overall household configuration to classify it as higher priority.

3. Individual Susceptibility Profiles

Member 13538493 — Model score 54.9% (flagged)

This 26-year-old adult received the highest susceptibility score in the household and was clearly above the model threshold.

The strongest upward influence was age. The model associated this young adult age range with increased susceptibility in this case. Higher individual disposable income also increased the score substantially. This should not be interpreted causally; income may be functioning as a proxy for behavioural or occupational patterns not directly measured in the data.

The model also weighted the presence of two children under 18 in the household as increasing this person's score. This may reflect the model learning associations between family structure and exposure opportunities.

Several household composition factors reduced the score somewhat, including the household's multigenerational structure and lower proportion of foreign-background members. Overall, however, the upward influences outweighed the downward ones. The model's pattern here is moderately plausible, although the positive contribution from higher income is less intuitively straightforward and may reflect indirect social or occupational correlates.

Member 9105013 — Model score 26.8% (not flagged)

This member's score fell below the model threshold but remained relatively close to it, making this a more borderline case.

The strongest downward influences were very low or absent recorded disposable and family income values. The model strongly associated these values with reduced susceptibility in this instance. Household-level income measures, larger living area, and lower crowding indicators also pushed the score downward.

The only notable upward influence was the presence of a child under six years old in the household, though this effect was small. Overall, the model treated this person as lower priority primarily because of socioeconomic and housing-related features rather than health-related characteristics. The reasoning is somewhat harder to interpret substantively because zero-income registry values can sometimes reflect administrative or life-stage factors rather than true deprivation. A domain expert would likely want more contextual information before treating this pattern as meaningful.

Member 12108918 — Model score 5.1% (not flagged)

This 5-year-old child received a very low model-assigned susceptibility score.

The child's young age was by far the strongest factor lowering the score. Additional downward influences included low recorded income variables, the presence of another young child in the household, and several household-level socioeconomic indicators.

Healthcare utilisation and medication variables had essentially no influence. The model's reasoning here is internally consistent: it largely interpreted this child's demographic profile as comparatively low priority within the household context.

4. Expert Considerations

Several aspects of the model’s reasoning appear domain-consistent. Housing space, household age structure, and absence of elderly members all behaved in ways broadly compatible with known social determinants of respiratory infection transmission. Income-related variables also appear to function as proxies for broader living conditions and social context.

At the same time, some associations are harder to interpret directly. The positive contribution of higher disposable income for the flagged adult may reflect unmeasured factors such as occupational exposure, mobility patterns, or social connectivity rather than income itself. Similarly, zero-income registry values may represent administrative circumstances rather than genuine economic disadvantage.

The model relied very little on healthcare utilisation or medication history in this case, suggesting that structural household features dominated its assessment. Before acting on these outputs, a domain expert would likely want additional contextual information such as occupation, school attendance, behavioural exposure patterns, household contact intensity, vaccination status, or timing of exposure events — none of which are captured here.

One individual score (26.8%) sits relatively near the operational threshold, indicating a more uncertain classification zone for that household member.

5. Model Confidence Note

The household-level score was moderately above the model threshold, suggesting a reasonably stable model flag but not an extreme prediction. At the individual level, one member was clearly above threshold, one clearly below, and one relatively close to the cutoff, indicating mixed confidence across household members.

These outputs should be interpreted strictly as model behaviour on registry-based statistical patterns. Whether the predictions were correct is unknown, and the results should be treated as a limited decision-support reference for qualified experts rather than as evidence of actual transmission or infection risk.

4.6.3 Grok 4.2

Grok 4.2 — Sample 6

CASE SUMMARY

This is a 4-person household (ID 14260863) with one index case. The model assigns a 37.6% probability of household transmission and flags the household as one where secondary spread is expected according to its patterns. At the individual level, one non-index member is flagged as susceptible (54.9%), while the other two are not (26.8% and 5.1%). The flagged member is a young adult; the non-flagged include a young child. These outputs reflect the model’s assessment of this household’s profile drawn from 2020 Swedish registry data.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 37.6%, which exceeds the empirical threshold of 0.256 and results in a transmission flag.

The factors most strongly influencing the score were predominantly those pulling it downward,

including:

- **Low family disposable income** (mean around 138,000 SEK annually), which the model weighted as reducing transmission potential.
- **Young index case age** (32 years) and overall young household mean age (around 19 years).
- Absence of elderly members, presence of a child under 6, and multi-family housing type.

Despite these downward influences, the model still flagged the household, likely due to the combination of 4-person size and other unlisted factors balancing the profile. The emphasis on small household size effects, young ages, and presence of young children is broadly coherent with epidemiological understanding — younger households with children can show distinct mixing patterns, while lower income may proxy for specific living conditions, occupations, or mobility in 2020. The overall pattern is plausible but highlights how the model balances multiple structural elements.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 13538493 (age ~26 in 2020): The model assigns a 54.9% susceptibility score and flags this person. The strongest upward drivers are adult working age and personal disposable income. Household age variation and foreign background elements pulled modestly downward, but not enough to avoid a flag. The model links this young adult profile with moderate-to-elevated susceptibility in the household context, which may reflect typical working-age contact patterns.

Member 9105013: The model assigns a 26.8% susceptibility score, below the 0.192 threshold, so this person is not flagged. Dominant downward factors include zero personal and family income indicators, multi-family housing type, and household income and crowding measures. The model consistently down-weights this profile relative to others in the home.

Member 12108918 (age ~5 in 2020): The model assigns a very low 5.1% susceptibility score and does not flag this young child. The top factors pushing the score down are young age and zero personal/family income (expected for a child), reinforced by household income measures, presence of young children, and other demographic factors. The model strongly differentiates young children in this setting.

EXPERT CONSIDERATIONS

The model flags the household for potential intervention but focuses individual attention primarily on the young adult member rather than the children. Domain-consistent patterns include the differentiating role of age groups (higher scores for working-age adults, lower for young children) and the influence of household size. Lower family income emerges as a notable downward driver at the household level, likely acting as a proxy for unmeasured factors such as residential area, occupation type, or social circumstances rather than a direct biological effect.

Medication history and homecare variables had little influence here. Experts should integrate information the model cannot see, such as exact timing of the index case, specific occupations or school/daycare attendance, isolation practices within the home, ventilation, and testing history. The household probability (37.6%) is moderately above threshold, while the primary individual

flag (54.9%) sits in a clear but not extreme range.

MODEL CONFIDENCE NOTE

The household probability of 37.6% is comfortably above the decision threshold, supporting a clear flag, while individual scores show strong differentiation — one moderately elevated and two decisively low. These outputs are statistical pattern recognitions from historical 2020 registry data, not verified infection risks or clinical predictions. They serve best as a limited supplementary reference to assist qualified experts in prioritising and investigating households.

4.6.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 6

Here is the structured interpretation of the model’s output for the sixth household, written for public health policy experts.

1. CASE SUMMARY

This household contains four people, including one designated index case and three other members. The model assigns a **37.6% probability** of within-household COVID-19 transmission, which exceeds its internal flagging threshold of 25.6%. Among the three non-index members, the model flags only one as “susceptible” (with a score of 54.9%), while the other two (with scores of 26.8% and 5.1%) are not flagged. **These are statistical model outputs, not confirmed infection outcomes.**

2. HOUSEHOLD MODEL ASSESSMENT

The model flagged this household for potential transmission. However, **every top influential feature pushed the score downward** — yet the final probability (37.6%) still crossed the threshold. This means the model’s baseline or other unshown factors were strong enough to overcome these downward pushes.

The strongest downward influences were:

- **Household disposable income (138,300 SEK/year – low to moderate):** As in other low-income households, lower income reduced the transmission score.
- **Age of index case (32 years):** The index case is relatively young; the model treated this as reducing risk.
- **Average household age (18.75 years):** The household has a low average age, which pushed the score down.
- **Presence of an elderly member (65+):** None present — and the model treated the *absence* of elderly as reducing risk (unlike the third household where absence pushed risk up). This inconsistency across households is worth noting.
- **Presence of a child under 6:** Yes, a young child is present. The model treated this as reducing household-level transmission risk.
- **Total living area (64 m²):** Relatively small total floor area also pushed the score down.

Coherence check: This is a paradoxical case. All visible features point toward lower risk (low income, young ages, small dwelling, presence of young child), yet the model still flags the household for transmission. This suggests that **unshown features** — possibly the individual-level characteristics of the index case or non-index members that are not captured in the household aggregates — are driving the score upward. Experts should be cautious: the model’s household-level flag rests on factors not visible in the top-20 list.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 13538493 (age 26) – Flagged (54.9%) – the only flagged individual Key upward drivers:

- **Age 26 (strongest upward):** Young-adult age strongly increased susceptibility.
- **Personal disposable income (286,200 SEK/year – relatively high):** Higher individual income pushed the score up, contrasting with the household’s lower average income.
- **Oldest age in household (pushed up):** The presence of older members increased this young adult’s score.
- **Number of children under 18 in household (2 children, upward):** Having children in the household increased this person’s susceptibility score.

Member 9105013 – Not flagged (26.8%) All top factors pushed downward:

- **Household disposable income (zero value):** A zero-income flag strongly reduced the score (likely data artifact).
- **Personal disposable income (zero):** Reinforced the downward push.
- **Housing type (row house – *FlerBo*, downward):** The dwelling form reduced the score.
- **Crowding index (downward):** Less crowding reduced susceptibility.
- **Homecare alarm sum (downward):** Any homecare presence pushed the score down.

Member 12108918 (age 5) – Not flagged (5.1%) – very low score Strong downward drivers:

- **Age 5 (strongest downward):** Very young age strongly reduced susceptibility.
- **Household disposable income (zero):** Again, a zero-income artifact pushed the score down.
- **Personal disposable income (zero):** Reinforced.
- **Presence of a child under 6 (downward):** Being part of a household with a young child reduced this child’s own score further.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Young children receiving very low scores, and young adults receiving higher scores, are consistent across multiple households. The 26-year-old being the only flagged individual is plausible.

- **Uncertain / surprising patterns:**
 - **Household flagged despite all top features being negative** is the most striking finding. This indicates the model’s decision is driven by features not appearing in the top-20 list — possibly interaction effects, medication history, or specific index case attributes. Experts cannot fully interpret this flag without access to those hidden drivers.
 - **Inconsistent treatment of “absence of elderly”:** In household 3, absence of elderly increased risk; here, absence of elderly decreased risk. This suggests the model’s weighting is context-dependent (e.g., interacts with income or household size).
 - **Zero-income values** appear again for two members, likely data artifacts.
- **Proxy variables:** Income remains a dominant proxy — here, individual income for the 26-year-old was high and pushed risk up, while household average income was low and pushed risk down. This suggests the model distinguishes between individual and household-level economic signals.
- **Additional information needed:** To understand why this household was flagged, experts would need to examine the index case’s individual features (e.g., medication history, healthcare contacts) and any interaction terms the model may be using. The zero-income values should be verified.
- **Uncertain zone:** The household score (37.6%) is moderately above the threshold (25.6%) — not borderline, but not extremely confident either. The individual flagged member (54.9%) is well above the individual threshold (19.2%). The two non-flagged members (26.8% and 5.1%) are below threshold, though 26.8% is close enough to warrant attention (only 7.6 percentage points above the 19.2% threshold? *Correction: 26.8% is above 19.2%, so this person is actually flagged? Let me re-check — the summary table shows 26.8% as “not flagged” but that is above 0.192. There may be a discrepancy in the report or a different threshold for this household. Experts should verify.*)

5. MODEL CONFIDENCE NOTE

The household-level prediction (37.6%) is moderately above the threshold, while the individual scores range from very low (5.1%) to clearly flagged (54.9%). The most important limitation is that the household-level flag appears to be driven by features not shown in the top-20 list, making the model’s reasoning partially opaque in this case. Correctness is unknown. These outputs should be treated as a reference that raises questions — not as a clear signal for action without further investigation into the model’s hidden drivers.

4.6.5 Llama 4

Llama 4 — Sample 6

COVID-19 Household Transmission Model Analysis

Household ID: 14260863

Case Summary

This is a 4-person household—likely a young mother (age 32) with three children including a 26-year-old young adult and a 5-year-old—where the model assigns a 37.6% household transmission probability, above the flagging threshold. At the individual level, the model shows sharp differentiation: the 26-year-old receives a 54.9% susceptibility score (flagged), one member scores 26.8% (just below threshold, not flagged), and the 5-year-old receives a very low 5.1% score (not flagged). This case demonstrates how the model can flag a household for intervention while recognizing that individual risk varies dramatically within it.

Household Model Assessment

The model assigns a 38% household transmission probability—modestly above the 25.6% threshold. Interestingly, every top feature pushes the score *downward*, yet the household is still flagged:

Low family income (138,325 SEK annually): The strongest protective factor, this below-average income would typically reduce transmission scores. At roughly 138,000 kronor per year for a 4-person household, this suggests limited financial resources.

Young index case (age 32): A 32-year-old index case—likely the mother—provides protection in the model’s learned patterns.

Low mean household age (18.75 years): The average age across all members is less than 19 years, indicating this is a predominantly young household. Youth typically reduces scores.

No elderly members (65+): The absence of elderly household members provides a protective effect.

Young child present (under age 6): Having a child under 6 reduces the score, consistent with young children being associated with lower transmission in the training data.

Small total floor area (64 m²): Modest living space, typical of lower-income housing, contributes protection.

Apartment dwelling (FlerBo): Multi-family housing rather than single-family home provides a slight protective effect.

Female index case: The index case being female reduces the transmission score.

Here’s the puzzle: **every** identifiable household feature reduces risk, yet the model still assigns a 38% transmission probability—well above the threshold. This suggests the baseline probability for this household configuration starts relatively high, and while protective factors lower it substantially, they don’t lower it below the flagging threshold. The model appears to be responding to unmeasured aspects of the household structure (perhaps the presence of a young adult alongside young children, creating a mixed-age dynamic) that aren’t explicitly visible in the top feature list.

Individual Susceptibility Profiles

Member 13538493 (Age 26) — 54.9% susceptibility, FLAGGED

The young adult receives by far the highest individual score:

- **Age (26 years):** Working age drives the score upward—the dominant risk factor for this person. The model consistently treats young-to-middle-aged adults as higher-risk than children or elderly.
- **Personal income (286,200 SEK):** Individual economic activity increases the score, potentially reflecting mobility, work contacts, and external exposure patterns.
- **Two children under 18 in household:** Having two children in the household pushes the score upward, possibly reflecting childcare responsibilities or household contact patterns.
- **Predominantly male household (75% male):** The gender composition increases this person’s score.

Protective factors include low age variance in the household and the multigenerational structure being absent. This young adult—possibly a grown child still living at home or a partner—receives a score suggesting the model sees them as having elevated external exposure and contact patterns.

Member 9105013 (Age unknown) — 26.8% susceptibility, NOT FLAGGED

This household member scores just below the 30% threshold:

- **Zero personal and family income:** The strongest protective factors—no independent economic activity means reduced external exposure in the model’s logic.
- **Low household income structure:** Multiple household income metrics pull the score downward.
- **Apartment dwelling:** The housing type provides protection.
- **Young index case (max age 32):** Having a relatively young index case reduces this person’s score.
- **Lower crowding:** More space per person than typical reduces transmission risk.

Every major factor reduces this person’s score, placing them just below the flagging threshold. The borderline nature (27% vs. 30% threshold) suggests this is an uncertain assessment—a small change in features would flip the decision. This might be an older child or teenager without independent income.

Member 12108918 (Age 5) — 5.1% susceptibility, NOT FLAGGED

The youngest child receives an extremely low score, similar to other young children we’ve seen:

- **Young age (5 years):** The dominant protective factor—early childhood age dramatically reduces susceptibility scores throughout this model’s learned patterns.
- **Zero personal and family income:** Children without economic profiles receive strong protection.
- **Presence of young child in household:** The household having a child under 6 reduces this specific child’s score.

- **Young index case:** A 32-year-old mother as index case provides protection for young children.

All factors align to give this 5-year-old one of the lowest possible scores. The model sees virtually no risk for this young child despite household-level flagging.

Expert Considerations

Household-individual disconnect: The household is flagged (38%), but only one of three members is flagged individually (55%). This reflects the model’s dual logic: the household structure creates enough overall risk to warrant attention, but within that household, risk concentrates in the young adult while the children receive protection.

Baseline elevation mystery: The most striking pattern is that all household-level protective factors are present—low income, young members, no elderly, small space, female index case—yet the household is still flagged. This suggests the model learned that certain household compositions (possibly young single parent with adult child and young children) had elevated transmission rates in 2020 for reasons not fully captured by the explicit features. Mixed-age households with both young adults and young children may have had complex contact and exposure dynamics.

Age stratification: The 51-point gap between the young adult (55%) and the 5-year-old (5%) demonstrates how powerfully age drives individual scores. The model has learned very distinct patterns for different age groups.

Threshold clustering: One member sits right at the boundary (27% vs. 30% threshold), indicating borderline confidence for that individual. The young adult (55%) is clearly flagged, the 5-year-old (5%) is clearly not, but the middle member represents genuine model uncertainty.

Income paradox: Low household income provides strong protection in individual assessments (zero income = major protective factor for two members), yet the household is still flagged. This suggests household structure trumps socioeconomic factors at the household level for certain demographic configurations.

Missing context: The model has no information about actual childcare arrangements, school attendance, work-from-home capability, or support networks—factors that could substantially affect real transmission risk for a young single parent household with children spanning 5 to 26 years old.

Model Confidence Note

The model shows mixed confidence: the household score (38%) sits comfortably above its threshold, suggesting confident household-level flagging. Individual scores are more varied—the young adult (55%) is decisively flagged, the 5-year-old (5%) is decisively not flagged, but one member (27%) hovers just below threshold, indicating uncertainty. The counterintuitive pattern where all protective factors are present yet the household is still flagged suggests the model is responding to demographic patterns that aren’t fully transparent in the top feature list. Experts should note that while the model flags this household, most members (2 of 3) are not individually flagged, and all explicit risk factors actually point toward protection—making this a case where model reasoning may warrant additional scrutiny against domain knowledge.

Note: This analysis highlights where the model’s reasoning appears counterintuitive and where confidence levels vary across household and individual assessments.

Evaluation Scores — Sample 6

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}\text{-R@2}$	B6 $\overline{\text{IND}}\text{-R@3}$	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.7 Sample 7

4.7.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 7

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 14392369 | Model Analysis

1. CASE SUMMARY

This household comprises two individuals: one confirmed COVID-19 index case and one exposed member. The model assigned a transmission score of 29% to this household—just below the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model narrowly declines to flag this household as high-risk. However, the single non-index member, a 13-year-old (member 12708868), received a susceptibility score of 53%, well above the individual-level flagging threshold of 30%, creating a discordant assessment where the household is not flagged but the individual within it is. This represents another borderline case with conflicting signals between household and individual model outputs.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 29%, falling just 1 percentage point below the 30% threshold—placing this case at the very edge of the model’s decision boundary. This is one of the narrowest non-flagging margins observed. Nearly every influential feature pushed the score downward, suggesting baseline risk was higher before protective factors were applied.

Key factors driving the model’s household-level assessment:

Moderate household income (mean family income of approximately 142,000 SEK/year) was the most influential factor, substantially reducing the model’s score. This income level falls in the lower-middle range and continues the pattern where lower-to-moderate income acts protectively in the model’s learned associations.

Small household size (2 persons) reduced the score considerably, consistent with the model’s treatment of smaller households as lower transmission risk due to fewer contact opportunities.

High index-to-household ratio (50%—one of two members is the index case) substantially reduced the score, continuing the counterintuitive pattern where having a high proportion of the household already infected acts protectively rather than increasing exposure risk for the remaining member.

Absence of index case homecare service use (zero homecare contacts for the index case) reduced the score, suggesting the model associates index case frailty or care needs with increased transmission likelihood.

High income variance (standard deviation of approximately 201,000 SEK/year across just two members indicates substantial income disparity) reduced the score, as did household age variance (suggesting age diversity acts protectively), apartment dwelling (FlerHy), mean family income in non-transformed form, presence of gender diversity in the household, and absence of

children under 18.

The pattern is partially coherent: small household size aligns with reduced transmission opportunity. However, the protective effects of high index-to-household ratio and moderate-low income remain epidemiologically questionable and likely reflect training data artifacts.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12708868 (Age 13)

The model assigned a susceptibility score of 53%, well above the 30% flagging threshold, creating clear discordance with the household-level assessment. This teenager is flagged as susceptible despite the household not being flagged for transmission. The feature influences show a complex interplay of competing signals.

Absence of personal family income (recorded as zero, as expected for a 13-year-old dependent) was the single strongest factor, driving the score dramatically downward. This protective effect is substantial but apparently insufficient to prevent flagging.

Age (13 years) was the strongest risk-increasing factor, pushing the score substantially higher. This represents a notable departure from the strong protective effects assigned to younger children (ages 3-7) in previous cases. The model appears to weight early adolescents as moderately higher risk compared to young children, though still lower risk than adults.

High index-to-household ratio (50%) elevated the score considerably—in contrast to its protective effect at the household level. This reveals that the two model levels learned different associations for the same feature.

Household male count (number of males in the household) pushed the score higher, as did household age variance (working in the opposite direction from the household level), sum of homecare alarms across the household, log-transformed crowding index, and index case foreign background proportion. Working against flagging were the standard (non-log) crowding index and age variance.

The pattern is paradoxical: the strongest protective feature (zero income) is overwhelmed by age and household composition factors that push toward flagging. The model’s treatment of 13-year-olds as moderately elevated risk—positioned between the strong protection assigned to young children and the elevated risk assigned to young adults—suggests it learned distinct age-band patterns from the training data.

4. EXPERT CONSIDERATIONS

Critical discordance: This case exemplifies the most severe household-individual conflict observed: the household score (29%) sits just below threshold while the individual score (53%) sits well above it. In a two-person household where the sole exposed member is flagged as susceptible, the logical expectation would be household-level transmission flagging. This tension arises because the two models operate independently and learned different feature associations.

Threshold hypersensitivity: The household score of 29% is the closest non-flagging margin encountered—a single percentage point from reversal. This extreme proximity to the decision boundary means trivial variations in feature measurement, encoding, or model randomness could flip the outcome.

Age transition zone: The model’s treatment of this 13-year-old reveals a transition in age-based risk assessment. Unlike young children (receiving strong protection) or young adults (receiving elevated risk), early adolescents appear to occupy middle ground in the model’s learned patterns. This may reflect genuine epidemiological patterns from 2020 data about adolescent transmission, or it may be an artifact of limited training examples in this age range.

Income-ratio paradox: Zero personal income acts as the strongest protective feature for the individual, while moderate household income acts protectively at the household level—yet the high index-to-household ratio (50%) acts protectively at the household level but as a risk factor at the individual level. These opposing directional effects for the same underlying feature suggest the two model components learned contradictory associations.

Information gaps: Experts would need: the relationship between household members (parent-child, siblings with external guardian, other); school attendance patterns for the 13-year-old during the 2020 period; index case symptom severity and isolation feasibility in a two-person household; actual spatial configuration within the 77 m² dwelling; and behavioral factors like meal sharing and bedroom arrangements.

5. MODEL CONFIDENCE NOTE

The household score of 29% sits at the absolute edge of the decision boundary—just 1 percentage point from flagging—while the individual score of 53% sits comfortably above threshold. This creates maximum uncertainty: the household assessment could reverse with the slightest perturbation, yet the individual assessment appears moderately confident. The directional contradiction between the two levels (household says "no transmission," individual says "this person is susceptible") reveals fundamental architectural tension in the dual-model system. For a two-person household, these conflicting outputs provide minimal actionable guidance and should be treated with extreme caution. As always, the model’s actual predictive accuracy remains unknown, and these outputs serve only as limited, uncertain reference points for qualified expert judgment—not clinical decisions.

Sample 7sec:out7

4.7.2 GPT-5.3

GPT-5.3 — Sample 7

1. Case Summary

This case involves a two-person household with one index case and one exposed non-index household member. The household-level model assigned a 29.2% probability of within-household transmission. Although this score was slightly above the stated operational threshold of 25.6%, the report indicates that the household was ultimately not flagged by the model, suggesting this case sits close to the model’s decision boundary and should be interpreted cautiously.

At the individual level, the single non-index household member received a susceptibility score of 52.9%, which is above the model’s individual threshold and therefore triggered a model flag.

The model’s reasoning relied primarily on demographic structure, household composition, and crowding-related variables rather than medical history or healthcare utilisation.

As with all outputs in this framework, these scores reflect the behaviour of a statistical model trained on historical registry patterns. They should not be interpreted as verified infection probabilities or evidence that transmission did or did not occur.

2. Household Model Assessment

The household-level model assigned a transmission score of 29.2%, placing the household near the operational threshold. This suggests the model viewed the household as neither clearly low-risk nor strongly high-risk, but rather as a borderline case.

The strongest factor lowering the model’s score was relatively high household disposable income. The model associated higher family income with lower transmission propensity in this case, possibly reflecting indirect associations with housing quality, reduced crowding, or other socio-economic conditions. Household size also substantially reduced the score. Because the household contains only two people, the model may have interpreted the potential contact network as more limited than in larger households.

Another important downward influence was the index-to-household ratio. With one index case in a two-person household, the model treated this configuration as lowering the transmission score overall. Absence of healthcare utilisation indicators — particularly no recorded homecare service use — also reduced the model’s assessment.

The dwelling type, classified as apartment housing (“FlerHy”), modestly lowered the score in this specific prediction. This is somewhat less intuitive epidemiologically, since apartment living is often associated with greater density, but the effect here was relatively small and may reflect correlations in the underlying training data rather than a direct interpretation of housing type itself.

Overall, the model’s reasoning forms a partly coherent structural pattern: higher income, smaller household size, and lack of frailty-related indicators all contributed toward a lower household transmission score. However, because the final probability remained close to the threshold, the model appears uncertain and sensitive to relatively small shifts in feature weighting.

3. Individual Susceptibility Profiles

Member 12708868 — Model score 52.9% (flagged)

The single non-index household member, a 13-year-old adolescent, received a comparatively high susceptibility score and was flagged by the model.

The strongest upward influence was age. In this case, the model associated early adolescence with increased susceptibility relative to other demographic profiles in the training data. Another major upward influence was the household index-to-household ratio. Because half the household consisted of the index case, the model interpreted the level of exposure concentration as increasing susceptibility for the remaining member.

The model also weighted household male composition as increasing the score, although this effect is harder to interpret mechanistically and may reflect indirect associations learned from

the training data rather than a meaningful causal pattern. Mild crowding-related indicators also pushed the score upward, suggesting the model associated closer shared living conditions with greater susceptibility.

At the same time, several factors strongly reduced the score. The largest downward influence was low or absent recorded family income values. As in many registry-based models, income may function as a proxy for multiple unmeasured social conditions, and zero-income values can sometimes reflect administrative or life-stage circumstances rather than true deprivation. Greater age variation within the household also lowered the score modestly.

Healthcare utilisation and medication history had essentially no meaningful influence on this prediction. The model’s reasoning was therefore dominated by demographic and household-structure factors rather than indicators of underlying health burden.

From a domain perspective, the combination of concentrated household exposure and crowding-related signals is plausible. However, the influence of income-related variables and household gender composition should be treated more cautiously because these may represent indirect statistical associations rather than interpretable epidemiological mechanisms.

4. Expert Considerations

Several elements of the model’s reasoning align with established public health concepts. Household size, exposure concentration, and crowding-related measures are all structurally relevant to respiratory transmission dynamics, suggesting the model may be capturing meaningful social-contact patterns.

At the same time, some influential variables likely function as proxies for unmeasured factors. Income measures may indirectly reflect housing quality, occupational patterns, or access to resources rather than financial status itself. The effect of household gender composition is less clearly interpretable and may represent correlations specific to the training data rather than robust epidemiological signals.

This case would benefit from additional contextual information before any operational use. Relevant missing factors include school attendance, timing and intensity of contact with the index case, ventilation conditions, behavioural isolation practices, and vaccination status — none of which are available in the registry features used by the model.

Both the household-level score and the household classification itself appear close to the operational decision boundary, making this an uncertain or borderline case from the model’s perspective.

5. Model Confidence Note

The household-level prediction sits only slightly above the stated threshold, indicating relatively low certainty in the model’s overall household assessment. The individual-level score for the adolescent household member is more clearly elevated, although it is not an extreme value.

These outputs should be interpreted strictly as model-generated prioritisation signals derived from historical registry patterns. Whether the model’s assessments correspond to actual transmission or susceptibility is unknown, and the results should be treated as a limited reference for expert interpretation rather than as actionable conclusions.

4.7.3 Grok 4.2

Grok 4.2 — Sample 7

CASE SUMMARY

This is a small 2-person household (ID 14392369) with one index case. The household-level model assigns a 29.2% probability of transmission, which is slightly above the empirical threshold but results in no overall transmission flag. At the individual level, the single non-index member — a 13-year-old — receives a 52.9% susceptibility score and is flagged as susceptible. This represents another example of the dual-level model identifying potential individual risk even when aggregate household transmission probability remains borderline.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 29.2%. Although this value is marginally above the 0.256 threshold, the model does not raise a transmission flag for the household.

The strongest factors pulling the score downward include:

- **Low family disposable income** (mean around 142,000 SEK annually), which the model associates with lower transmission potential.
- **Small household size of 2**, which naturally limits opportunities for spread.
- **Index-to-household ratio of 50%**, indicating half the household members are already index cases.
- Absence of TRYGG homecare indicators for the index case, multi-family housing type, and moderate age variation within the home.

The model’s reasoning is largely coherent with epidemiological expectations for small households: limited size and a relatively high proportion of known cases reduce the scope for additional spread. Lower income and housing type likely serve as proxies for broader living conditions or behavioral patterns observed in the 2020 data.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12708868 (age ~13 in 2020): The model assigns a 52.9% susceptibility score and flags this adolescent as susceptible. While low family income pulled the score downward, this was offset by upward influences including the child’s age (13 years), the high index-to-household ratio, male household composition, presence of homecare alarm indicators in the household, and crowding measures.

The model appears to weigh the adolescent’s age and specific household structural factors (such as the index case proportion and care-related variables) as increasing susceptibility, even in a small low-income household. This pattern is plausible from a domain perspective, as adolescents may have distinct contact patterns (e.g., school or social activities) that differ from both younger children and adults, and homecare indicators may proxy for underlying household health or support needs.

EXPERT CONSIDERATIONS

This case highlights the complementary value of the two model levels: the household is not

prioritised overall, yet the model directs attention to the teenage member. Domain-consistent elements include the risk-reducing effects of small household size and the differentiating role of age. The divergence between household and individual outputs is noteworthy and may reflect how the individual model captures person-specific dynamics (such as adolescent age and household care indicators) more sensitively.

Income, housing type, and TRYGG homecare variables likely function as proxies for unmeasured factors like geographic location, family support needs, or social circumstances rather than direct causes. Medication history variables had minimal impact. Public health experts should consider additional details unavailable to the model, including the exact timing of the index case, school attendance, extracurricular activities, isolation measures within the home, and any symptom or testing history for the 13-year-old. The household probability sits near the decision threshold, placing it in a somewhat uncertain zone.

MODEL CONFIDENCE NOTE

The household probability (29.2%) is close to the decision threshold, creating mild boundary uncertainty at the aggregate level. The individual susceptibility score (52.9%) is more clearly above its threshold, giving the model moderate confidence in flagging the adolescent. As with all cases, these are statistical assessments based on patterns in 2020 Swedish registry data — not verified infection risks or clinical predictions. They should be treated as one limited reference to support expert judgment and further investigation.

4.7.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 7

Here is the structured interpretation of the model's output for household 14392369, written for public health policy experts.

1. CASE SUMMARY

This household consists of two people: one designated index case and one other member. The model assigns a **29.2% probability** of within-household COVID-19 transmission, which is just above its internal flagging threshold of 25.6% — yet the summary table indicates the household was **not flagged** (a borderline or possible reporting inconsistency). The single non-index member receives a susceptibility score of **52.9%** and is clearly flagged as susceptible at the individual level. **These are statistical model outputs, not confirmed infections.**

2. HOUSEHOLD MODEL ASSESSMENT

The model's household-level prediction (29.2%) is very close to the threshold (25.6%) — a borderline case. Every top influential feature pushed the score downward:

- **Household disposable income (142,350 SEK/year – low):** The strongest downward driver, consistent with other low-income households.
- **Household size (2 persons):** Very small household size reduced the transmission score.

- **Index-to-household ratio (0.5):** Having one index case in a two-person household pushed the score down — the opposite of what one might expect, but consistent with the previous two-person household (report 5) where the same pattern appeared.
- **No homecare alarms for the index case (`index_mean_TRYGG_total` = 0):** The absence of homecare use reduced the score.
- **Income spread within the household (`sd_DisInk04`):** Variation in individual incomes pushed the score down.
- **Housing type (row house – *FlerHy*):** This dwelling form also reduced the score.

Coherence check: This is another borderline case where all visible features point toward lower risk, yet the final probability (29.2%) still exceeds the threshold (according to the raw output, though the summary says “no transmission flagged” — this inconsistency needs resolution). The pattern is broadly consistent with other small, low-income households, but the discordance between the numerical output and the summary flag is a concern.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 12708868 (age 13) – Flagged (52.9%) – clearly above the individual threshold (19.2%)

Key factors influencing the model’s score:

- **Household disposable income (zero value – strongest downward):** A zero-income value (likely a data artifact) strongly pushed the score down — yet the final score remained high (52.9%), indicating other factors strongly overrode this.
- **Age 13 (strongest upward):** Adolescent age strongly increased susceptibility. This is notable — earlier households showed very young children (ages 3–7) with very low scores, but this 13-year-old receives a high score. The model appears to draw a sharp distinction between young children and adolescents.
- **Index-to-household ratio (upward):** Unlike at the household level where this ratio pushed the score down, at the individual level it pushed this member’s score up — an interesting cross-level difference.
- **Male count in household (upward):** Being male or having more males in the household increased the score.
- **Homecare alarm sum (upward):** The presence of homecare alarms in the household (likely from another member) pushed this adolescent’s score higher.
- **Crowding index (upward):** Higher crowding (more persons per square metre) increased susceptibility.

Plausibility: The model’s assignment of a high susceptibility score to a 13-year-old, while young children (ages 3–7) received very low scores, is a striking pattern. This may reflect registry data patterns where adolescents are more often recorded as secondary cases than young children — possibly due to differences in testing, symptoms, or social mixing. Experts should note this age inflection point as a specific model behaviour.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Small household size and low income pointing toward lower transmission risk are consistent across examples. The high susceptibility score for an adolescent (vs. very low for young children) is a notable and potentially plausible pattern that merits further investigation.
- **Uncertain / surprising patterns:**
 - **Household-level inconsistency:** The model output shows $P=0.2917$ (>0.256) but the summary says “no transmission flagged.” This is either a rounding issue (29.17% vs 25.6% — still above) or a reporting error. Experts should clarify which is correct.
 - **Index-to-household ratio had opposite effects** at household level (downward) and individual level (upward) — a reminder that the same feature can influence the two model levels differently.
 - **Zero-income value** appears again for family income, likely a data artifact. Despite this strong downward push, the individual score remained high (52.9%), showing how other factors (age, crowding, homecare) can dominate.
- **Proxy variables:** Income again acts as a proxy — here, zero income likely indicates missing data rather than true economic status. Homecare alarms (TRYGG) appear as an upward driver at the individual level, possibly proxying for household frailty or healthcare contact intensity.
- **Additional information needed:** Experts would need to resolve the household flag inconsistency, verify whether the zero-income value is genuine or imputed, and understand why the model treats adolescents so differently from younger children.
- **Uncertain zone:** The household score (29.2%) is very close to the threshold — a true uncertain zone. The individual score (52.9%) is comfortably above its threshold. The household-level prediction could easily flip with small changes in input features or threshold calibration.

5. MODEL CONFIDENCE NOTE

The household-level prediction (29.2%) is extremely close to the empirical threshold (25.6%), placing this case in a low-confidence uncertain zone for the household model — and the inconsistency between the numerical output and the summary flag further reduces confidence. In contrast, the individual-level prediction (52.9%) is more clearly above its threshold. Correctness is unknown for both levels. These outputs should be treated as a borderline reference: the model is essentially uncertain about the household, while clearly flagging the adolescent individual based on age, crowding, and homecare signals.

4.7.5 Llama 4

Llama 4 — Sample 7

COVID-19 Household Transmission Model Analysis

Household ID: 14392369

Case Summary

This is a 2-person household—likely a parent and 13-year-old child—where the model assigns a 29.2% household transmission probability. This sits just 3.6 percentage points above the 25.6% flagging threshold, making it a marginal case. The report indicates "no transmission flagged" despite the score exceeding the threshold, which may reflect either a reporting inconsistency or a different decision rule at play. At the individual level, the story is clearer: the 13-year-old receives a 52.9% susceptibility score, decisively above the 30% individual threshold. This creates another household-individual divergence where household assessment suggests borderline risk while individual assessment clearly flags the exposed teenager.

Household Model Assessment

The model assigns a 29% household transmission probability—barely above the 25.6% threshold. All major factors push *downward*:

Low family income (142,350 SEK annually): The strongest protective factor, this below-median income provides substantial risk reduction. At roughly 142,000 kronor per year for a two-person household, this suggests limited economic resources.

Small household size (2 people): Two-person households minimize transmission pathways—with only one person exposed after the index case, the mathematical opportunity for "additional" transmission is inherently limited.

High index-to-household ratio (50%): Half the household is already infected, which reduces the score. This reflects the model's specific task: predicting whether *additional* members will be flagged when 50% are already index cases.

No homecare utilization: The index case has zero homecare service contacts, indicating no documented frailty or care dependency.

Income variance: High standard deviation in individual income (different income levels between household members) provides protection, possibly reflecting economic diversity or different life stages.

Age variance: Substantial age spread between members reduces transmission risk in the model's logic, potentially reflecting different activity spheres and contact patterns across generations.

Apartment dwelling (FlerHy): Multi-family housing provides a protective effect, consistent with lower-income household profiles.

All-female household (proportion male = 0.0): Both household members are female, which the model treats as protective.

The pattern is coherent: small, lower-income, two-generation female household with high age diversity. Every structural factor points toward lower risk. The 29.2% score barely exceeds the threshold—a difference of just 3-4 percentage points—making this one of the most borderline

household-level decisions possible.

Individual Susceptibility Profiles

Member 12708868 (Age 13) — 52.9% susceptibility, FLAGGED

This teenager receives a surprisingly high individual score despite household-level protective factors:

- **Zero family income** (reported at individual level): The strongest protective factor, pulling the score substantially downward. As a 13-year-old without independent income, this would normally provide major protection.
- **Age (13 years)**: Teenage age *increases* the score significantly—a notable departure from the strong protection we’ve seen for younger children (ages 3-7). At 13, this person sits in early adolescence where the model begins treating them more like adults than children. This is the first teenager we’ve seen in these cases, and the model clearly differentiates teenage risk from childhood risk.
- **50% index-to-household ratio**: Having half the household already infected increases this teenager’s score—high exposure concentration.
- **Male household count**: Despite the household being all-female, the presence of males in the household composition scoring increases risk (this may be a data artifact or proxy for other factors).
- **Homecare service sum**: Some household homecare utilization (possibly for the parent) increases the teenager’s score, potentially indicating caregiver contact or household vulnerability.
- **Crowding indicators**: Various crowding metrics show mixed effects, with log-crowding increasing risk while standard crowding decreases it.

The striking pattern: despite zero income providing massive protection (-0.109 SHAP value), teenage age (+0.071) and household exposure factors (+0.044 for index ratio, +0.039 for male count) overwhelm it, yielding a 53% susceptibility score—well above the threshold.

Expert Considerations

Household-individual disconnect again: The household score (29%) barely exceeds its threshold and the report indicates "no flag," while the individual score (53%) decisively flags the teenager. This inconsistency mirrors the earlier case where two model levels reached different conclusions.

Teenage risk elevation: The 13-year-old’s high score (53%) despite zero income protection reveals that the model learned teenage age carries elevated transmission risk compared to younger children. Whether this reflects biological susceptibility, behavioral patterns, school exposure, testing practices, or social activity in 2020 Sweden isn’t directly observable. This is a crucial finding—the model’s protective stance toward young children (ages 3-7) does not extend to teenagers.

Threshold hypersensitivity: The household score sits just 3-4 percentage points above its threshold. Small perturbations in any feature—slightly different income, slightly older teenager, different housing—would likely flip the household decision. Combined with the "no flag" indication despite numerical threshold crossing, this suggests extreme model uncertainty.

All-female household pattern: Both members are female, which typically provides protection in the model's logic, yet the teenager is still flagged. This suggests age and exposure factors override gender effects for this demographic.

Income paradox redux: Zero income provides the strongest individual-level protection, yet isn't strong enough to bring the teenager below the flagging threshold. This parallels the earlier borderline case (report 5) where all protective factors were present but the person was still flagged.

Single-parent dynamics: A two-person household with one parent and one teenager likely represents different isolation and supervision dynamics than nuclear families with multiple children. The model may have learned specific patterns about these household configurations that aren't fully captured by explicit features.

Missing behavioral context: The model has no information about school attendance (especially relevant for 2020 with remote learning), peer socialization patterns, after-school activities, or parental supervision—all crucial factors for teenage transmission risk.

Model Confidence Note

This case epitomizes model uncertainty. The household score (29.2%) sits barely above its threshold, the report indicates "no flag" despite numerical threshold crossing, and the individual score (52.9%) contradicts the household assessment by decisively flagging the teenager. When scores cluster this close to thresholds (3-4 percentage points) and when household and individual models disagree, these outputs represent statistical noise rather than confident predictions. The model is essentially saying "I can't tell"—which is honest but provides minimal actionable guidance. Experts should treat this as a high-uncertainty case requiring substantial human judgment rather than model-driven decision-making. The teenager's score suggests elevated risk, but the household-level ambiguity indicates this isn't a clear-cut scenario.

Note: This analysis highlights a borderline case where model confidence is extremely low and household/individual assessments conflict. These scores reflect statistical uncertainty at decision boundaries, not validated infection predictions.

Evaluation Scores — Sample 7

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.8 Sample 8

4.8.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 8

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 18583221 | Model Analysis

1. CASE SUMMARY

This household comprises three individuals, including one confirmed COVID-19 index case and two exposed members. The model assigned a transmission score of 21% to this household—well below the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model does not consider within-household transmission likely. Both non-index members were not flagged: a 6-year-old child (member 10795703, score 15%) and a 1-year-old infant (member 12472756, score 5%). This represents one of the model’s strongest protective assessments, with all scores sitting comfortably below threshold, driven primarily by the presence of very young children and low household income.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 21%, sitting 9 percentage points below the 30% threshold, indicating a clear non-flagging decision. Every influential feature pushed the score downward, creating a uniformly protective profile.

Key factors driving the model’s low household-level score:

Very low household income (mean family income of approximately 51,000 SEK/year, among the lowest observed across all cases) was overwhelmingly the most influential factor, driving the score dramatically downward. This continues the persistent pattern where lower income acts protectively in the model’s learned associations.

Young index case age (both maximum and minimum of 28 years, indicating a single young adult index case) substantially reduced the score, consistent with the model’s treatment of younger index cases as lower transmission risk.

Absence of elderly members (no member aged 75 or older) and **presence of young children** (child under 6 in the household) both contributed strong protective effects. With two children under 6 in this three-person household, the young age composition dominates the household profile.

High proportion of children (67% of household members are under 18) further reduced the score, as did index case living area per person (29.3 m²), high proportion born in Sweden, mean family income (in non-transformed form), and individual living area per person (27.7 m²).

The pattern is relatively coherent: presence of very young children, absence of vulnerable elderly, young index case, and small household size all align with factors that could plausibly reduce transmission risk. The dominant protective role of very low income remains the primary interpretive challenge, likely reflecting training data artifacts rather than causal mechanisms. This household represents the opposite extreme from the high-income, high-flagging cases

observed earlier.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 10795703 (Age 6)

The model assigned a susceptibility score of 15%, well below the flagging threshold, and did not flag this child. Age was overwhelmingly the most influential factor—the model assigns very strong protective weight to young children, with this 6-year-old receiving massive protective influence. Absence of personal income (both family and disposable income recorded as zero) further drove the score substantially downward. Index case maximum age (young at 28 years), presence of a child under 6 (this child or their younger sibling), mean household family income (very low), absence of personal disposable income, household age range, absence of respiratory medication history (bronchodilators), proportion of males in household, and absence of homecare alarms all contributed additional protective effects. Every influential feature pushed the score lower, with no risk-increasing factors visible. The model expresses high confidence in not flagging this child.

Member 12472756 (Age 1)

The model assigned a susceptibility score of 5%, far below the flagging threshold, representing one of the lowest individual scores observed across all cases. This infant receives the model’s strongest protective assessment. Age was again the dominant factor—at just 1 year old, this is the youngest household member encountered, receiving even stronger protective weighting than the 6-year-old. Absence of personal income (both family and disposable income at zero) drove the score dramatically downward, nearly as influential as age itself. Mean household family income (very low, in both log-transformed and standard form), household gender diversity, number of index cases (just one), number of children aged 0-17 in the household (two), income range across household members, and absence of foreign background across all members all contributed additional protective effects. Every influential feature pushed the score lower. The model expresses maximum confidence in not flagging this infant—this represents the strongest protective assessment observed across all individual cases.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s extremely strong protective weighting of very young age—particularly for the 1-year-old infant—aligns with documented early-pandemic patterns showing lower documented transmission among young children. The absence of elderly household members eliminates the most vulnerable demographic group. The young index case (28 years) combined with a household dominated by young children (67% under 18) creates a demographic profile the model associates with low transmission risk.

Income paradox at extreme: This household has the lowest income level observed (approximately 51,000 SEK/year mean family income, roughly one-twentieth the highest-income case). The model treats this as overwhelmingly protective, reinforcing the persistent income-as-risk gradient that appears throughout the model’s learned patterns. This likely reflects systematic differences in 2020 Swedish training data—perhaps related to testing access, occupational

exposure patterns, or household reporting structures—rather than genuine causal mechanisms.

Age composition dominance: With a 28-year-old index case, a 6-year-old, and a 1-year-old, this household’s average age is very young. The model appears to have learned that households with this demographic structure—young adult parent(s) with very young children—show systematically lower transmission patterns in the training data, possibly reflecting behavioral factors (isolation precautions with infants present), reduced external contacts (childcare closures in 2020), or selection effects in testing and reporting.

Proxy features: Very low income may proxy for single-parent households, reduced mobility, smaller social networks, or other unmeasured factors specific to the 2020 Swedish context. The absence of medication histories indicates generally healthy household members without chronic conditions requiring pharmaceutical management.

Information gaps: Experts would benefit from knowing: the exact household relationship structure (single parent with two children, two parents with one child, other arrangement); childcare or preschool attendance patterns during 2020; whether the 28-year-old index case could effectively isolate from a 1-year-old infant in practical terms; actual living arrangement within the 83 m² space; and timing of index case confirmation relative to infant exposure.

Confidence across all assessments: All three scores (household 21%, 6-year-old 15%, 1-year-old 5%) sit well below the 30% threshold, with no borderline cases. The model expresses uniform confidence in protective assessment.

5. MODEL CONFIDENCE NOTE

The household score of 21% sits comfortably below the 30% threshold, and both individual scores (15% and 5%) are even further from the flagging boundary, indicating the model is expressing strong confidence in protective assessment across all levels. The 1-year-old’s score of 5% represents one of the lowest individual scores observed, approaching the floor of the model’s output range. However, the model’s reliance on very low income as the dominant protective factor—contradicting typical health gradients where lower socioeconomic status correlates with worse outcomes—introduces interpretive uncertainty about whether these patterns would generalize beyond the specific 2020 Swedish training population. These outputs provide a reference point suggesting low transmission risk according to the model’s learned patterns, but as always, actual predictive accuracy remains unknown and these should serve only as limited inputs to qualified expert judgment, not standalone clinical decisions.

Sample 8sec:out8

4.8.2 GPT-5.3

GPT-5.3 — Sample 8

1. Case Summary

This case involves a three-person household with one identified index case and two non-index household members, both of whom are young children. The household-level model assigned a

20.6% probability of within-household transmission, which falls below the operational threshold of 25.6%, so the household was not flagged by the model.

At the individual level, both non-index members also received relatively low susceptibility scores: 15.2% for a 6-year-old child and 4.8% for a 1-year-old child. Neither child crossed the model's individual threshold for flagging. Across both household- and individual-level assessments, the model consistently weighted young age and several socioeconomic indicators toward lower model-assigned risk.

These outputs describe how the statistical model interpreted this household profile based on registry-derived patterns. They do not indicate whether transmission actually occurred or whether the model's assessment was correct.

2. Household Model Assessment

The household-level model assigned a transmission score of 20.6%, below the threshold used for model flagging. Compared with previous examples, this represents a relatively low model-assigned transmission propensity.

The strongest factor lowering the score was household disposable income. The model associated this household's income profile with reduced transmission likelihood. As in many registry-based models, income is likely acting as a proxy for broader structural conditions such as housing quality, household stability, or differences in occupational exposure rather than income itself being biologically meaningful.

Several demographic features also lowered the model's assessment. The index case was approximately 28 years old, and the model associated this relatively young adult age profile with lower transmission propensity. The household included two children, including one under six years old, and these features also pushed the score downward. In this case, the model appears to interpret a child-heavy household composition as comparatively lower risk.

Housing-related variables modestly reduced the score as well. The household had approximately 27.7 square metres of living area per person and a total dwelling area of around 83 square metres, both of which the model associated with lower transmission propensity. The absence of elderly members and the absence of healthcare dependency indicators also contributed modestly toward a lower score.

Overall, the model's reasoning forms a fairly coherent structural pattern. Lower crowding, absence of elderly or frail members, and relatively young household composition all align with factors that could plausibly influence transmission opportunities or observed case patterns in registry data. However, some associations — particularly the strong protective effect assigned to income-related variables — should still be interpreted cautiously because they may reflect indirect correlations rather than direct mechanisms.

3. Individual Susceptibility Profiles

Member 10795703 — Model score 15.2% (not flagged)

This 6-year-old child received a relatively low susceptibility score and was not flagged by the model.

The strongest downward influence was age. The model strongly associated early childhood with lower susceptibility in this case. Household income-related variables also substantially reduced the score, including both family income and individual disposable income measures. Again, these variables likely function as indirect proxies for broader social or household conditions. The presence of another young child in the household and the relatively young age of the index case also lowered the model's assessment. A mild respiratory medication signal (bronchodilator use) contributed slightly downward as well, though this effect was small. Overall, the model's reasoning appears internally consistent: young age, relatively stable housing conditions, and absence of strong medical vulnerability indicators all contributed toward a lower model-assigned susceptibility profile.

Member 12472756 — Model score 4.8% (not flagged)

This 1-year-old child received the lowest score in the household and was very far below the model threshold.

As with the older child, age was the dominant factor. The model strongly associated infancy with reduced susceptibility in this household context. Income-related variables again substantially lowered the score, including both household-level and individual-level income measures.

Several household composition variables also contributed modestly downward, including gender diversity within the household, the number of children in the household, and the number of index cases. None of the healthcare utilisation or medication variables meaningfully affected the prediction.

From a domain perspective, the model appears to be consistently treating very young children in relatively stable, non-crowded households as comparatively low priority for secondary susceptibility. Whether that pattern reflects meaningful epidemiological structure or artefacts of the underlying training data would require further validation.

4. Expert Considerations

The model's reasoning in this case is relatively consistent across both levels of analysis. Young household age structure, absence of elderly members, lack of healthcare dependency indicators, and moderate living space per person all contributed toward lower scores. These patterns broadly align with plausible structural determinants of household transmission opportunity.

At the same time, the dominance of income-related variables deserves careful interpretation. Disposable income may be functioning as a proxy for multiple unmeasured factors such as housing density, employment type, access to remote work, or broader social stability. The model cannot distinguish between these mechanisms directly.

The strong downward influence of very young age is also notable. This may reflect observed reporting patterns, testing practices, or exposure structures in the 2020 registry data rather than a direct biological interpretation. Experts would likely want additional contextual information — such as childcare attendance, behavioural exposure patterns, household isolation practices, and timing of exposure — before drawing operational conclusions.

None of the household or individual scores sit especially close to the decision thresholds, suggesting relatively consistent model behaviour in this case.

5. Model Confidence Note

Both the household-level and individual-level scores fall clearly below their respective operational thresholds, particularly for the 1-year-old child. This suggests the model’s internal assessment was relatively stable toward the lower end of its scoring range for this household configuration. However, the correctness of these predictions is unknown. These outputs should be treated as statistical model behaviour derived from historical registry patterns rather than as verified evidence about true transmission dynamics or individual susceptibility.

4.8.3 Grok 4.2

Grok 4.2 — Sample 8

CASE SUMMARY

This is a small 3-person household (ID 18583221) with one index case and two young children. The model assigns a low 20.6% probability of household transmission and does not flag the household. At the individual level, both non-index members receive very low susceptibility scores (15.2% for the 6-year-old and 4.8% for the 1-year-old) and are not flagged. These outputs reflect the model’s assessment of this young, low-income household profile based on patterns in 2020 Swedish registry data.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 20.6%, below the empirical decision threshold of 0.256, resulting in no transmission flag.

The strongest factors pushing the model’s score downward include:

- **Very low family disposable income** (mean around 51,300 SEK annually), which the model strongly associates with reduced transmission potential.
- **Young index case age** (28 years).
- Presence of a child under 6, high proportion of children in the household (two-thirds), and absence of elderly members (no one aged 75+).

The model’s reasoning forms a coherent and epidemiologically plausible pattern. Small households with young children and a young adult index case generally present fewer opportunities for sustained secondary transmission. Lower income and the specific age composition (young family) likely serve as proxies for living conditions, mobility patterns, or social circumstances observed in the training data, all of which the model associates with lower risk in this context.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 10795703 (age ~6 in 2020): The model assigns a low 15.2% susceptibility score and does not flag this child. The dominant downward drivers are young age and zero personal/family income (typical for a child), reinforced by household income measures, presence of a young child in the home, and index case characteristics. The model strongly down-weights this young child profile, which aligns with domain expectations regarding age-related contact and exposure patterns in registry data.

Member 12472756 (age ~1 in 2020): The model assigns an even lower 4.8% susceptibility score and does not flag this toddler. The strongest factors pulling the score down are the very young age and zero income indicators, along with household income averages and gender diversity measures. The model consistently assigns the lowest scores to the youngest children in this type of household.

EXPERT CONSIDERATIONS

The model does not prioritise this household for intervention and shows clear differentiation toward low risk across both levels. Domain-consistent patterns include the strong protective effects (in model terms) of small household size, very young children, and young adult index case. Low family income emerges as the most influential downward factor — likely acting as a proxy for specific residential, occupational, or geographic circumstances rather than income itself being protective.

Housing and demographic variables (proportion of children, birth country indicators) further support the low-risk profile. Medication and homecare variables had minimal influence. Experts should consider contextual information unavailable to the model, such as exact timing of the index case, caregiving arrangements, any isolation or hygiene practices in the home, and actual testing or symptom data for the children. All scores are well below their respective thresholds, with no members in uncertain boundary zones.

MODEL CONFIDENCE NOTE

Both the household transmission probability (20.6%) and the individual susceptibility scores (15.2% and 4.8%) are substantially below their decision thresholds, indicating the model is quite decisive in assigning low risk to this young family household. These outputs represent statistical patterns learned from 2020 Swedish registry data and are not verified infection risks or clinical predictions. They provide a structured reference to support, but never replace, expert public health judgment and on-the-ground assessment.

4.8.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 8

Here is the structured interpretation of the model's output for household 18583221, written for public health policy experts.

1. CASE SUMMARY

This household contains three people: one designated index case and two other members. The model assigns a **20.6% probability** of within-household COVID-19 transmission, which is below its internal flagging threshold of 25.6%. Neither of the two non-index members is flagged as susceptible, with scores of 15.2% and 4.8%. **These are statistical model outputs, not confirmed absence of infection.**

2. HOUSEHOLD MODEL ASSESSMENT

The model did **not** flag this household for transmission. Every top influential feature pushed

the score downward, and the final probability (20.6%) is well below the threshold. The strongest downward drivers were:

- **Household disposable income (51,300 SEK/year – very low):** This was by far the strongest factor. This is the lowest income seen across all example households, and the model treated it as a very strong downward signal.
- **Age of index case (28 years):** The index case is young, which reduced the transmission score.
- **No member aged 75+ (has__member__75plus = false):** The absence of a very elderly person reduced risk.
- **Presence of a child under 6:** A young child is present, which pushed the score down.
- **High proportion of children (2 out of 3 members, 66.7%):** A household dominated by children reduced the transmission estimate.
- **Index case’s living area per person (29.3 m² – relatively spacious):** More space per person for the index case reduced the score.

Coherence check: This is the most internally consistent low-risk household among the examples. Very low income, young index case, presence of young children, and no elderly members all point in the same direction. The pattern aligns with the intuitive idea that households with very young children and very low income (which may correlate with lower mobility or different testing patterns) are assessed as lower risk by the model. Experts may find this pattern plausible as a statistical association, even if causation is unclear.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 10795703 (age 6) – Not flagged (15.2%) All top factors pushed the score downward:

- **Age 6 (strongest downward):** Young childhood strongly reduced susceptibility — consistent with earlier households (ages 3, 5, 6, 7 all had low scores).
- **Household disposable income (zero value, downward):** A zero-income artifact pushed the score down further.
- **Oldest age in household (index case age 28, downward):** The presence of a relatively young oldest member reduced this child’s score.
- **Presence of a child under 6 (downward):** Being in a household with a younger child (the 1-year-old) reduced this 6-year-old’s score.

Member 12472756 (age 1) – Not flagged (4.8%) – lowest score in all examples
Overwhelmingly downward drivers:

- **Age 1 (strongest downward):** The model assigns the lowest susceptibility to the youngest children. Age 1 produced an even stronger negative influence than age 3 or 5.
- **Household disposable income (zero, downward):** Again, a data artifact.
- **Personal disposable income (zero, downward):** Reinforced.

- **Household mean income (zero/low, downward):** Consistent downward signal.

Plausibility: The model’s clear gradient — the younger the child, the lower the susceptibility score — is a striking and consistent pattern across all households with young children. This likely reflects patterns in the Swedish registry data (e.g., young children being tested less frequently, or being less often recorded as secondary cases) rather than a known biological gradient in infection risk. Experts should interpret this as a model behaviour, not a clinical fact.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Very low income, young index case, and households dominated by young children all pointing toward lower model-assigned risk form a coherent pattern. The age gradient (1-year-old lowest, then 5, then 6, then 13 much higher) is remarkably consistent across households.
- **Uncertain / surprising patterns:**
 - **Zero-income values** appear for family income again. Given that this household has three members, a zero-income value is almost certainly a data artifact (e.g., missing or imputed data). The model treats these zeros as real, which may artificially deflate scores.
 - **No positive drivers anywhere** in this household — all top features at both levels are negative. This is the cleanest example of a “low-risk according to the model” household.
- **Proxy variables:** Income is again the dominant proxy — here, extremely low income is associated with very low model risk. Homecare alarms (TRYGG) are absent and did not appear. Living area per person (spaciousness) pushed risk down in this household, unlike some earlier examples where more space increased risk — highlighting that the model’s treatment of space depends on other household characteristics (income, age composition, etc.).
- **Additional information needed:** Experts would need to verify whether the zero-income values are genuine (extremely unlikely for a household of three) or imputation artifacts. They would also benefit from understanding whether the model’s very low scores for young children align with any external validation data on actual secondary attack rates by age in Swedish households.
- **Uncertain zone:** Not applicable. The household score (20.6%) is comfortably below the threshold, and the individual scores (15.2% and 4.8%) are well below their thresholds. The model is statistically confident in its low-risk classification.

5. MODEL CONFIDENCE NOTE

The model’s probability estimates for this household (20.6% for transmission, individual scores of 15.2% and 4.8%) are well below the empirical thresholds, indicating high statistical confidence in

its low-risk classification. However, the pervasive zero-income values raise data quality concerns that could affect the reliability of these estimates. Correctness is unknown. These outputs should be treated as a limited reference — the consistent low-risk pattern is notable, but experts should prioritise verifying the income data before placing any weight on the model’s assessment.

4.8.5 Llama 4

Llama 4 — Sample 8

COVID-19 Household Transmission Model Analysis

Household ID: 18583221

Case Summary

This is a 3-person household—likely a young mother (age 28) with two very young children (ages 1 and 6)—where the model assigns just a 20.6% household transmission probability, comfortably below the 25.6% flagging threshold. Both children receive extremely low individual susceptibility scores: 15.2% for the 6-year-old and 4.8% for the 1-year-old, both well below the individual threshold. This case represents the model’s clearest low-risk profile: small household, very low income, young single parent with young children, and no elderly members. Every major factor aligns toward minimal transmission risk in the model’s learned patterns.

Household Model Assessment

The model assigns a 21% household transmission probability—clearly below the threshold. Every top feature pushes the score *downward*:

Extremely low family income (51,333 SEK annually): This is the dominant protective factor and represents one of the lowest household incomes we’ve seen. At roughly 51,000 kronor per year for three people (about 17,000 SEK per person), this household sits well below poverty thresholds and likely qualifies for significant social support. The model learned that such low-income households had substantially reduced observed transmission rates in 2020 Swedish data.

Young index case (age 28): A 28-year-old index case—likely a young single mother—provides strong protection. Young adult parents with very young children appear to have had lower transmission rates in the training data.

No elderly members (75+): The complete absence of elderly household members reduces the score, reinforcing that this is a young household unit.

Young child present (under 6): Having children under 6 provides substantial protection, consistent with the strong pattern we’ve seen throughout these cases where young children are associated with lower transmission.

High proportion of children (66.7%): Two-thirds of the household are children under 18, which the model treats as strongly protective. Young family units with predominantly child composition received lower transmission scores in the training patterns.

Small living area (index case: 29.3 m², mean per person: 27.7 m²): Modest living

space, typical of low-income housing, contributes additional protection. Smaller homes may correlate with reduced external mobility or different social patterns.

Swedish-born household: The proportion of household members born in Sweden provides a slight protective effect in this context.

The pattern is exceptionally coherent and represents the inverse of high-risk households: very low income, very young parent, very young children, small space, no elderly, high child concentration. Every structural factor points unambiguously toward low transmission risk based on the model’s learned associations from 2020 Swedish data.

Individual Susceptibility Profiles

Member 10795703 (Age 6) — 15.2% susceptibility, NOT FLAGGED

The older child receives a low score with all factors pointing downward:

- **Young age (6 years):** The dominant protective factor—early childhood age dramatically reduces susceptibility in the model’s learned patterns. Consistent across all cases examined.
- **Zero family income:** Children without economic profiles receive strong protection, reflecting reduced external exposure or activity patterns associated with economic dependency.
- **Young index case (max age 28):** Having a young parent as the index case reduces this child’s score, possibly reflecting different household dynamics or isolation capabilities in young single-parent families.
- **Young child present (under 6):** The household having children under 6 (including this child themselves and their 1-year-old sibling) provides additional protection.
- **Low household income structure:** Multiple income metrics consistently pull the score downward.

All factors align to give this 6-year-old a 15% score—low but not the absolute minimum, possibly reflecting that age 6 sits at the edge of school age where external contacts begin.

Member 12472756 (Age 1) — 4.8% susceptibility, NOT FLAGGED

The toddler receives one of the lowest scores possible:

- **Very young age (1 year):** Infancy/toddlerhood provides the strongest possible protection—the model learned that children this young had minimal observed transmission in the training data, whether due to biology, reduced testing, symptom presentation, or limited external contacts.
- **Zero family income:** Massive protective effect, even stronger than for the 6-year-old.
- **Zero personal income:** Complete economic dependency reinforces the protective pattern.
- **Extremely low household income:** Multiple household income metrics compound the protection.
- **Gender diversity in household:** Having both males and females provides a slight protective effect.

- **High child count:** Having two children under 18 in a three-person household reduces this toddler’s score.

Every single factor drives this 1-year-old’s score downward. The 5% score represents the model’s lowest-risk profile for this demographic: infant in a very low-income, young single-parent household.

Expert Considerations

Perfect low-risk alignment: Unlike several cases where protective and risk factors mixed, this household shows unanimous protection signals. Young mother, very young children, very low income, small household, no elderly—every measured factor points toward low transmission risk.

Income as dominant protector: The 51,333 SEK household income is the strongest driver, suggesting that at extremely low income levels, the associated lifestyle constraints (limited mobility, reduced social activity, constrained external contacts) create substantial transmission barriers in the model’s learned patterns.

Age stratification clarity: The age gradient is crystal clear: 1-year-old (5%) < 6-year-old (15%) < 28-year-old mother (index case). The model has learned very distinct risk profiles across ages, with infancy providing maximum protection.

Young single-parent pattern: This household profile—young mother with multiple very young children, very low income—appears to be a prototypical low-risk scenario in the model’s training data. Whether this reflects 2020-specific factors (lockdown policies, school closures, reduced childcare) or more generalizable patterns isn’t clear.

No conflicting signals: Unlike borderline cases where household and individual models disagreed, or where protective factors couldn’t overcome baseline risk, this case shows complete model consensus: low risk at household level (21%), low risk for both children individually (15% and 5%).

Missing socioeconomic context: The model has no information about social support systems, government assistance programs, childcare arrangements, or actual isolation capabilities for a very low-income single parent—factors that could substantially affect real transmission dynamics.

2020 temporal specificity: This low-risk profile may be highly specific to 2020 Sweden (pandemic lockdowns, school closures, economic support programs, healthcare access patterns) and might not generalize to other time periods or contexts.

Model Confidence Note

The model expresses high confidence in the low-risk assessment: the household score (21%) sits well below the 25.6% threshold, and both individual scores (15% and 5%) are decisively below the 30% individual threshold. The unanimous alignment of protective factors—very low income, young mother, very young children, small household—creates the clearest possible low-risk signal in the model’s learned patterns. However, as with all model outputs, this represents statistical associations from 2020 Swedish registry data, not a guarantee that transmission didn’t occur. A low score means the model sees this household profile as similar to patterns that had low transmission rates in the training data—it doesn’t provide certainty about this specific case’s outcome. These outputs offer one reference point for expert assessment, particularly valuable

for resource prioritization where identifying genuinely low-risk households can help focus limited public health capacity on higher-risk scenarios.

Note: This analysis explains why the model assigned a confident low-risk assessment based on patterns in 2020 training data, not actual infection outcomes or contextual factors specific to this household’s circumstances.

Evaluation Scores — Sample 8

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.9 Sample 9

4.9.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 9

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 22954508 | Model Analysis

1. CASE SUMMARY

This household comprises four individuals, including one confirmed COVID-19 index case and three exposed members. The model assigned a transmission score of 66% to this household—well above the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model considers within-household transmission likely. All three non-index members were flagged as susceptible: a 19-year-old (member 9806355, score 58%), a 51-year-old (member 420689, score 72%), and a 44-year-old (member 265342, score 63%). This represents uniform flagging across all exposed members, driven primarily by moderate-to-high household income, presence of a teenage index case, homecare service use, and middle-aged adult household composition.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 66%, substantially exceeding the 30% threshold, indicating the model considers this household at elevated risk for within-household transmission. Unlike protective-profile cases, this household shows predominantly risk-increasing features.

Key factors driving the model’s household-level score:

Young teenage index case (minimum index case age of 16 years) was the most influential factor, pushing the score higher. This contrasts with the protective effects assigned to young adult index cases (ages 26-32) in other cases, suggesting the model treats teenage index cases distinctly—possibly learning that adolescent index cases in 2020 Swedish data showed different transmission patterns.

Low proportion of children (25% of household members are under 18—just one child among four people) elevated the score, inverting the protective effect seen when children dominate household composition. This suggests adult-heavy households receive higher risk assessment.

Moderate-high household income (mean family income of approximately 543,000 SEK/year) pushed the score higher, continuing the consistent but counterintuitive pattern where higher income correlates with increased model-assigned risk.

Small house/attached dwelling (Småhus Ag—small house or townhouse, rather than apartment) elevated the score modestly, as did total floor area (140 m², or 35 m² per person—spacious but apparently not protective), sum of homecare alarms across household (24 total, indicating significant care needs), median family income (approximately 610,000 SEK/year), presence of elderly members (65+), index case foreign background, and index case income.

The pattern presents mixed epidemiological coherence: presence of elderly members and substantial homecare alarm usage align with genuine vulnerability indicators. However, the risk-increasing effects of moderate-high income and spacious living conditions contradict typical protective expectations. The teenage index case as a primary risk driver is notable and may reflect age-specific transmission patterns in the training data.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 9806355 (Age 19)

The model assigned a susceptibility score of 58%, well above the flagging threshold. High family income (approximately 804,000 SEK/year, the highest in the household) was the strongest driver elevating the score, reinforcing the income-as-risk pattern. Index case maximum age worked against flagging—the only protective feature visible—reducing the score substantially. Age (19 years) elevated the score moderately, as did personal disposable income (approximately 51,000 SEK/year), the index case being a child/teenager, minimum household family income, absence of homecare alarms (paradoxically scored as risk-increasing with value 0), NSAID medication history, and index case minimum age. The model shows moderate confidence in flagging this young adult, with competing signals but net elevation above threshold.

Member 420689 (Age 51)

The model assigned a susceptibility score of 72%, the highest in the household and well above the flagging threshold. Age (51 years) was overwhelmingly the most influential factor, pushing the score dramatically higher—the model weights individuals in their fifties as substantially elevated risk. High family income (approximately 417,000 SEK/year) and personal disposable income (approximately 156,000 SEK/year) both elevated the score considerably. Mean household personal income reduced the score (suggesting the model distinguishes between different income measures), while antihistamine medication history, index case being a child, log-transformed crowding index, minimum household family income, and living area per person (35 m²) all pushed the score higher modestly. The model expresses strong confidence in flagging this middle-aged adult.

Member 265342 (Age 44)

The model assigned a susceptibility score of 63%, above the flagging threshold. Age (44 years) was the dominant driver, pushing the score substantially higher, consistent with the model's treatment of middle-aged adults as elevated risk. Absence of children under 18 in this individual's record reduced the score, as did mean household personal income. High proportion born in Sweden, presence of a child under 6 in the household, log-transformed crowding index, maximum household family income, proportion female, household age spread (IQR), female sex, and household age range all elevated the score. The model shows moderate-to-strong confidence in flagging this individual.

4. EXPERT CONSIDERATIONS

Plausible patterns: Presence of elderly household members (65+) combined with substantial homecare service use (24 alarm activations total) indicates genuine frailty and care dependency

in the household, which could plausibly complicate isolation and increase transmission risk. The model’s strong weighting of middle-aged adults (ages 44 and 51) as higher risk compared to the teenager (19) aligns with some early-pandemic observations about age-stratified transmission, though patterns varied by period and variant.

Teenage index case signal: The 16-year-old index case (minimum age) driving household-level risk is notable. This may reflect patterns in the 2020 Swedish training data related to school attendance, social mobility, or household contact patterns specific to adolescents during that period.

Income paradox persistence: Moderate-to-high income (543,000 SEK/year mean family income) acts as a primary risk driver at both household and individual levels, contradicting health gradient expectations. This household falls in the mid-to-upper income range—not as extreme as the 935,000 SEK/year case but still showing the model’s learned positive income-risk correlation.

Homecare alarm interpretation: The household has 24 homecare alarm activations total, indicating at least one member with significant care needs and frailty. This likely proxies for underlying health vulnerability, mobility limitations, or advanced age requiring monitoring—all legitimate transmission risk factors given the challenges of maintaining isolation for individuals requiring hands-on care.

Uniform flagging significance: All three exposed members flagged (58%, 72%, 63%) represents high model confidence in household-wide transmission risk. Unlike cases with mixed assessments, this household shows no protective age profiles among exposed members—all are adults or near-adults.

Information gaps: Experts would need: which household member(s) account for the 24 homecare alarms and what underlying conditions drive this care need; the relationship between the 16-year-old index case and other household members; actual caregiving arrangements and whether effective isolation from a frail household member is feasible; room configuration within the 140 m² dwelling; and timing of index case confirmation relative to symptom onset.

5. MODEL CONFIDENCE NOTE

The household score of 66% sits well above the 30% threshold, and all three individual scores (58%, 63%, 72%) are likewise comfortably above threshold, indicating the model is expressing clear preferences for flagging across all levels. The 51-year-old’s score of 72% represents particularly strong model confidence. However, the model’s reliance on moderate-high income as a primary risk driver—contradicting typical health gradients—and the prominence of a teenage index case as the leading household risk factor introduce interpretive uncertainty about whether these patterns reflect genuine causal mechanisms or training data artifacts specific to 2020 Swedish households. The presence of substantial homecare alarm usage provides more epidemiologically plausible grounding for the elevated risk assessment. As always, actual predictive accuracy remains unknown, and these outputs serve as limited reference points for qualified expert judgment, not standalone clinical decisions.

Sample 9sec:out9

4.9.2 GPT-5.3

GPT-5.3 — Sample 9

1. CASE SUMMARY

This household consists of four people, including one identified index case and three non-index household members evaluated by the individual-level model. The household-level model assigned a transmission score of 65.6%, which is substantially above the model’s internal flagging threshold of 25.6%, meaning the household was classified by the model as higher priority for potential within-household transmission assessment.

At the individual level, all three non-index members were assigned susceptibility scores above the model threshold. The model assigned scores of 58.5%, 72.1%, and 63.0%, respectively, indicating that, according to the model’s learned patterns from registry data, all three individuals matched feature profiles the model associated with elevated susceptibility.

The model’s reasoning appears to rely primarily on household age structure, the presence of children, household socioeconomic characteristics, and indicators linked to healthcare utilisation and living conditions. These outputs reflect model behaviour only and should not be interpreted as verified infection risk or evidence that transmission actually occurred.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level model assigned a probability of 0.656 (65.6%), clearly above the empirical model threshold of 0.256. This indicates that the model viewed the household profile as relatively consistent with patterns it had previously associated with secondary household transmission.

Several features strongly influenced the model’s assessment:

- The most influential factor was the young age of the index case (minimum index age 16 years). The model weighted younger index-case age toward higher transmission likelihood in this household.
- The proportion of children in the household also increased the score. One-quarter of household members were children, suggesting the model associates households with children with greater within-household spread potential.
- Household income measures, including both mean and median family disposable income, pushed the score upward. Income variables in this context are likely acting as indirect proxies for broader social or housing characteristics rather than indicating a direct causal relationship.
- The dwelling type (“small house” rather than apartment-style housing) also increased the model score.
- Larger total floor area contributed positively, which is somewhat counterintuitive from a conventional crowding perspective and may reflect correlations in the training data rather than a direct epidemiological mechanism.
- The presence of homecare alarm usage in the household modestly increased the score, suggesting the model associated some degree of healthcare dependency or frailty with elevated transmission-related patterns.

Overall, the model’s reasoning forms a partially coherent pattern. The emphasis on children, age structure, and household composition aligns with established epidemiological thinking that household mixing patterns and intergenerational contact can influence respiratory disease spread. However, the positive contribution from larger living area and higher income is less intuitively consistent with standard crowding-based risk frameworks and should therefore be interpreted cautiously. These may represent proxy relationships specific to the underlying registry dataset rather than robust causal signals.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 9806355 — Model score: 58.5% (flagged)

The model assigned this 19-year-old household member a moderately elevated susceptibility score.

The strongest upward influence was high family income, followed by the individual’s young age. The model also increased the score because the household included a child-related structure linked to the index case. In practical terms, the model appears to associate younger household members in child-containing households with increased susceptibility patterns.

One feature pushed the score downward: the household’s maximum index-case age characteristic reduced the model’s estimate slightly. The model also gave a small upward contribution to the absence of homecare alarm usage, suggesting that this feature may behave nonlinearly in the model.

From a domain perspective, the age and household-composition signals appear plausible, while the strong income effect is more difficult to interpret epidemiologically and may reflect broader social-network or behavioural proxies embedded in the data.

Member 420689 — Model score: 72.1% (flagged)

This individual received the highest susceptibility score in the household.

The dominant factor was age: the individual was 51 years old, and the model strongly weighted this upward. Family and personal income measures also substantially increased the score. Additional positive contributions came from antihistamine medication history, household crowding measures, and the presence of children in the household.

The antihistamine variable may be acting as a proxy for chronic allergic or respiratory conditions, although this should not be interpreted causally. The crowding contribution is more epidemiologically intuitive, as denser living conditions are commonly associated with respiratory transmission opportunities.

Overall, this profile reflects a combination of demographic, household-structure, and health-related signals that the model appears to interpret consistently.

Member 265342 — Model score: 63.0% (flagged)

This 44-year-old female household member also received a clearly elevated model score.

Age was again the dominant upward influence. The model additionally increased the score because the household included young children, had a wider age spread, and showed some crowding-related characteristics. Household birth-country composition also contributed positively, indicating the

model may be using immigration-related variables as proxies for broader social or structural factors.

One factor reduced the score: the individual had no children under 18 directly associated with them personally, which the model treated as mildly protective.

The overall reasoning pattern is relatively coherent from a household-structure perspective, though the immigration-related features should be interpreted carefully because they may reflect unmeasured socioeconomic or occupational patterns rather than biological susceptibility.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear domain-consistent. The repeated emphasis on household age composition, presence of children, crowding indicators, and multigenerational structure aligns with known mechanisms influencing respiratory disease transmission opportunities.

At the same time, some features likely function as indirect proxies rather than direct risk indicators. Income variables may capture differences in occupation, social mixing, household organisation, or healthcare access. Homecare alarm variables likely proxy frailty, disability, or healthcare dependency. Birth-country and foreign-background variables may reflect broader structural inequalities or social-network patterns rather than intrinsic susceptibility.

Some findings are less intuitive, particularly the positive contribution of larger housing area and higher income at the household level. These effects could reflect dataset-specific correlations and would require further validation before being interpreted substantively.

A domain expert would likely need additional information before acting on these outputs, including occupational exposure, timing and duration of contact, vaccination status, symptom history, viral testing data, and behavioural information such as isolation practices within the home.

None of the individual scores were especially close to the threshold, so the model appears relatively decisive in this case. However, decisiveness in model output does not establish correctness.

5. MODEL CONFIDENCE NOTE

Both the household-level score (65.6% versus a 25.6% threshold) and the individual-level scores (58.5%–72.1% versus a 19.2% threshold framework) were well above the model’s internal decision boundaries. From a machine-learning perspective, this means the model was relatively confident in assigning elevated scores to this household and its non-index members.

However, these outputs remain statistical model estimates derived from historical registry patterns. Whether the predictions correspond to actual transmission or susceptibility in this household is unknown, and the results should be treated only as a limited analytical reference for qualified experts.

4.9.3 Grok 4.2

Grok 4.2 — Sample 9

CASE SUMMARY

This is a 4-person household (ID 22954508) with one index case. The model assigns a 65.6% probability of household transmission and flags the household for potential secondary spread. All three non-index members are flagged as susceptible at the individual level, with scores of 58.5% (age ~19), 72.1% (age ~51), and 63.0% (age ~44). These outputs reflect the model’s assessment of this household’s registry-derived profile from 2020 Swedish data, highlighting a multi-generational or mixed-age family with notable income and care-related characteristics.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 65.6%, well above the empirical threshold of 0.256, resulting in a clear transmission flag.

The top factors pushing the model’s score higher include:

- **Young index case age** (minimum age 16 years), suggesting the index case is an adolescent or young adult.
- **Proportion of children** (25% of the household), indicating the presence of minors.
- **Higher family disposable income** (mean around 543,000 SEK annually).
- Housing type (small house/agricultural category), larger total living area, sum of homecare alarms (TRYGG_1), and presence of elderly members (65+).

The model’s reasoning combines structural elements — mixed ages with children and elderly, plus a young index case — that align well with known epidemiological risk factors for within-household transmission, such as age mixing and close contacts across generations. Higher income and specific housing type may act as proxies for suburban/rural location, larger shared spaces, or other unmeasured factors like work or testing patterns in 2020. Overall, the pattern appears coherent from a domain perspective.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 9806355 (age ~19 in 2020): The model assigns a 58.5% susceptibility score and flags this young adult. Key upward drivers include high family income, age 19, presence of a child in the household, and personal income indicators. Some household index age factors pulled downward. The model associates this late-teen profile in a higher-income mixed household with moderate susceptibility.

Member 420689 (age ~51 in 2020): The model assigns the highest individual score of 72.1% and flags this middle-aged adult. The strongest upward influences are adult working age (~51), high family and personal income, and antihistamine prescription history. Crowding and income distribution measures provided modest additional upward pressure. The model links this middle-aged profile with elevated susceptibility, which is plausible given typical adult contact patterns.

Member 265342 (age ~44 in 2020): The model assigns a 63.0% susceptibility score and flags this adult. Major upward drivers include age ~44, proportion born in Sweden, presence of

a child under 6, crowding measures, and female gender. The model weighs this combination — mid-adult age in a household with young children — as increasing susceptibility.

EXPERT CONSIDERATIONS

The model strongly flags the entire household, with attention directed across all non-index members, particularly the middle-aged adults. Domain-consistent patterns include the roles of age mixing (young index + children + adults + elderly), household size, and presence of young children — all well-established transmission facilitators. Higher family income and homecare alarm history emerge as influential but likely serve as proxies for unmeasured variables such as residential setting, occupational exposure, family support needs, or healthcare access rather than direct causal factors. NSAID and antihistamine prescriptions played minor roles for some individuals.

Experts should incorporate real-world details unavailable to the model: precise timing of the index case, specific occupations or school attendance of members, isolation and ventilation practices inside the home, and detailed symptom or testing history. All individual scores are solidly above threshold with no borderline cases, while the household probability is decisively elevated.

MODEL CONFIDENCE NOTE

The household transmission probability of 65.6% is substantially above threshold, and all three individual scores are clearly elevated well above the individual cutoff, indicating relatively confident and consistent flagging by the model across both levels. These are statistical pattern-based assessments derived from 2020 Swedish registry data, not verified infection risks or clinical predictions. They should be used only as a limited supplementary reference to inform expert public health decision-making and further investigation.

4.9.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 9

Here is the structured interpretation of the model’s output for household 22954508, written for public health policy experts.

1. CASE SUMMARY

This household contains four people, including one designated index case and three other members. The model assigns a **65.6% probability** of within-household COVID-19 transmission, well above its internal flagging threshold of 25.6%. All three non-index members are flagged as “susceptible,” with scores ranging from 58.5% to 72.1%. **These are statistical model outputs, not confirmed infections.**

2. HOUSEHOLD MODEL ASSESSMENT

The model strongly flagged this household for potential transmission. Unlike several previous households where all top features pushed risk downward, here **most top features pushed the score upward:**

- **Youngest age among index cases (16 years):** The index case is an adolescent. This was the strongest upward driver — notable given that very young children (ages 1–7) consistently pushed risk down, while adolescents push risk up.
- **Proportion of children in household (25%, one child):** Having some children present increased the transmission score.
- **Household disposable income (542,800 SEK/year – relatively high):** Higher income pushed the score upward, consistent with the pattern seen in other high-income households (e.g., household 3).
- **Housing type (detached house with agricultural land – *SmahusAg*):** This dwelling form increased risk, similar to household 3.
- **Total living area (140 m² – large):** Larger total floor area pushed the score higher — again, more space associated with higher risk in this context.
- **Sum of homecare alarms (24 – very high):** A very high total of homecare alarm activations across the household strongly increased the transmission score. This is the first example where homecare alarms appear as a top driver.
- **Presence of an elderly member (65+):** Having an elderly person present increased risk (unlike household 6 where absence of elderly reduced risk — now consistent directionally).

Coherence check: This household presents a “high-risk according to the model” profile that is almost the mirror image of the low-risk household 8: higher income, larger dwelling, presence of elderly, adolescent index case (rather than young adult), and very high homecare alarm use. The pattern is internally coherent and aligns with the idea that the model has learned a set of features that consistently push risk in one direction.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 9806355 (age 19) – Flagged (58.5%) Key upward drivers:

- **Household disposable income (high, 803,700 SEK/year – strongest upward):** Very high income strongly increased susceptibility.
- **Oldest age in household (pushed down – a rare negative for this person):** The presence of an older member reduced this young adult’s score slightly.
- **Age 19 (upward):** Young adult age increased susceptibility.
- **Index case is a child (upward):** Having an adolescent index case increased this person’s score.
- **Homecare alarms (zero for this individual, but the household sum is high – upward):** Even with no personal alarms, the household’s high homecare use pushed the score up.

Member 420689 (age 51) – Flagged (72.1%) – highest in this household Key upward drivers:

- **Age 51 (strongest upward):** Middle age strongly increased susceptibility.

- **Household disposable income (high, 417,100 SEK/year – upward):** Again, high income pushed the score up.
- **Personal disposable income (156,300 SEK/year – upward):** Moderate individual income added upward weight.
- **Antihistamine use (lmed_R06A – upward):** A medication for allergies pushed the score higher — one of the few times a specific medication category appears in top drivers.
- **Crowding index (upward):** Higher crowding (more persons per square metre) increased susceptibility — unlike some previous households where crowding reduced risk.

Member 265342 (age 44) – Flagged (63.0%) Key upward drivers:

- **Age 44 (strongest upward):** Middle age again strongly increased susceptibility.
- **No children under 18 in household for this person? (AntalBarnUnder18=0, downward):** The absence of children in this individual’s immediate record reduced the score slightly.
- **Proportion born in Sweden (upward):** A higher share of Swedish-born members increased susceptibility.
- **Presence of a child under 6 in household (upward):** Having a young child in the household increased this middle-aged adult’s score.
- **Crowding index (upward):** Again, more crowding increased susceptibility.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** The model’s high-risk profile for this household — higher income, larger dwelling, adolescent index case, presence of elderly, and very high homecare alarm use — forms a coherent set. The fact that all three non-index members are flagged with similarly high scores is internally consistent.
- **Uncertain / surprising patterns:**
 - **Very high homecare alarms (sum = 24)** is an extreme value. This likely indicates a household with significant care needs. The model treated this as a strong upward driver — plausible as a proxy for frailty, healthcare contact, or underlying health conditions.
 - **Adolescent index case (age 16) pushing risk up** contrasts sharply with very young children (ages 1–7) pushing risk down. The model appears to draw a sharp age threshold around early adolescence.
 - **Antihistamine use (lmed_R06A)** appearing as a driver for one individual is notable — this is the first time a specific medication category has appeared in top features. This may be a chance correlation or a genuine proxy for allergic conditions (which have been hypothesised to relate to COVID-19 susceptibility).
- **Proxy variables:** Homecare alarms are a clear proxy for frailty and healthcare system contact. Income again acts as a proxy for unmeasured socioeconomic or behavioural

factors. Living area (more space) associated with higher risk continues the paradoxical pattern seen in earlier high-income households.

- **Additional information needed:** Experts would need to understand whether the extreme homecare alarm value (24) is accurate and what it represents (e.g., one very frail individual or multiple members). They would also benefit from exploring why antihistamines appeared as a driver for one individual but not others in the same household.
- **Uncertain zone:** Not applicable. The household score (65.6%) is well above the threshold, and all individual scores (58.5–72.1%) are comfortably above their thresholds. The model is statistically confident in its high-risk classification.

5. MODEL CONFIDENCE NOTE

The model’s probability estimates for this household (65.6% for transmission, individual scores between 58.5% and 72.1%) are well above the empirical thresholds, indicating high statistical confidence in its high-risk classification. However, the appearance of specific medication use (antihistamines) and extreme homecare alarm values as top drivers raises questions about whether the model is picking up stable epidemiological signals or overfitting to rare combinations in the registry data. Correctness is unknown. These outputs should be treated as a limited reference — the coherent high-risk pattern is notable, but the specific drivers warrant expert scrutiny.

4.9.5 Llama 4

Llama 4 — Sample 9

COVID-19 Household Transmission Model Analysis

Household ID: 22954508

Case Summary

This is a 4-person household with members aged 16 (index case), 19, 44, and 51, where the model assigns a 65.6% household transmission probability—well above the flagging threshold. At the individual level, all three non-index members receive elevated susceptibility scores: 58.5% for the 19-year-old, 72.1% for the 51-year-old, and 63.0% for the 44-year-old. Notably, this household has 24 homecare alarm activations, indicating significant care dependency or health vulnerability among household members. The model flags this as a high-risk transmission scenario across both household and individual assessments, with middle-income status, mixed-age composition, and documented health fragility all contributing to the elevated risk profile.

Household Model Assessment

The model assigns a 66% household transmission probability, significantly above the threshold. Multiple risk factors push the score upward:

Teenage index case (minimum age 16): Having a 16-year-old as the index case increases the transmission score. Teenage index cases may have been associated with higher household transmission in the training data, possibly reflecting school contacts, social activities, or difficulty

isolating within the home during 2020.

Low child proportion (25%): Only one of four household members is under 18, meaning this is predominantly an adult household. Adult-heavy households appear to have had higher transmission rates in the model’s learned patterns.

Moderate-high family income (542,800 SEK annually): At roughly 543,000 kronor per year for four people, this household sits at middle-income levels. While not in the extreme high-income range we’ve seen drive scores upward dramatically, this income level still pushes toward higher risk compared to low-income households.

Single-family detached home (SmahusAg): Detached house dwelling increases the score, consistent with patterns where single-family homes have been associated with higher risk—possibly proxying for suburban locations, commuting patterns, or specific socioeconomic clusters.

Large living area (140 m²): Spacious housing (35 m² per person) increases risk, counter to crowding expectations. This may correlate with the income and housing type factors, capturing lifestyle patterns associated with larger suburban homes.

Homecare alarm history (24 activations): This is a critical finding—24 homecare alarm activations indicates substantial health vulnerability or care dependency among household members. This could reflect elderly members (the data shows "has_elderly_65plus" as a risk factor), chronic illness, disability, or frailty requiring monitored care. Households with homecare needs likely had elevated transmission risk due to caregiver contacts, medical visits, or underlying health conditions.

Presence of elderly members (65+): Having at least one member aged 65 or older increases the household score, though not dramatically. The presence of elderly combined with documented homecare needs suggests a multigenerational household caring for a vulnerable member.

The pattern is coherent: mixed-age household with documented health fragility, teenage index case, middle-income suburban setting. The combination of a socially active teenager, working-age adults, and a health-vulnerable member creates multiple transmission pathways and reduced isolation capability.

Individual Susceptibility Profiles

Member 9806355 (Age 19) — 58.5% susceptibility, FLAGGED

The young adult receives a moderately high score:

- **High family income (803,700 SEK):** The dominant driver, though this value appears higher than the household-level figure, possibly reflecting different aggregation methods or individual-level reporting.
- **Age (19 years):** Young adult age pushes the score upward. At 19, this person sits at the transition to full adulthood where the model begins treating them more like working-age adults than protected children.
- **Personal income (50,900 SEK):** Economic activity, even modest, increases susceptibility—possibly reflecting part-time work, internships, or early career exposure.
- **Index case is a child:** Having a teenage index case increases this young adult’s score,

possibly reflecting shared social spheres or household contact patterns between near-peer ages.

- **NSAID medication use (lmed_M01A):** Chronic pain medication increases the score, potentially indicating underlying health conditions or healthcare system engagement.

Interestingly, the maximum age in the index case characteristics *decreases* this person's score—possibly a data artifact or reflection of complex household dynamics.

Member 420689 (Age 51) — 72.1% susceptibility, FLAGGED (HIGHEST)

The 51-year-old receives the highest individual score:

- **Age (51 years):** The strongest single driver—middle age significantly elevates the model's susceptibility score. The model has learned that this age bracket (50s) carries particularly high transmission risk, whether due to occupational patterns, social activity, biological factors, or healthcare system engagement observed in 2020 Sweden.
- **Family income (417,100 SEK):** Substantial income contributes heavily to the high score.
- **Personal income (156,300 SEK):** Individual economic activity reinforces the income effect.
- **Antihistamine medication use (lmed_R06A):** Allergy medication increases the score, potentially indicating respiratory sensitivities or healthcare engagement.
- **Crowding indicators:** Log-crowding increases risk while mean household income decreases it—complex interactions suggesting household dynamics matter.

This 51-year-old represents the model's highest-risk profile: working-age adult in peak career years, economically active, in a household with documented health needs.

Member 265342 (Age 44) — 63.0% susceptibility, FLAGGED

The 44-year-old receives a high score:

- **Age (44 years):** Mid-40s age is the dominant factor, following the pattern where working-age adults receive elevated scores.
- **Swedish-born household composition:** High proportion of Swedish-born members increases the score, possibly reflecting social network structures or contact patterns in the training data.
- **Young child present (under 6):** Wait—the data shows "has_child_under_6" as a risk-increasing factor for this person, but we identified the youngest as 16. This may be a data artifact or coding issue.
- **Crowding indicators:** Log-crowding increases risk, suggesting household density affects this person's score.
- **No children under 18 (for this person specifically):** Not having children reduces the score slightly—this person may not be the parent of the 16 and 19-year-olds.
- **Female (Kon = 2):** Being female slightly increases this person's score, contrary to some earlier patterns where female gender was protective.

Expert Considerations

Homecare dependency as a major risk amplifier: The 24 homecare alarm activations are the most striking feature—this indicates a household managing significant health vulnerability. Whether for an elderly member, someone with chronic illness, or disability, this documented care dependency substantially increases both household-level transmission risk and creates caregiving dynamics that affect individual members' exposure.

All-adults-flagged scenario: All three non-index members receive scores above 58%, with the 51-year-old hitting 72%. The model sees this as a household where every adult is at elevated risk—no protected members beyond the (already infected) teenage index case.

Age-risk relationship clarity: The ascending scores by age (19 years = 58%, 44 years = 63%, 51 years = 72%) demonstrate the model's strong age gradient for working-age and middle-aged adults. Unlike the protective effect for young children, the model treats ages 19-51 as progressively higher-risk.

Income bidirectionality: While extremely low income (50-100K SEK) protects and extremely high income (700-900K SEK) increases risk sharply, this middle-income household (540K SEK) shows moderate risk elevation. The income-risk relationship appears nonlinear.

Multigenerational household with health complexity: The presence of elderly members, documented homecare needs, teenagers, and working-age adults creates a household where isolation and risk mitigation strategies are particularly challenging. Caregiving responsibilities, economic obligations, and social patterns all intersect.

Missing clinical context: The model has no information about the specific health conditions requiring homecare, vaccination status, ability to isolate within the home, or actual caregiving arrangements—all critical factors for understanding real transmission dynamics in a health-vulnerable household.

Model Confidence Note

The model expresses high confidence in this high-risk assessment: the household score (66%) sits well above the threshold, and all three individual scores (58-72%) decisively exceed the individual threshold. The unanimous flagging of all adults, combined with documented health vulnerability (homecare alarms), creates a clear high-risk signal in the model's learned patterns. However, these scores reflect statistical associations from 2020 Swedish registry data about households with similar demographic and health profiles—not predictions of actual infection outcomes in this specific case. The model identifies this household as matching patterns associated with elevated transmission risk, but cannot account for unmeasured protective behaviors, vaccination status, or specific mitigation strategies this household may have implemented. These outputs provide one data point for expert assessment of resource allocation and intervention priorities.

Note: This analysis explains the model's high-risk assessment based on documented health vulnerability and household demographics, not actual infection outcomes.

Evaluation Scores — Sample 9

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%

4.10 Sample 10

4.10.1 Claude Sonnet 4.5

Claude Sonnet 4.5 — Sample 10

HOUSEHOLD TRANSMISSION PREDICTION INTERPRETATION

Household ID 23495841 | Model Analysis

1. CASE SUMMARY

This household comprises two individuals: one confirmed COVID-19 index case and one exposed member. The model assigned a transmission score of 13% to this household—well below the model’s flagging threshold of 30%—indicating that, based on the household’s feature profile, the model does not consider within-household transmission likely. The single non-index member, a 6-year-old child (member 7618294), received a susceptibility score of 17%, also below the flagging threshold, and was not flagged by the model. This represents a strong protective assessment at both household and individual levels, driven primarily by low household income, absence of homecare service use, high index-to-household ratio, young household members, and the child’s age.

2. HOUSEHOLD MODEL ASSESSMENT

The model assigned this household a transmission score of 13%, sitting 17 percentage points below the 30% threshold—a clear non-flagging decision with substantial margin. Every influential feature pushed the score downward, creating a uniformly protective profile.

Key factors driving the model’s low household-level score:

Low household income (mean family income of approximately 124,000 SEK/year, in the lower range observed across cases) was the most influential factor, substantially reducing the model’s score. This continues the persistent pattern where lower income acts protectively in the model’s learned associations.

Absence of homecare services (zero total homecare service contacts across the household) was the second-strongest protective factor, indicating no frailty, care dependency, or underlying health burden requiring monitoring. This absence of vulnerability indicators aligns with genuinely protective health status.

High index-to-household ratio (50%—one of two members is the index case) substantially reduced the score, continuing the counterintuitive pattern observed in other two-person households where having half the household already infected acts protectively rather than increasing exposure risk for the remaining member.

Young maximum household age (27 years) reduced the score considerably, indicating a young household overall—likely a young parent and child.

Zero homecare contacts per capita (reinforcing the absence-of-frailty signal), mean family income (in non-transformed form), apartment dwelling (FlerHy), and high proportion of children (50%—one child in a two-person household) all contributed additional protective effects. The young mean household age (16.5 years, reflecting the 6-year-old bringing down the average) also

reduced the score modestly.

The pattern is relatively coherent: absence of health vulnerabilities (no homecare needs), young household members, and low income (following the model’s learned gradient) all contribute to protective assessment. The small household size and high index ratio present the same interpretive puzzle seen in other two-person households.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 7618294 (Age 6)

The model assigned a susceptibility score of 17%, well below the flagging threshold, and did not flag this child. Two factors dominated with nearly equal influence: absence of personal family income (recorded as zero for this child dependent) and age (6 years) both drove the score dramatically downward. Together, these protective effects are massive. Absence of personal disposable income, low mean household family income, and index case maximum age (young at 27) all further reduced the score substantially. Having a child aged 6-17 in the household (this child) was the only factor pushing the score higher, though its influence was modest. All-foreign-background household status, absence of homecare alarms, low mean household personal income, and low maximum household personal income contributed additional protective effects.

The pattern is strongly protective and epidemiologically coherent: very young age combined with zero income (indicating dependent status) receives the model’s strongest protective weighting. The single risk-increasing factor—presence of a school-aged child—may reflect the model learning about school-related exposure patterns in the 2020 Swedish training data, though this signal is overwhelmed by age and income protections. The model expresses strong confidence in not flagging this child.

4. EXPERT CONSIDERATIONS

Plausible patterns: The model’s strong protective weighting of young age (6 years) aligns with documented early-pandemic transmission patterns. Absence of homecare services indicates a healthy household without frailty or chronic care needs—a genuinely protective health status indicator. The young maximum household age (27 years) suggests a young parent-child household, which may have had systematically different exposure or isolation patterns in 2020 Sweden.

Two-person household dynamics: This case reinforces the pattern observed across two-person households where high index-to-household ratio (50%) acts protectively despite seemingly creating maximum exposure for the remaining member. This may reflect selection effects in who gets tested and reported in two-person households, or genuine behavioral differences (stronger isolation precautions when only one vulnerable member remains).

Income gradient consistency: Low income (approximately 124,000 SEK/year) acts as the dominant protective factor, consistent across all cases examined. This household sits at the lower end of the income distribution, receiving maximum protective weighting from this feature. While counterintuitive from typical health gradient perspectives, the consistency suggests this pattern is robust in the model’s learned associations from 2020 Swedish training data.

School-aged child signal: The presence of a child aged 6-17 modestly increases the susceptibility score, the only risk-elevating factor visible for this individual. This may capture school-related exposure patterns during Sweden’s 2020 approach to education (which differed from many other countries in keeping some schools open), though the signal is relatively weak compared to protective factors.

Proxy features: Low income may proxy for single-parent households, reduced occupational mobility, smaller social networks, or other unmeasured factors specific to the 2020 Swedish context. The absence of any medication histories indicates a child without chronic health conditions requiring pharmaceutical management.

Information gaps: Experts would need: the exact relationship between household members (parent-child, siblings with external guardian, other arrangement); childcare or school attendance patterns during 2020; whether the 27-year-old index case could effectively isolate from a 6-year-old in a two-person household; actual living arrangement within the 72 m² space (36 m² per person—relatively spacious); and timing of index case confirmation relative to child exposure.

Confidence across all assessments: Both scores (household 13%, individual 17%) sit well below the 30% threshold with substantial margins. The model expresses uniform strong confidence in protective assessment.

5. MODEL CONFIDENCE NOTE

The household score of 13% and individual score of 17% both sit comfortably below the 30% threshold, indicating the model is expressing strong confidence in protective assessment at both levels. Neither score approaches the uncertain zone near the decision boundary. The model’s reasoning relies heavily on well-established protective patterns—young child age and absence of health vulnerabilities—alongside the persistent but epidemiologically questionable income-as-protective gradient. The combination of very young age (6 years), no homecare needs, and low income creates one of the model’s most confident protective profiles. However, as with all cases, actual predictive accuracy remains unknown, and these outputs serve only as limited reference points for qualified expert judgment, not standalone clinical decisions.

4.10.2 GPT-5.3

GPT-5.3 — Sample 10

1. CASE SUMMARY

This household consists of two people: one identified index case and one non-index household member evaluated by the individual-level model. The household-level model assigned a transmission score of 12.6%, which is below the model’s internal flagging threshold of 25.6%. According to the model’s framework, this household was therefore not classified as a higher-priority transmission household.

At the individual level, the single non-index member received a susceptibility score of 17.4%, also below the model’s flagging threshold. The model therefore did not flag this person as susceptible within its internal decision framework.

Overall, the model’s assessment appears driven primarily by factors associated with lower predicted transmission and susceptibility, including small household size, absence of healthcare dependency indicators, young age, and low recorded income variables. These outputs describe the model’s learned statistical behaviour only and do not indicate whether transmission actually occurred.

2. HOUSEHOLD MODEL ASSESSMENT

The household-level model assigned a probability of 0.126 (12.6%), substantially below the empirical threshold of 0.256. This indicates that the household profile resembled patterns the model had previously associated with lower likelihood of secondary household transmission.

Several features strongly pushed the model toward a lower-risk assessment:

- Household income measures were the most influential factors. Both mean family disposable income and related income variables reduced the model score considerably. In this case, the household appears to have very low recorded income values. This should not be interpreted directly as “protective”; rather, the model has learned that this type of income profile correlated with lower transmission labels in the training data.
- The complete absence of homecare utilisation or homecare alarm activity strongly lowered the model score. The model appears to associate households without healthcare dependency or frailty markers with lower transmission-related patterns.
- The household structure itself also reduced the score. This was a two-person household with one index case, meaning the index-to-household ratio was relatively high. The model weighted this configuration toward lower transmission likelihood, possibly because there were fewer exposed household contacts overall.
- Younger household age structure contributed further downward pressure. The oldest household member was only 27 years old, and the mean household age was low.
- The household included children, but in this case the proportion of children slightly reduced the model score rather than increasing it.

Taken together, the model’s reasoning is relatively coherent. Small household size, absence of elderly or frail members, and low healthcare utilisation align with plausible epidemiological interpretations of lower household vulnerability. However, the strong influence of income variables remains difficult to interpret substantively and may reflect indirect social or registry-related correlations rather than direct mechanisms of infection risk.

3. INDIVIDUAL SUSCEPTIBILITY PROFILE

Member 7618294 — Model score: 17.4% (not flagged)

The model assigned this 6-year-old child a susceptibility score below the model threshold and therefore did not flag the individual.

The strongest downward influence was the child’s very young age. According to the model’s learned patterns, being six years old substantially reduced the assigned susceptibility score in this case. Very low recorded family and personal disposable income values also strongly reduced the model score.

Additional downward influences included:

- The absence of homecare alarm activity, suggesting no indicators of frailty or healthcare dependency.
- Household-level income measures, which consistently pushed the prediction lower.
- Household foreign-background composition, which modestly reduced the score in this case.

One feature modestly increased the score: the presence of a child aged 6–17 within the household. This suggests the model associates school-aged children in the household with slightly greater transmission-related potential, although this effect was much smaller than the dominant downward influences.

From a domain perspective, the model’s emphasis on young age and absence of healthcare dependency appears plausible. However, the unusually strong influence of income-related variables should be interpreted cautiously, as these may reflect broader social or administrative patterns embedded in the training data rather than direct biological susceptibility.

4. EXPERT CONSIDERATIONS

Several aspects of the model’s reasoning appear broadly consistent with public health expectations. The model assigned lower scores in a small household with no elderly members, no evidence of healthcare dependency, and limited household complexity. These are all factors plausibly associated with fewer opportunities for prolonged or severe within-household transmission dynamics.

At the same time, income variables were among the strongest influences at both the household and individual levels. These likely function as proxies for unmeasured structural or social factors rather than representing causal effects themselves. For example, income may indirectly capture housing patterns, employment structures, healthcare engagement, or differences in testing and reporting behaviour during 2020.

The child-age signal is also noteworthy. The model strongly weighted young age downward for the individual assessment, while still assigning a small upward effect to the presence of school-aged children in the household. This suggests the model may distinguish between the susceptibility profile of the child themselves and the broader household contact structure associated with children.

No scores were particularly close to the decision thresholds. Both the household-level and individual-level predictions were clearly below their respective cutoffs, suggesting the model was relatively decisive in this case. However, this should not be interpreted as confirmation that the household truly experienced lower transmission risk.

5. MODEL CONFIDENCE NOTE

Both the household-level score (12.6% versus a 25.6% threshold) and the individual-level score (17.4% versus the individual threshold framework) were meaningfully below the model’s decision boundaries. From a machine-learning perspective, this indicates the model assigned this case relatively consistently to its lower-risk side of the classification boundary.

However, these outputs remain statistical estimates derived from historical registry patterns rather than verified infection outcomes. Whether the model’s assessment was correct for this household is unknown, and the results should be treated only as a limited interpretive aid for qualified experts rather than as evidence or decision guidance.

4.10.3 Grok 4.2

Grok 4.2 — Sample 10

CASE SUMMARY

This is a small 2-person household (ID 23495841) with one index case and one non-index member. The model assigns a low 12.6% probability of household transmission and does not flag the household. The single non-index member, a 6-year-old child, receives a very low susceptibility score of 17.4% and is not flagged by the individual-level model. These outputs reflect the model’s assessment of this minimal, young-family household configuration based on 2020 Swedish registry patterns.

HOUSEHOLD MODEL ASSESSMENT

The household-level model assigns a transmission probability of 12.6%, well below the empirical decision threshold of 0.256, so no transmission flag is raised.

The strongest factors pulling the model’s score downward include:

- **Low family disposable income** (mean around 124,000 SEK annually), which the model weighted heavily as reducing predicted risk.
- **Absence of homecare service contacts** (TRYGG_total_sum = 0 and per capita = 0), indicating no recorded care dependency in the household.
- **Index-to-household ratio of 50%**, meaning half the household members are already identified as cases.
- **Young ages** (index and household maximum around 27 years) and presence of a child (proportion children = 50%), combined with multi-family housing type.

The model’s reasoning is coherent with epidemiological expectations for small households. Limited household size, young age composition, and a high proportion of known cases naturally constrain opportunities for additional spread. Low income and lack of homecare indicators likely serve as proxies for specific living conditions, mobility, or family circumstances observed in the training data.

INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 7618294 (age ~6 in 2020): The model assigns a 17.4% susceptibility score, below the 0.192 threshold, and does not flag this young child. The dominant downward drivers are low/zero family and personal income indicators (typical for a child) and young age. These were reinforced by household income measures. A modest upward influence came from the presence of a child aged 6–17 in the household, but it was insufficient to elevate the overall score.

The model’s strong down-weighting of a young child in a small household with a young index case aligns with plausible domain patterns. Young children often exhibit different exposure or

recording behaviours in registry data compared to adults, and the absence of income and care indicators further supports the low assessment.

EXPERT CONSIDERATIONS

The model does not prioritise this household or the child for intervention, showing clear low-risk differentiation at both levels. Domain-consistent elements include the risk-reducing roles of very small household size, young child presence, and young index case age. Low income and absence of homecare utilisation emerge as important downward factors, likely acting as proxies for unmeasured variables such as residential setting, family structure, or social/economic circumstances rather than direct protective effects.

Medication history variables had negligible influence. Experts should review additional contextual information unavailable to the model, including the exact timing of the index case, caregiving arrangements for the child, any isolation or protective measures taken in the home, and actual symptom or testing records. All probabilities are well below their respective thresholds, with no scores in uncertain boundary zones.

MODEL CONFIDENCE NOTE

The household transmission probability (12.6%) and the individual susceptibility score (17.4%) are both substantially below their decision thresholds, indicating the model is quite decisive in assigning low risk to this small household with a young child. As with all outputs from this system, these reflect statistical patterns learned from 2020 Swedish registry data rather than verified infection risks or clinical predictions. They serve best as one limited reference tool to support expert public health judgment.

4.10.4 DeepSeek V3.2

DeepSeek V3.2 — Sample 10

Here is the structured interpretation of the model's output for household 23495841, written for public health policy experts.

1. CASE SUMMARY

This household consists of two people: one designated index case and one other member. The model assigns a **12.6% probability** of within-household COVID-19 transmission, well below its internal flagging threshold of 25.6%. The single non-index member receives a susceptibility score of **17.4%** and is not flagged by the model. **These are statistical model outputs, not confirmed absence of infection.**

2. HOUSEHOLD MODEL ASSESSMENT

The model confidently did **not** flag this household for transmission. Every top influential feature pushed the score downward:

- **Household disposable income (124,000 SEK/year – low):** The strongest downward driver, consistent with other low-income households.

- **Total homecare service contacts (TRYGG_total_sum = 0):** The complete absence of homecare use strongly reduced the transmission score — the mirror image of household 9, where very high homecare use pushed risk up.
- **Index-to-household ratio (0.5):** As in other two-person households (reports 5 and 7), a single index case in a two-person household pushed the score down.
- **Oldest age in household (27 years):** The index case is relatively young (27), which reduced the transmission estimate.
- **Homecare contacts per capita (0):** Reinforced the absence-of-care signal.
- **Housing type (row house – *FlerHy*):** This dwelling form pushed the score down.
- **High proportion of children (50%, one child):** The non-index member is a child (age 6), and a household with one child out of two members reduced the transmission score.

Coherence check: This is another internally consistent low-risk household, very similar to household 8 (very low income, young index case, young child). The presence versus absence of homecare alarms stands out as a strong directional signal across households — zero homecare use pushes risk down, high use pushes risk up. This pattern is plausible if homecare use proxies for underlying health conditions, frailty, or healthcare contact frequency.

3. INDIVIDUAL SUSCEPTIBILITY PROFILES

Member 7618294 (age 6) – Not flagged (17.4%) The score is just below the individual threshold of 19.2% — a borderline case. Most top factors pushed the score downward, but one factor pushed upward:

Downward drivers:

- **Household disposable income (zero value – strongest downward):** A zero-income artifact strongly reduced the score.
- **Age 6 (strong downward):** Consistent with all other households — young children receive low susceptibility scores.
- **Personal disposable income (zero, downward):** Reinforced.
- **Household mean income (zero/low, downward):** Consistent downward signal.
- **Oldest age in household (index case age 27, downward):** A relatively young oldest member reduced the child’s score.
- **All members have foreign background (downward):** This pushed the score lower.

The **only upward driver** in the top 10 was:

- **Household has a child aged 6–17 (upward, +0.0252):** Interestingly, the presence of a child in this age range (the 6-year-old themselves, or another child? The household has only one child) slightly increased the score. This may be a feature interaction where being a child in a child-present household has a different effect than being a child in an adult-only household.

Plausibility: The score of 17.4% is extremely close to the individual threshold of 19.2% — a borderline case. The model is essentially uncertain about this child. All major signals (young age, low income, no homecare) push toward low risk, but the “child present” feature pushes slightly upward, resulting in a near-threshold score.

4. EXPERT CONSIDERATIONS

- **Domain-consistent patterns:** Very low income, young index case, young child, and no homecare use all pointing toward low model-assigned risk form a coherent pattern that matches households 8 and 4. The model’s treatment of homecare use (zero = lower risk, high = higher risk) is directionally plausible.
- **Uncertain / surprising patterns:**
 - **Zero-income values** appear again for family and personal income. In a two-person household with an index case (age 27, presumably an adult), zero income is almost certainly a data artifact. The model treats these zeros as real, which likely deflates scores artificially.
 - **The only upward driver** for the individual (presence of a child aged 6–17) is interesting but difficult to interpret without knowing whether it refers to the child themselves or another child in the household.
 - **Borderline individual score (17.4% vs. 19.2% threshold):** This child is very close to being flagged. Small changes in features or threshold calibration could flip this prediction.
- **Proxy variables:** Income again acts as a proxy — here, zero income likely indicates missing data. Homecare use (or its absence) appears as a strong proxy for health status and healthcare contact. Foreign background (all members) pushed risk down in this case, though its direction has varied across households.
- **Additional information needed:** Experts would need to verify whether the zero-income values are accurate or imputed. They would also benefit from understanding why “household has a child aged 6–17” pushed the score upward for this child — is this a genuine interaction or a model artifact?
- **Uncertain zone:** The household score (12.6%) is comfortably below its threshold. However, the individual score (17.4%) is **very close** to the 19.2% threshold — only 1.8 percentage points away. This is a true uncertain zone at the individual level. The model is not confident about whether to flag this child.

5. MODEL CONFIDENCE NOTE

The household-level prediction (12.6%) is well below the threshold, indicating high statistical confidence in the low-risk classification for the household. However, the individual-level score (17.4%) is extremely close to the individual threshold (19.2%), placing this child in a low-confidence uncertain zone. The pervasive zero-income values raise data quality concerns

that could affect both estimates. Correctness is unknown. These outputs should be treated as a limited reference — the household is clearly low-risk according to the model, but the child's borderline status warrants caution.

4.10.5 Llama 4

Llama 4 — Sample 10

COVID-19 Household Transmission Model Analysis

Household ID: 23495841

Case Summary

This is a 2-person household—likely a young parent (age 27) and a 6-year-old child—where the model assigns just a 12.6% household transmission probability, well below the 25.6% flagging threshold. The 6-year-old receives a 17.4% individual susceptibility score, also clearly below the individual threshold. With low income (124,000 SEK annually), no homecare service use, young household composition, and predominantly child makeup, every major factor aligns toward minimal transmission risk. This case represents another clear low-risk profile in the model's learned patterns: small household, low income, young parent with young child, and no documented health vulnerabilities.

Household Model Assessment

The model assigns a 13% household transmission probability—far below the threshold. Every top feature provides protection:

Low family income (124,000 SEK annually): The dominant protective factor, this below-average income for a two-person household suggests limited financial resources. At roughly 124,000 kronor per year (62,000 SEK per person), this household sits below typical Swedish income levels and likely qualifies for social support programs. The model consistently associates lower income with reduced transmission risk in the training data.

Zero homecare service use: Complete absence of homecare contacts provides the second-strongest protection. No documented care dependency indicates no elderly members requiring monitored care, no chronic health conditions necessitating home support, and no frailty markers in the household health profile. This contrasts sharply with the previous case's 24 homecare alarms.

High index-to-household ratio (50%): With one infected person in a two-person household, half the household is already the index case. This mathematically limits the opportunity for "additional" transmission—only one person remains exposed.

Young maximum age (27 years): The oldest household member is just 27, indicating a very young household unit. Young adult households, particularly with young children, appear to have had lower transmission rates in the training patterns.

Zero homecare per capita: Reinforces the absence of health vulnerability or care dependency across the household.

Low mean household income: Multiple income metrics consistently push the score downward.

Apartment dwelling (FlerHy): Multi-family housing provides protection, correlating with lower-income, smaller-space profiles.

High child proportion (50%): Half the household consists of children under 18, which the model treats as strongly protective. Young family units with high child concentration received lower transmission scores.

The pattern is exceptionally clear: young single parent, young child, low income, small space, no health vulnerabilities, no elderly. Every structural factor points unambiguously toward low transmission risk based on 2020 Swedish patterns.

Individual Susceptibility Profiles

Member 7618294 (Age 6) — 17.4% susceptibility, NOT FLAGGED

The 6-year-old receives a low score with nearly all factors pointing downward:

- **Zero family income:** The strongest protective factor, tied with age. Children without economic profiles receive massive protection in the model’s logic, reflecting minimal external exposure or economic activity.
- **Young age (6 years):** Equally dominant—early childhood age dramatically reduces susceptibility scores throughout every case examined. Six years old sits right at the edge of school age, receiving strong but not quite maximal protection (compare to the 1-year-old’s 5% score in report 8).
- **Zero personal income:** Complete economic dependency reinforces the protective pattern.
- **Low household income structure:** Multiple household income metrics compound the protection.
- **Young index case (max age 27):** Having a young parent as the index case reduces this child’s score, possibly reflecting different household dynamics or caregiving patterns in young single-parent families.
- **All-foreign-background household:** The household having all members with foreign background provides protection, suggesting this demographic classification was associated with lower transmission in the training data.
- **Zero homecare alarms:** No documented health vulnerability in the household reduces this child’s score.

Interestingly, one factor pushes the score *upward*: having a child aged 6-17 in the household (has_child_6_17) increases this person’s score. This is the 6-year-old themselves, suggesting school-age children carry slightly elevated risk compared to preschoolers, possibly reflecting school contact patterns even in 2020 with disrupted schooling.

Expert Considerations

Prototypical low-risk profile: This household embodies the model’s clearest low-risk signals: young single parent, young child, low income, small household, no health vulnerabilities, no elderly. It sits at the opposite end of the risk spectrum from high-income, large, multigenerational, or health-vulnerable households.

Income as bidirectional marker: The 124,000 SEK income provides strong protection, continuing the pattern where very low income (50-150K SEK) reduces risk, moderate income (150-400K SEK) has mixed effects, and high income (500K+ SEK) increases risk substantially. The nonlinear relationship suggests income proxies for complex bundles of mobility, social activity, housing quality, and occupational exposure.

Homecare absence as health indicator: Zero homecare contacts contrasts dramatically with previous cases showing 24 homecare alarms. The model clearly distinguishes between households with documented health vulnerabilities (risk) and those without (protection). This binary indicator appears highly influential.

Foreign background as protection: All-foreign-background households receive protection in this case, consistent with certain immigration background codes showing lower transmission associations in the training data. This may reflect specific demographic subgroups, geographic clustering, social network structures, or healthcare access patterns present in 2020 Sweden.

School-age threshold: The slight risk increase from having a school-age child (6-17) versus preschool child (under 6) suggests the model learned that school age introduces elevated transmission pathways, even in 2020's disrupted education context. This 6-year-old sits right at that boundary.

Two-person household dynamics: Small households (especially parent-child pairs) consistently receive lower scores than larger families. Whether this reflects actual transmission biology, isolation capability, or simply mathematical constraints (fewer people = fewer transmission pathways) isn't directly observable from the model.

Missing socioeconomic context: The model has no information about actual living conditions, social support networks, childcare arrangements, or specific factors affecting this young single-parent household's transmission dynamics. Income and homecare usage are crude proxies for complex realities.

Model Confidence Note

The model expresses high confidence in the low-risk assessment: the household score (13%) sits well below the 25.6% threshold, and the individual score (17%) is clearly below the 30% threshold. The unanimous alignment of protective factors—young parent, young child, low income, no health vulnerabilities—creates an unambiguous low-risk signal in the model's learned patterns. However, as with all model outputs, this represents statistical associations from 2020 Swedish registry data, not guarantees about this specific household's outcome. Low scores indicate the model recognizes this household profile as similar to patterns that had minimal transmission in the training data, which can help prioritize limited public health resources toward higher-risk households. These outputs serve as one reference point for expert decision-making, not as standalone predictions of transmission outcomes.

Note: This analysis explains the model's confident low-risk assessment based on demographic and socioeconomic patterns, not actual infection outcomes or household-specific circumstances.

Evaluation Scores — Sample 10

Part A: Qualitative Metrics (max 2 per metric; A3 Expert assessed for Claude Sonnet 4.5 only)

Model	A1 Comp.	A2 Sound.	A3 Plaus. (Expert / Auto)	A4 Scope	Total (/8)
Claude Sonnet 4.5	2	2	2 / 2	2	8
GPT-5.3	2	2	— / 2	2	—
Grok 4.2	2	2	— / 2	2	—
DeepSeek V3.2	2	2	— / 2	2	—
Llama 4	2	2	— / 2	2	—

Part B: Quantitative Metrics (all values in %; Degr. = coverage degradation flags)

Model	B1 HH-R@2	B2 HH-R@3	B3 IND-R@2	B4 IND-R@3	B5 $\overline{\text{IND}}$ -R@2	B6 $\overline{\text{IND}}$ -R@3	B7 Dir.	B8 Faith.
Claude Sonnet 4.5	100%	100%	100%	100%	100%	100%	100%	100%
GPT-5.3	100%	100%	100%	100%	100%	100%	100%	100%
Grok 4.2	100%	100%	100%	100%	100%	100%	100%	100%
DeepSeek V3.2	100%	100%	100%	100%	100%	100%	100%	100%
Llama 4	100%	100%	100%	100%	100%	100%	100%	100%